



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df020-2018-android
Operator:	Floris
Analysis Started:	2026-06-18 14:27:27
Analysis Completed:	2026-06-18 14:52:44
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2018\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2018\dfdof_output\ft-df020-2018-android
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	452.1 MB	2026-06-18 14:27:46
df020_external.001	drone_sd	physical	14.9 GB	2026-06-18 14:28:32
df020_internal.001	drone_flight_storage	physical	3.7 GB	2026-06-18 14:28:43

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2018\android

Cryptographic Hashes

android_logical.zip		controller_android	logical
SHA-256:	a27e204eb7a55ccab1ee2c40d7a02cd59bb2ebb8bc1a008832d77dc228473ff0		
SHA-1:	d2c76b389fa8d1eaed8346c90d63069e68e7805b		
df020_external.001		drone_sd	physical
SHA-256:	23f64d700f5a4122e8773851a87c170974b17205f9f4432c9155150ee7021a0c		
SHA-1:	db63665a82ed93ab48e036c2a547990d4cfe546c		
df020_internal.001		drone_flight_storage	physical
SHA-256:	e58c68472a5604102d757e3255119c88206d7b68c3ddc52e5b9977f0216c3830		
SHA-1:	2b33830ae5e04e46831105c998b36e412243afce		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone flight storage]: mmls: no partition table found in df020_internal.001; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
6. [p3 - controller android - account data]: no artefacts found
7. [p3 - controller android - databases]: no artefacts found
8. [p3 - controller ios]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: WPSettings.dat
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY007.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY011.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY013.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016.csv
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY018.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY019.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY029.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY030.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY033.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY040.csv
26. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY041.csv
27. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY042.csv
28. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY043.csv
29. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY044.csv
30. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY045.csv
31. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY046.csv
32. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY047.csv
33. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY048.csv
34. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY049.csv
35. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY050.csv
36. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY051.csv
37. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY052.csv
38. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY053.csv
39. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY055.csv
40. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY056.csv
41. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY057.csv
42. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY058.csv
43. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY059.csv

44. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY060.csv
 45. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY062.csv
 46. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY063.csv
 47. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY064.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	64	63	[~] Partial
exiftool	20	17	[~] Partial
fls	2	2	[OK]
icat	69	69	[OK]
mmls	2	1	[~] Partial
parser_android	1	1	[OK]
txtlogtocsvtool	3	3	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	3 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	1 / 28
Tools Succeeded	4 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Drone Serial Number	0670136230 08RDE5F001007H 0K1DF313BD4PR1	[!] Inconsistent	3
Drone Model	Matrice 600 Pro Mavic Air Mavic Pro MavicAir MavicPro	[!] Inconsistent	29
DJI Application Version	3.1.11 4.2.16	[!] Inconsistent	3

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: db0990a4eef4965870582a5cad5d4a9541d97c5b85dc8db9ef707a0f0fdd4203
Drone Log Name: FLY054.csv
Start: 2015-10-18T00:00:30.235000+00:00
End: 2015-10-18T00:04:29.552000+00:00
Duration: 239.3 s

Flight Metrics

Peak Height: 29.402990702335103 m
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

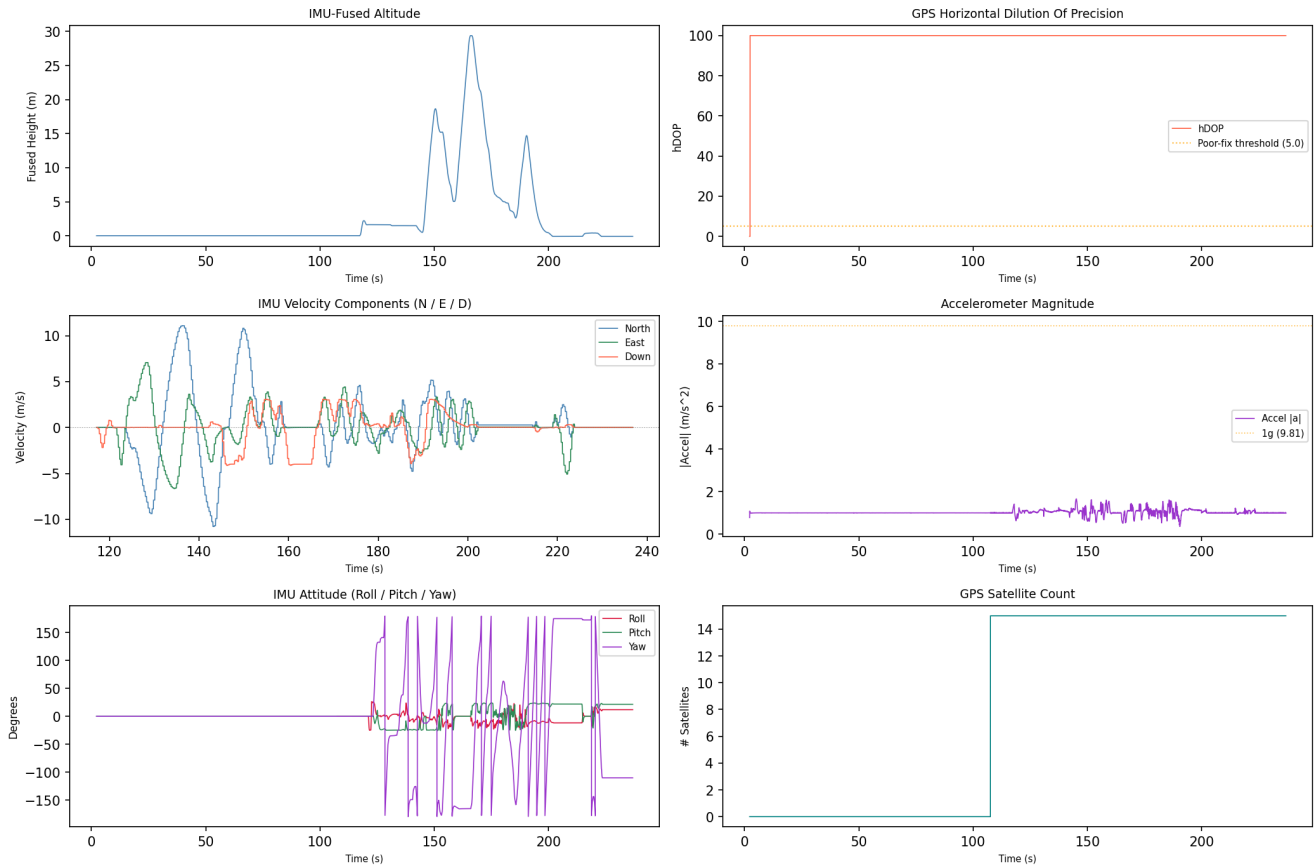
1. Atti
2. GPS_Atti
3. ASST_TAKEOFF
4. AssitedTakeoff
5. FORCE_LANDING
6. AutoLanding
7. Manual

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	29.4 m
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
2015-10-18 00:00:35	drone_flight_storage:dl	Log started	MavicPro
2015-10-18 00:00:35	drone_flight_storage:dl	Log ended	
2015-10-18 00:00:35	drone_flight_storage:dl	Motor turned off	off

Drone Log - Sensor Telemetry

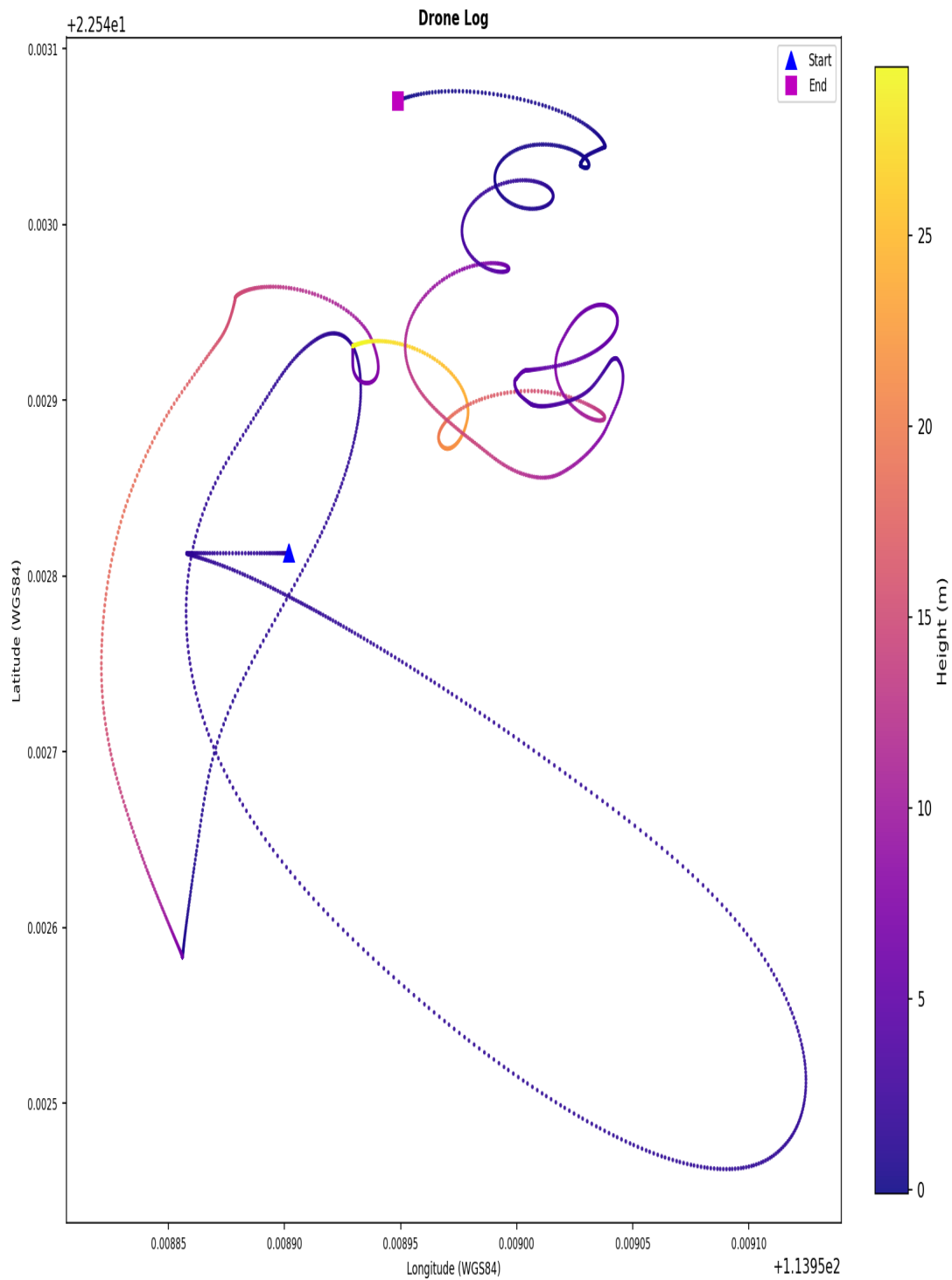
Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY054.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY054.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: d4fe793adabbb9839f6bede86dc31bfe1f5d74d6a3e4d6ee93f05291f6ba9dbe
Drone Log Name: FLY010.csv
Start: 2015-10-18T00:01:17.230000+00:00
End: 2015-10-18T00:01:41.589000+00:00
Duration: 24.4 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

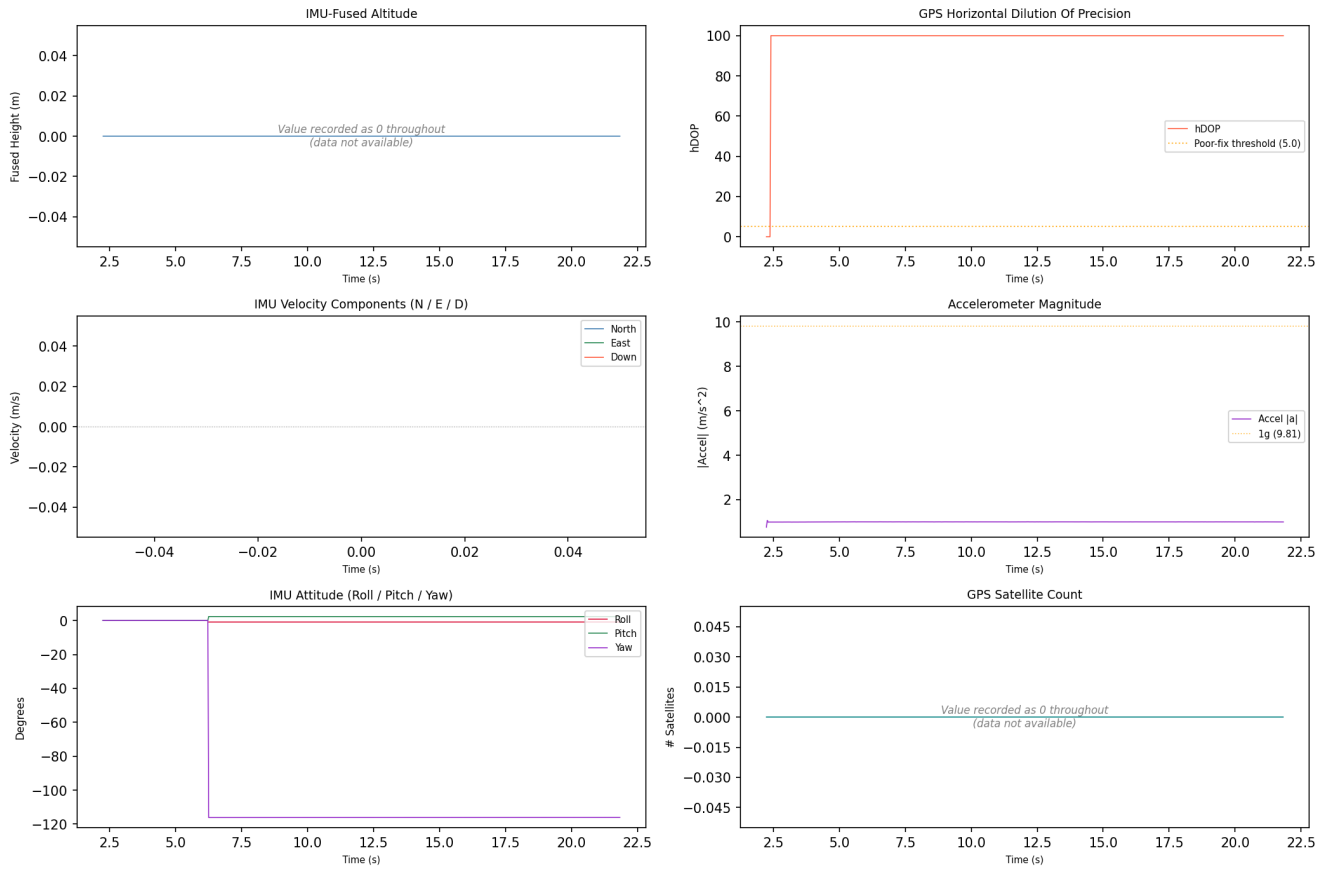
1. Atti
2. Manual

Key Events

Timestamp (UTC)	Source	Event	Details
2015-10-18 00:01:22	drone_flight_storage:dl	Log started	MavicPro
2015-10-18 00:01:22	drone_flight_storage:dl	Log ended	
2015-10-18 00:01:22	drone_flight_storage:dl	Reached peak height	-0.0 m
2015-10-18 00:01:22	drone_flight_storage:dl	Motor turned off	off

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: d4476d071663554e058704387b9a69290947ca02660896cde83823e404c41c55
Drone Log Name: FLY061.csv
Start: 2017-08-29T18:20:40.739000+00:00
End: 2017-08-29T18:21:52.068000+00:00
Duration: 71.3 s

Flight Metrics

Peak Height: 0.0836210596987712 m
Distance Travelled: 0.00317199515511295 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

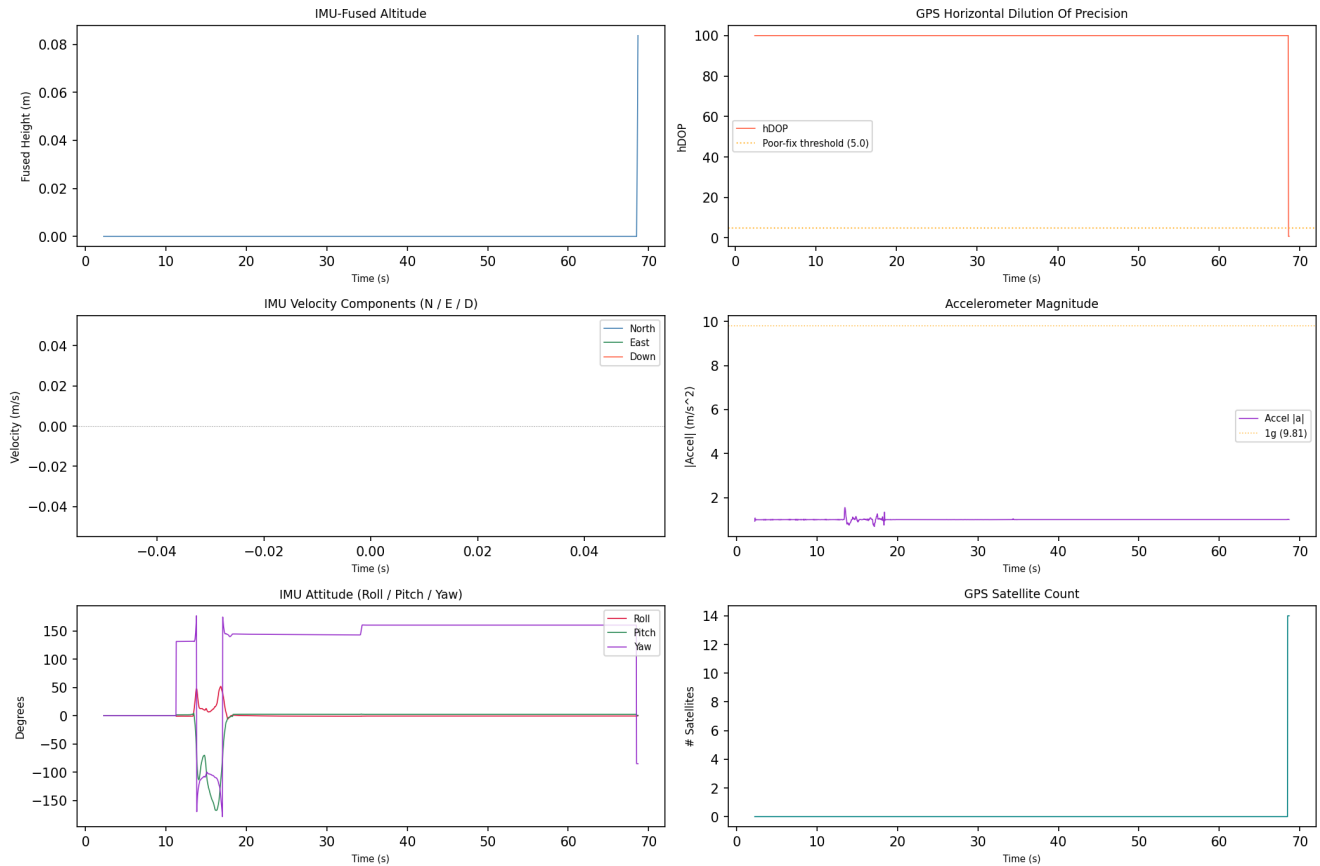
1. Manual
2. Atti
3. AutoTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2017-08-29 18:21:52	drone_flight_storage:dl	Log started	MavicPro
2017-08-29 18:21:52	drone_flight_storage:dl	Log ended	
2017-08-29 18:21:52	drone_flight_storage:dl	Reached peak height	0.1 m

Drone Log - Sensor Telemetry

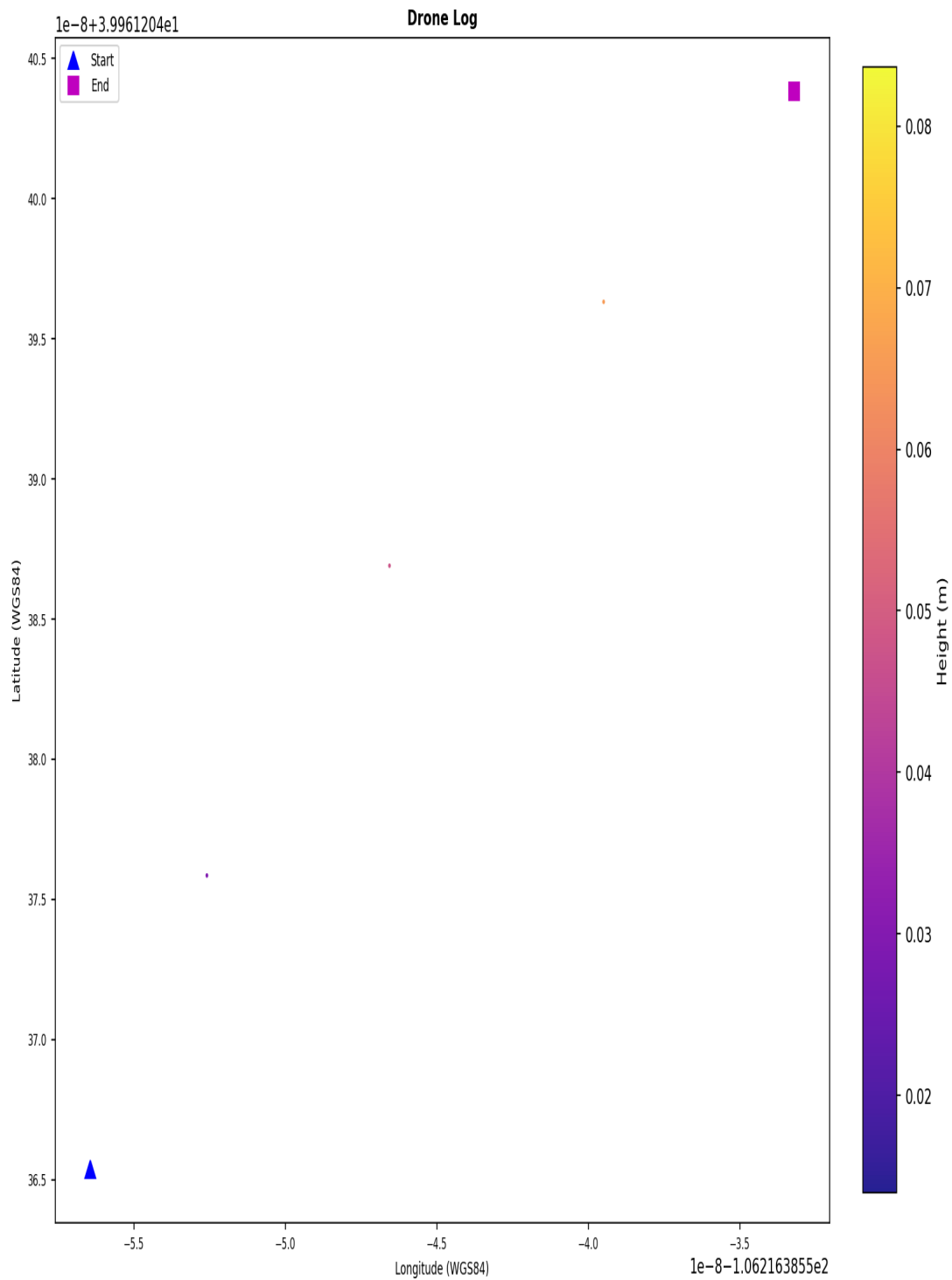
Drone Log Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_03



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY061.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY061.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: eb147a2263a033419d84a1ddf76ec224bb14d75afd3d7e60721913917a4be493
Flight Record Name: DJIFlightRecord_2017-10-10_[10-02-22].csv
Start: 2017-10-10T16:02:22.901000+00:00
End: 2017-10-10T16:07:59.866000+00:00
Duration: 337.0 s

Flight Metrics

Peak Height: 35.3 m
Distance Travelled: 2723.32 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-10-10 16:02:22	ctrl_and:fr (v3.1.11)	Log started	SN: 0670136230; App: 3.1.11; Matrice 600 Pro
2017-10-10 16:02:22	ctrl_and:fr (v3.1.11)	Motor turned on	on
2017-10-10 16:02:23	ctrl_and:fr (v3.1.11)	Record mode changed	No
2017-10-10 16:02:47	ctrl_and:fr (v3.1.11)	Record mode changed	Starting
2017-10-10 16:03:48	ctrl_and:fr (v3.1.11)	Reached peak height	35.3 m
2017-10-10 16:07:59	ctrl_and:fr (v3.1.11)	Log ended	Dist home: 2.0 m; Height: -1.5 m; Rec: 0.0 s

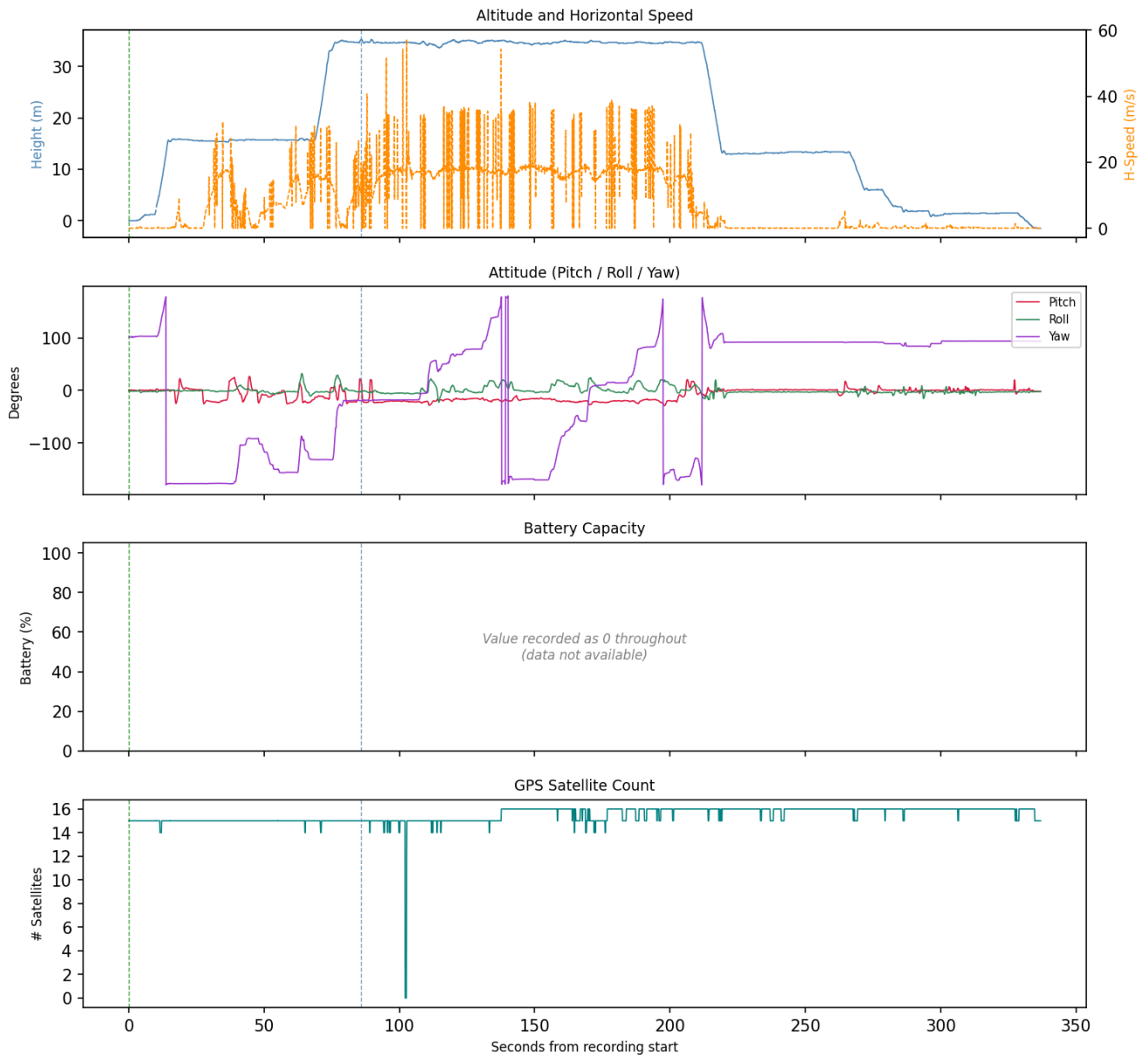
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	
controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cfc7b341bd62ed6b	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:0f0eb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24	

Flight Record - Sensor Telemetry

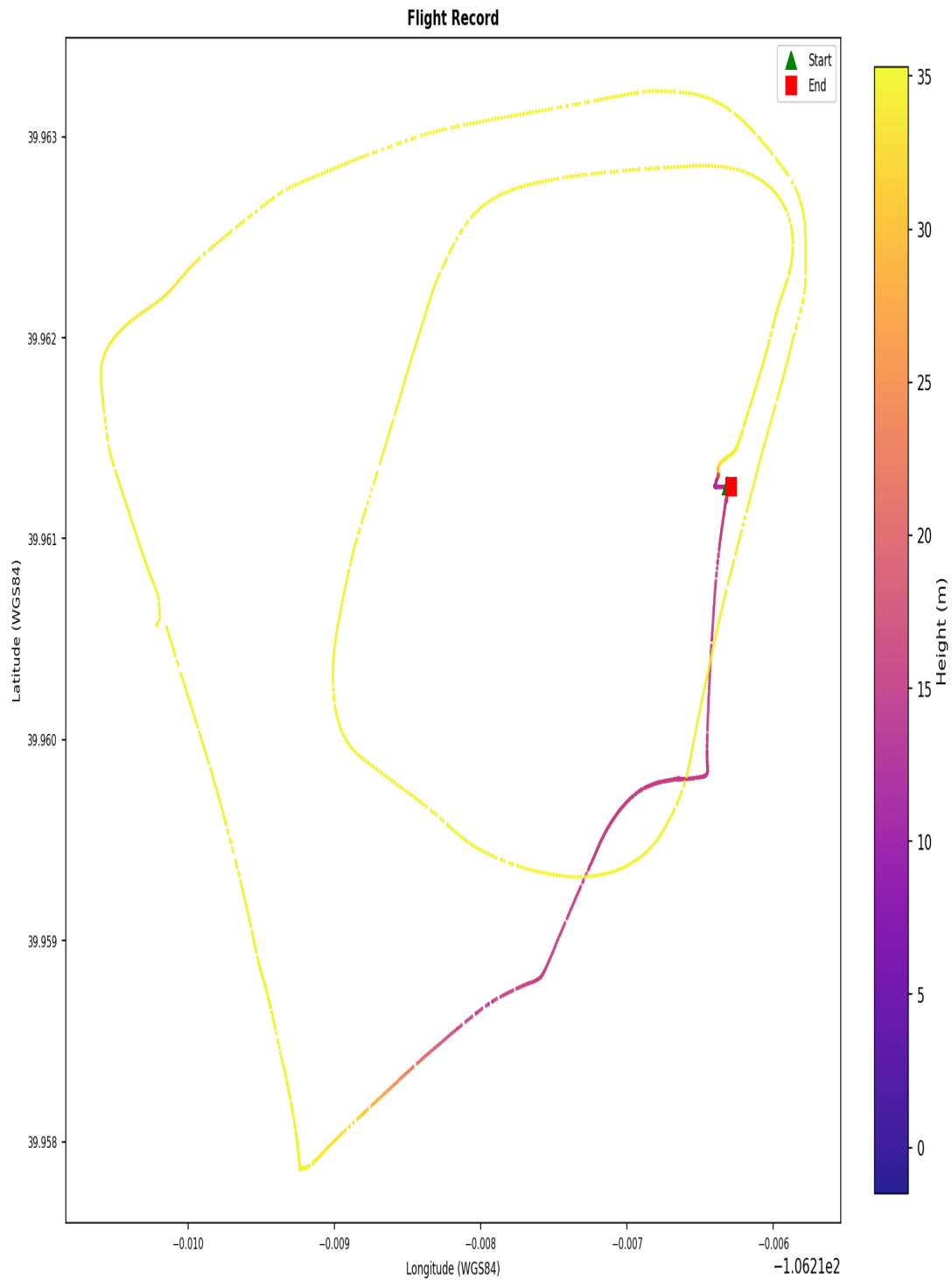
Flight Record Telemetry - flight_04



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_04



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 3f5f966fab1200781515fd6cc5fa59cdb0bae8e8008ced4aa76c4b027723494d
Drone Log Name: FLY004.csv
Start: 2017-10-10T16:41:33.653000+00:00
End: 2017-10-10T16:42:42.246000+00:00
Duration: 68.6 s

Flight Metrics

Peak Height: -
Distance Travelled: 1.6462191685826097 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

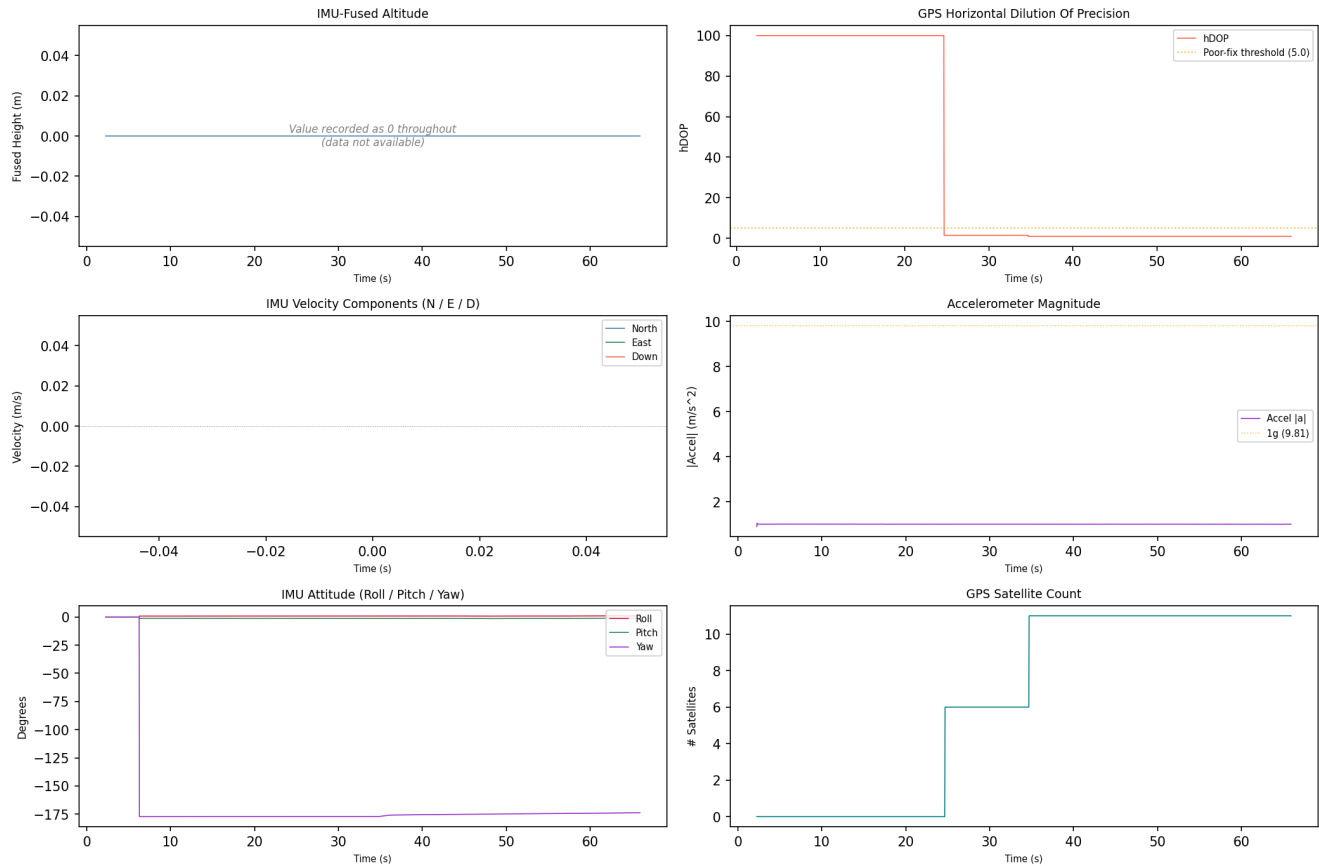
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-10-10 16:42:01	drone_flight_storage:dl	Log started	MavicPro
2017-10-10 16:42:41	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_05



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: a6905b47f17a20a94a0b0cad6d7d419964459cc8d75bc653eb01dfaea79ff4d3
Drone Log Name: FLY005.csv
Start: 2017-10-10T16:42:44.865000+00:00
End: 2017-10-10T16:48:46.173000+00:00
Duration: 361.3 s

Flight Metrics

Peak Height: 2.2283837525787042 m
Distance Travelled: 65.90613695941164 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

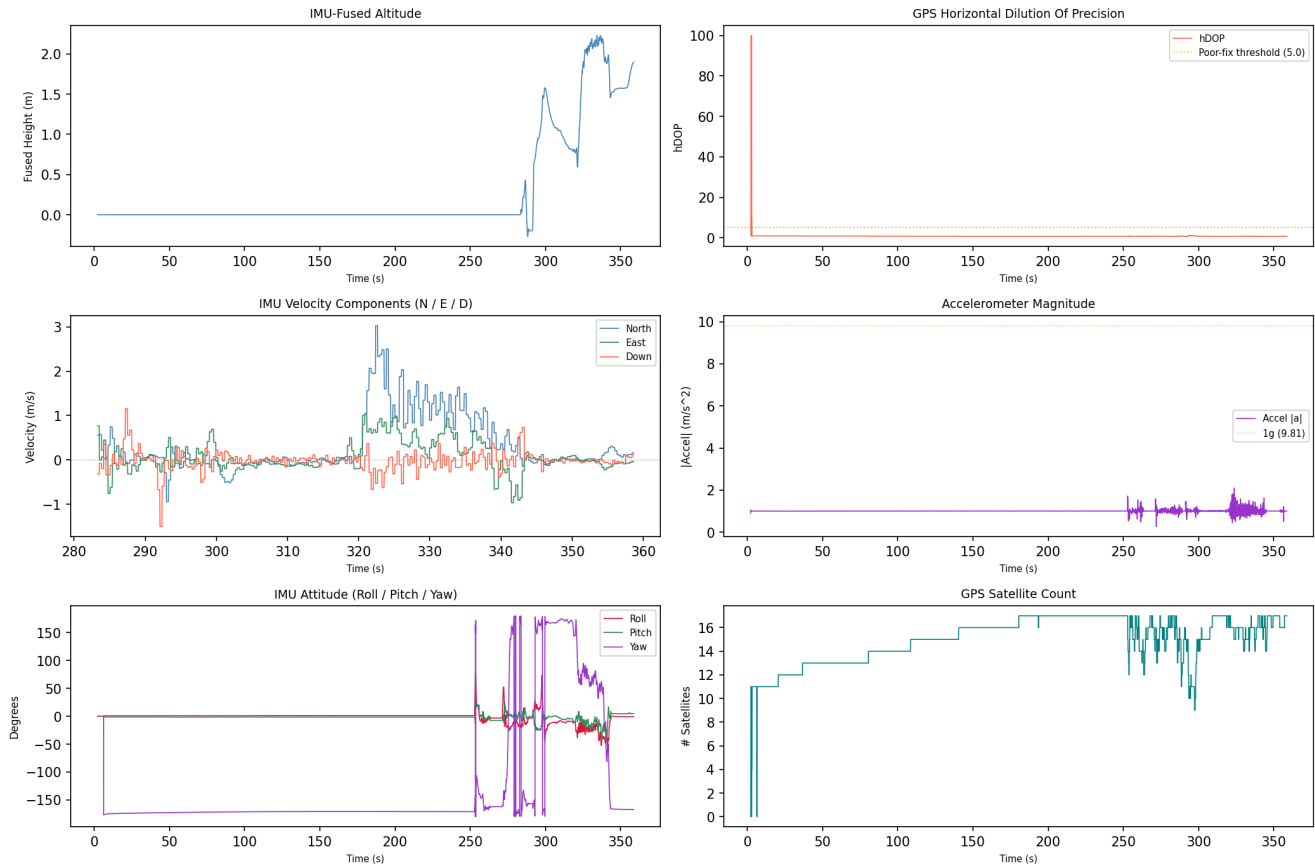
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-10-10 16:42:50	drone_flight_storage:dl	Log started	MavicPro
2017-10-10 16:48:24	drone_flight_storage:dl	Reached peak height	2.2 m
2017-10-10 16:48:46	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

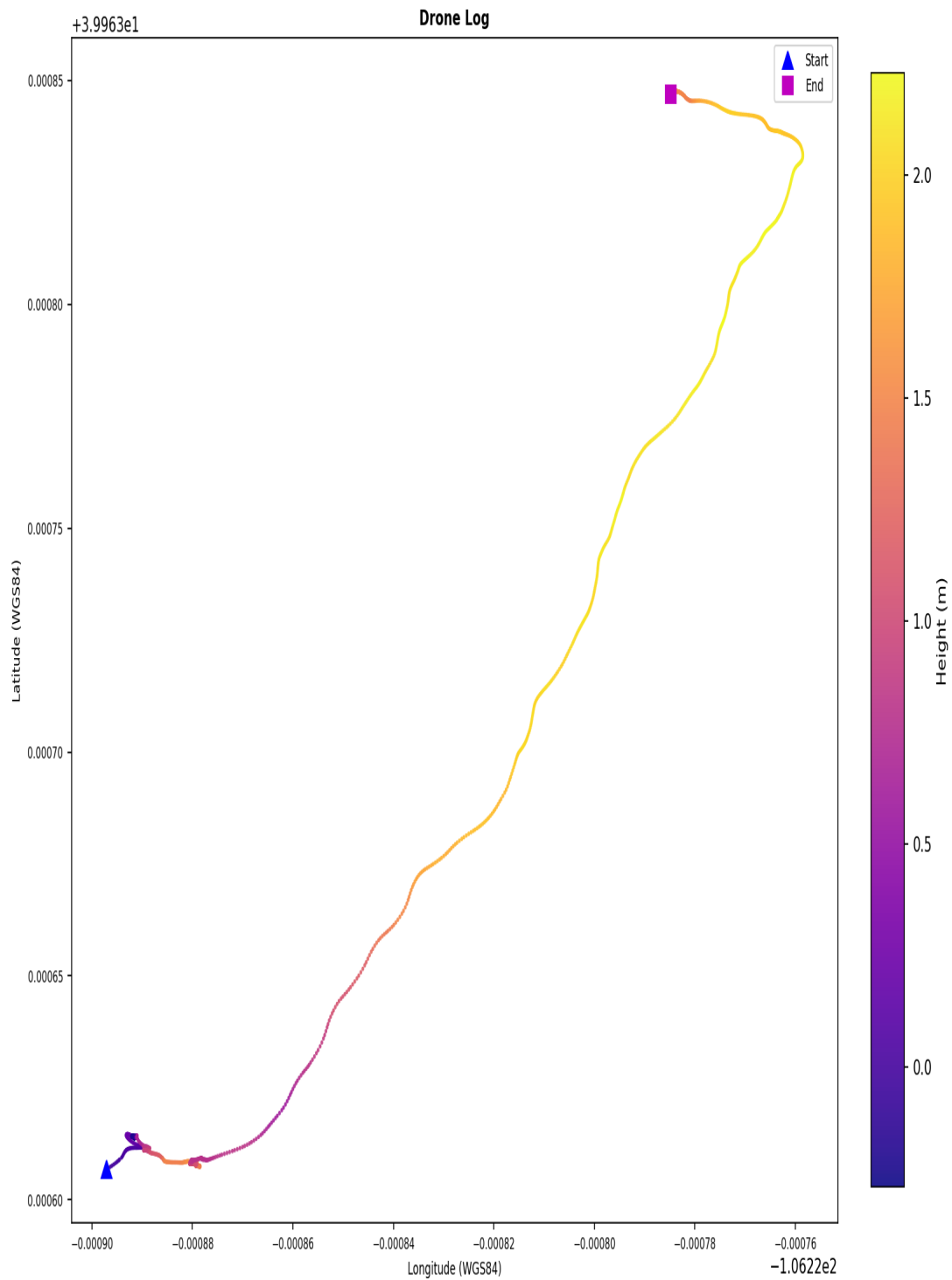
Drone Log Telemetry - flight_06



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_06



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY005.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: c305d9ec5ebd05865dec89b26e27fa960730f4603c5e9b44fa41d429562be7de
Drone Log Name: FLY006.csv
Start: 2017-10-10T16:48:54.959000+00:00
End: 2017-10-10T17:12:55.089000+00:00
Duration: 1440.1 s

Flight Metrics

Peak Height: 50.87775079822495 m
Distance Travelled: 3796.0917122962856 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

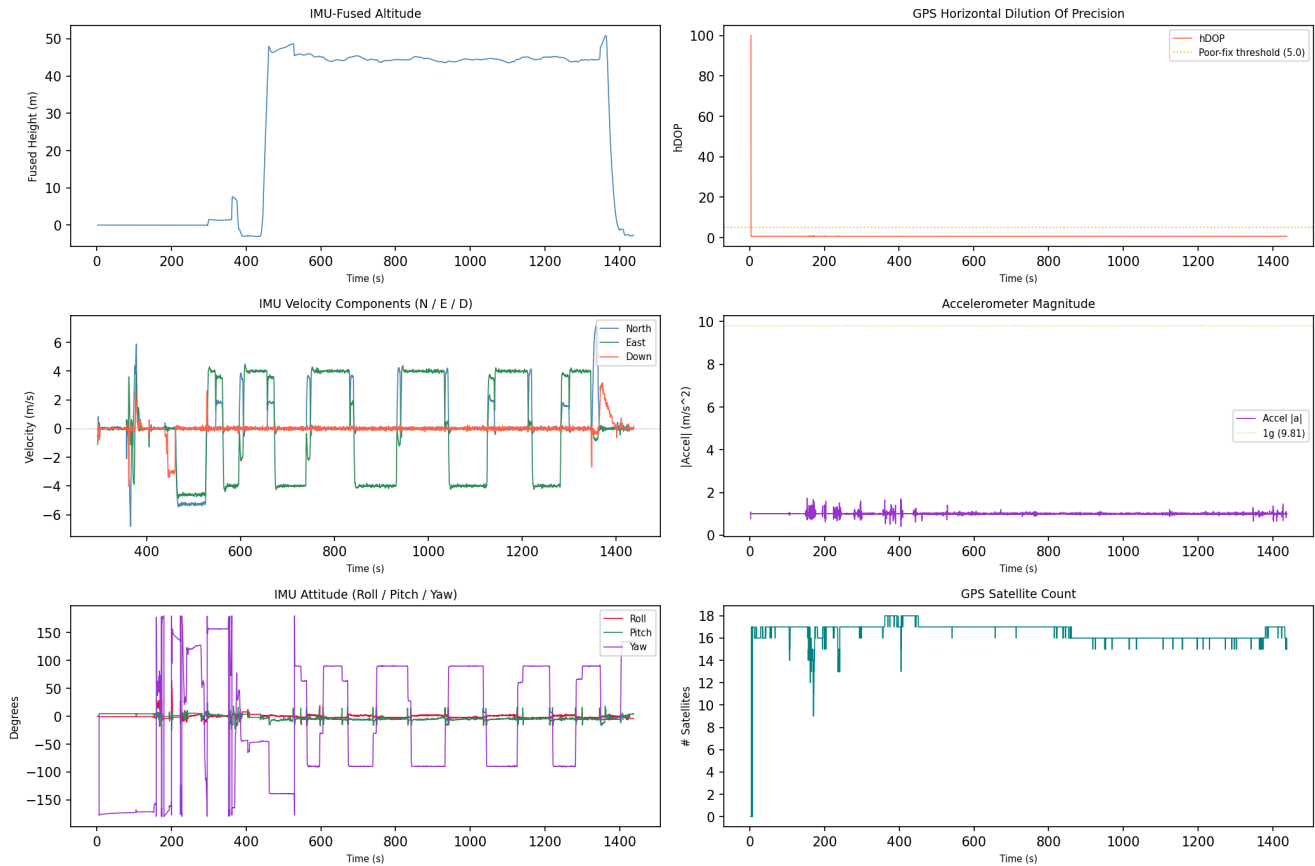
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. NaviGo
7. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-10-10 16:49:02	drone_flight_storage:dl	Log started	MavicPro
2017-10-10 16:53:53	drone_flight_storage:dl	Motor turned on	on
2017-10-10 16:55:28	drone_flight_storage:dl	Motor turned off	off
2017-10-10 16:56:15	drone_flight_storage:dl	Motor turned on	on
2017-10-10 17:11:41	drone_flight_storage:dl	Reached peak height	50.9 m
2017-10-10 17:12:46	drone_flight_storage:dl	Motor turned off	off
2017-10-10 17:12:55	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

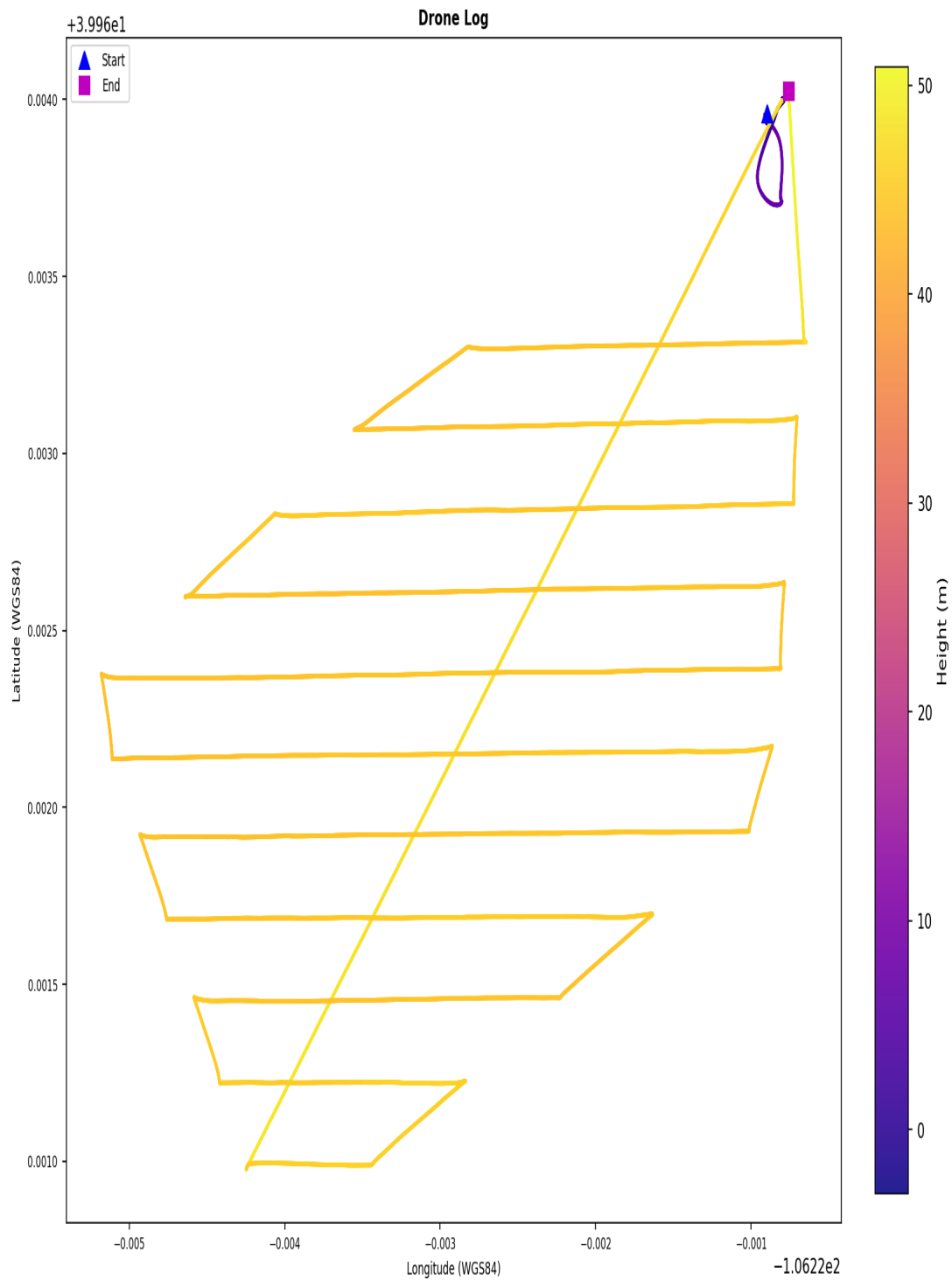
Drone Log Telemetry - flight_07



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_07



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY006.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 08

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: b654b343923bb048f9cb412fa6c873adab0a0a37833dcaff271672ff691b4430
Drone Log Name: FLY017.csv
Start: 2017-11-02T17:28:23.214000+00:00
End: 2017-11-02T17:37:53.480000+00:00
Duration: 570.3 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

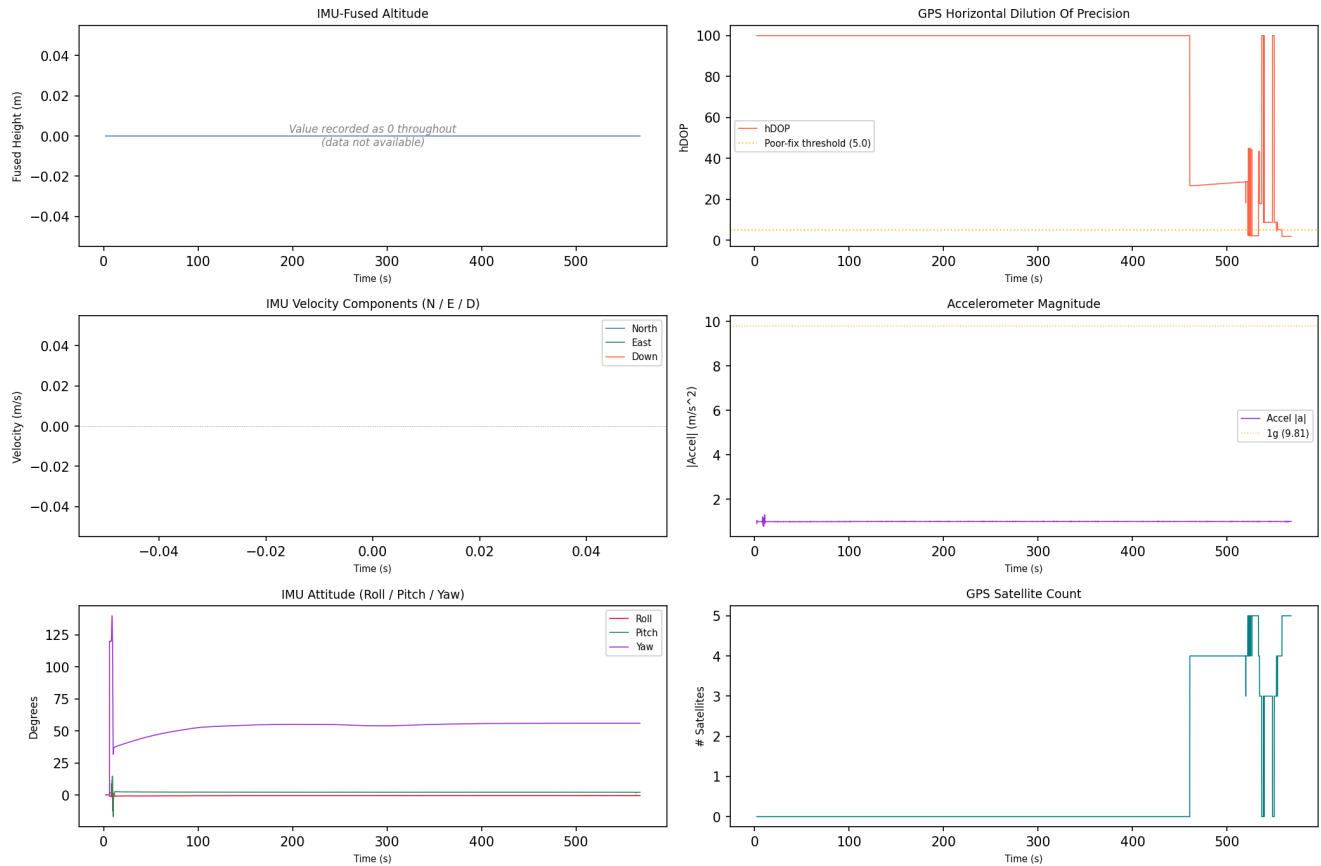
1. Manual
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-11-02 17:37:10	drone_flight_storage:dl	Log started	MavicPro
2017-11-02 17:37:53	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_08



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 09

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 67f0a04a43e15a3e3dbc8339f2176f0578fe61c1cf80f48c8fe4e1e1cb79a0e5
Drone Log Name: FLY020.csv
Start: 2018-01-30T21:03:17.177000+00:00
End: 2018-01-30T21:09:30.557000+00:00
Duration: 373.4 s

Flight Metrics

Peak Height: 16.080509364165643 m
Distance Travelled: 300.87057314900824 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

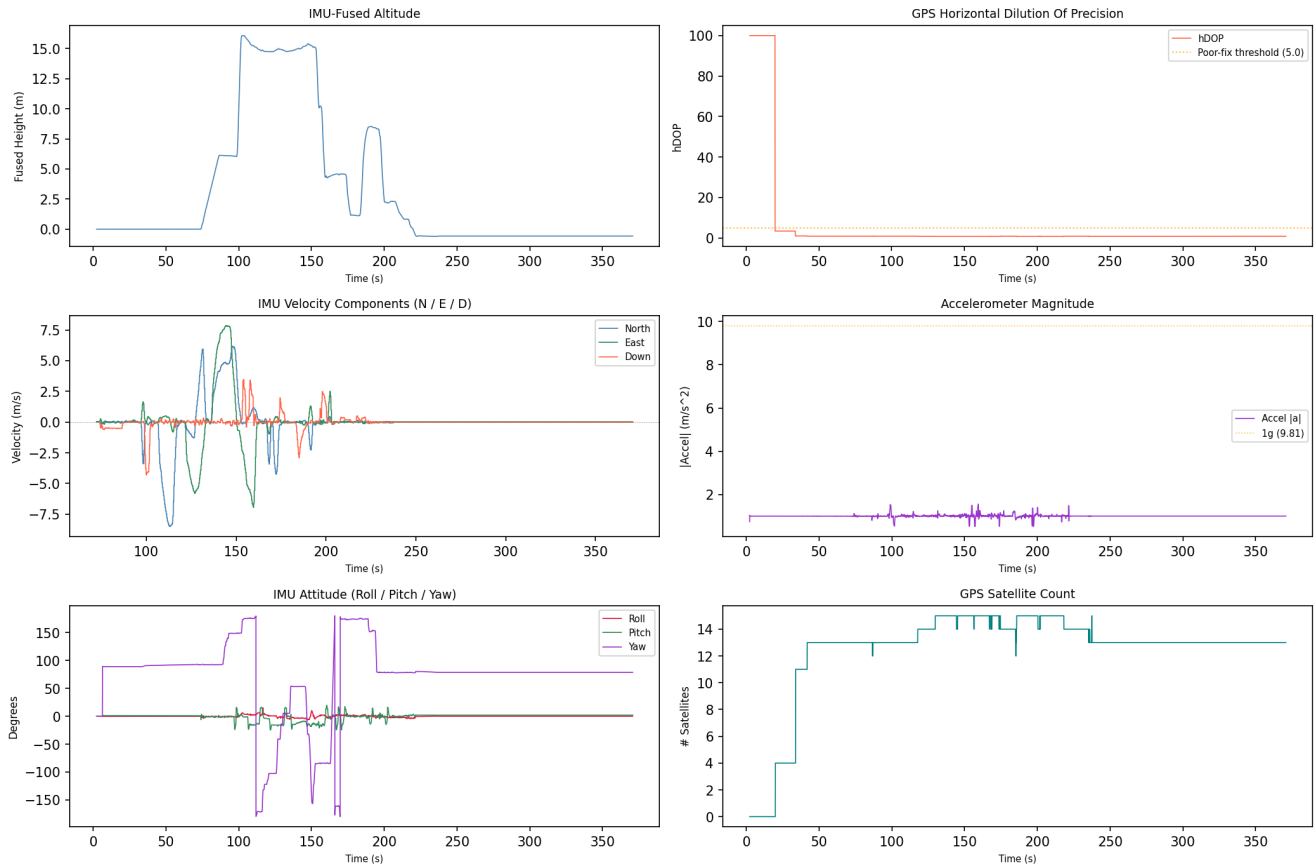
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. GoHome
7. AutoLanding
8. FORCE_LANDING

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:03:40	drone_flight_storage:dl	Log started	MavicPro
2018-01-30 21:04:32	drone_flight_storage:dl	Motor turned on	on
2018-01-30 21:05:03	drone_flight_storage:dl	Reached peak height	16.1 m
2018-01-30 21:07:02	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:07:17	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

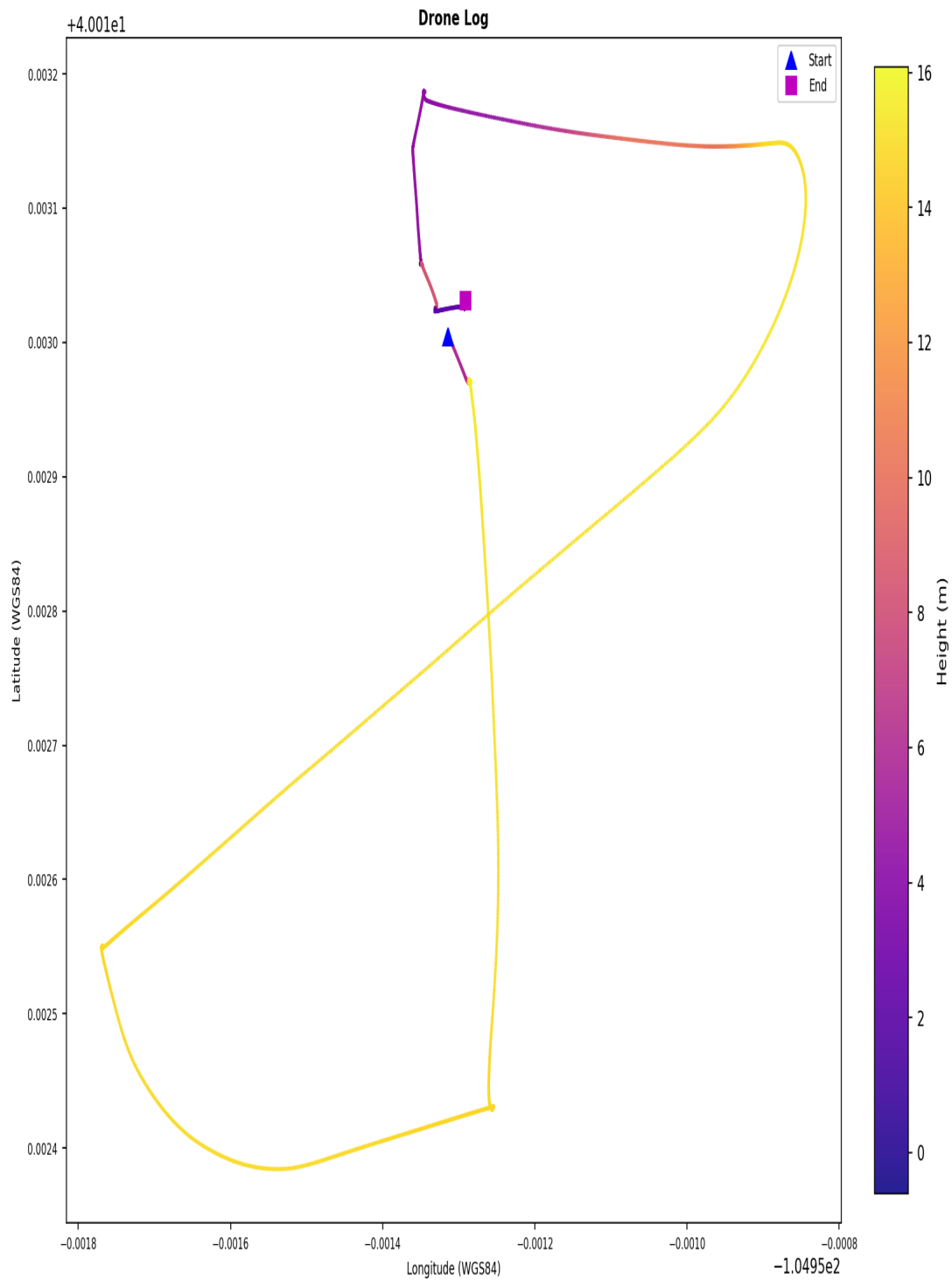
Drone Log Telemetry - flight_09



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_09



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY020.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 10

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 49ff942f9f350022e409017c8e7f026f88e046ba6e726c4d74b1bd9e9067b908
Drone Log Name: FLY021.csv
Start: 2018-01-30T21:23:44.817000+00:00
End: 2018-01-30T21:33:46.935000+00:00
Duration: 602.1 s

Flight Metrics

Peak Height: 14.36521400690044 m
Distance Travelled: 168.28624524116816 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

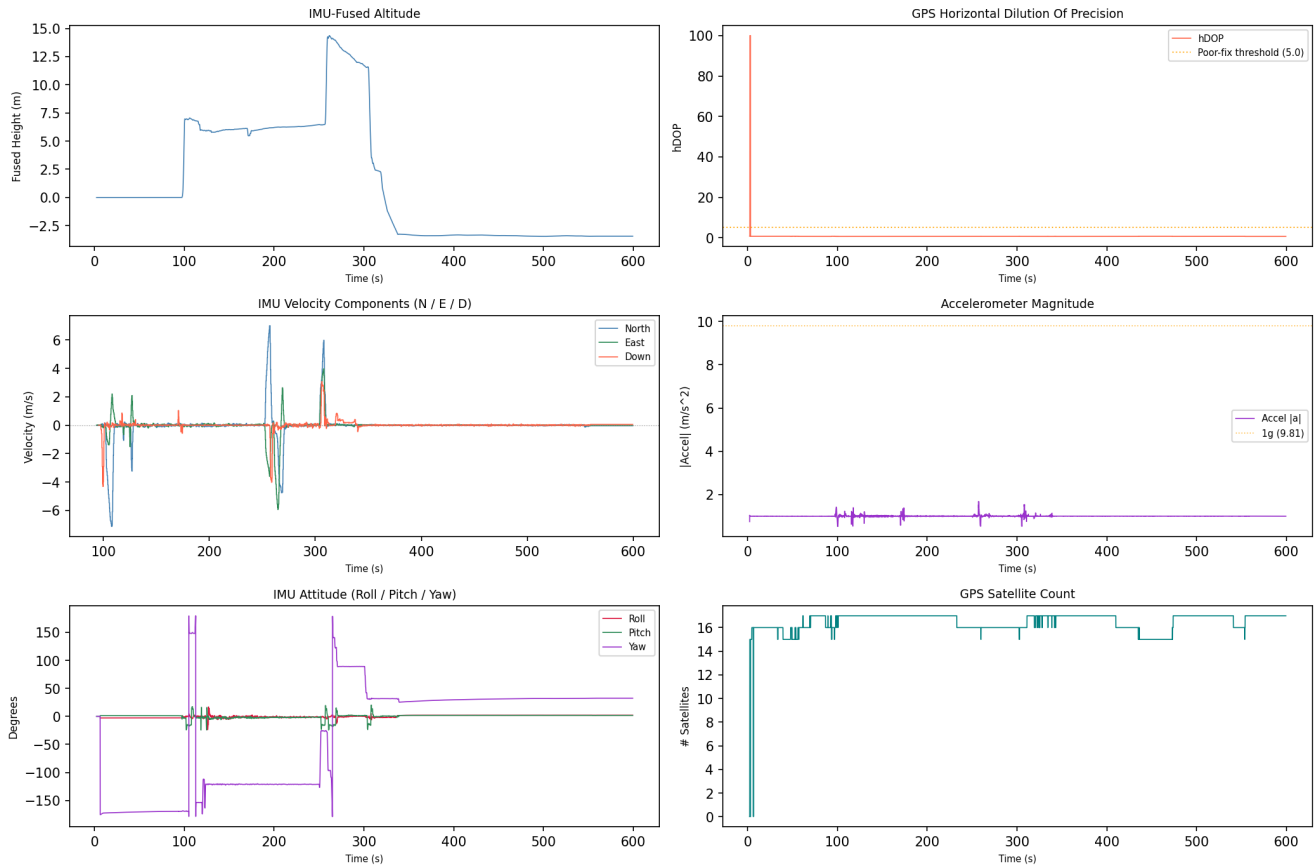
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:23:50	drone_flight_storage:dl	Log started	MavicPro
2018-01-30 21:25:21	drone_flight_storage:dl	Motor turned on	on
2018-01-30 21:28:09	drone_flight_storage:dl	Reached peak height	14.4 m
2018-01-30 21:29:27	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:33:08	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

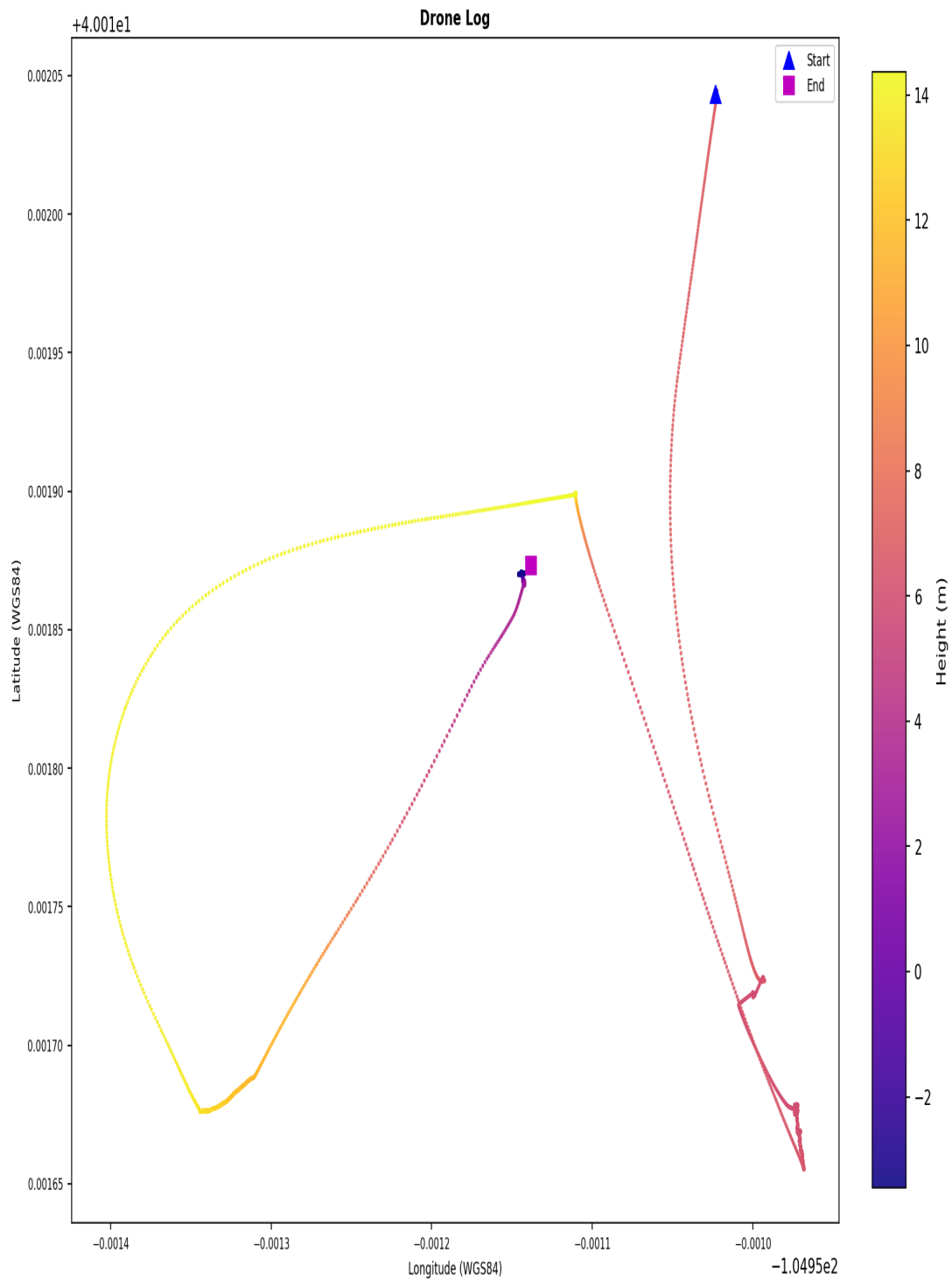
Drone Log Telemetry - flight_10



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_10



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY021.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY021.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 11

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: e77e18b395ef051c4128d70c0455776bba1056d6b8c7676d9808b76537919a08
Drone Log Name: FLY022.csv
Start: 2018-01-30T21:43:31.072000+00:00
End: 2018-01-30T21:55:30.571000+00:00
Duration: 719.5 s

Flight Metrics

Peak Height: 6.256511406987819 m
Distance Travelled: 307.5618577405231 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

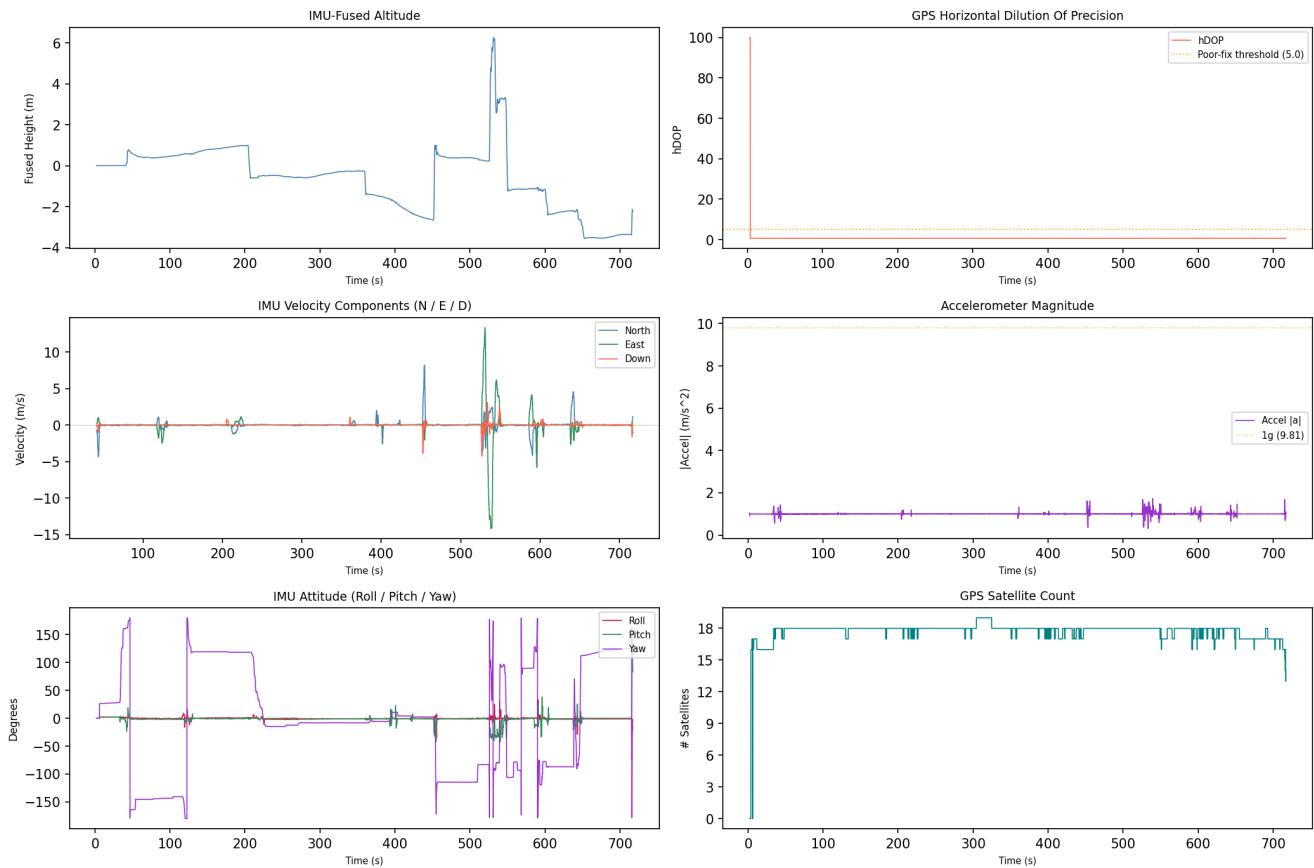
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. SPORT
7. FORCE_LANDING
8. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:43:37	drone_flight_storage:dl	Log started	MavicPro
2018-01-30 21:44:04	drone_flight_storage:dl	Motor turned on	on
2018-01-30 21:52:25	drone_flight_storage:dl	Reached peak height	6.3 m
2018-01-30 21:54:27	drone_flight_storage:dl	Motor turned off	off
2018-01-30 21:55:30	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

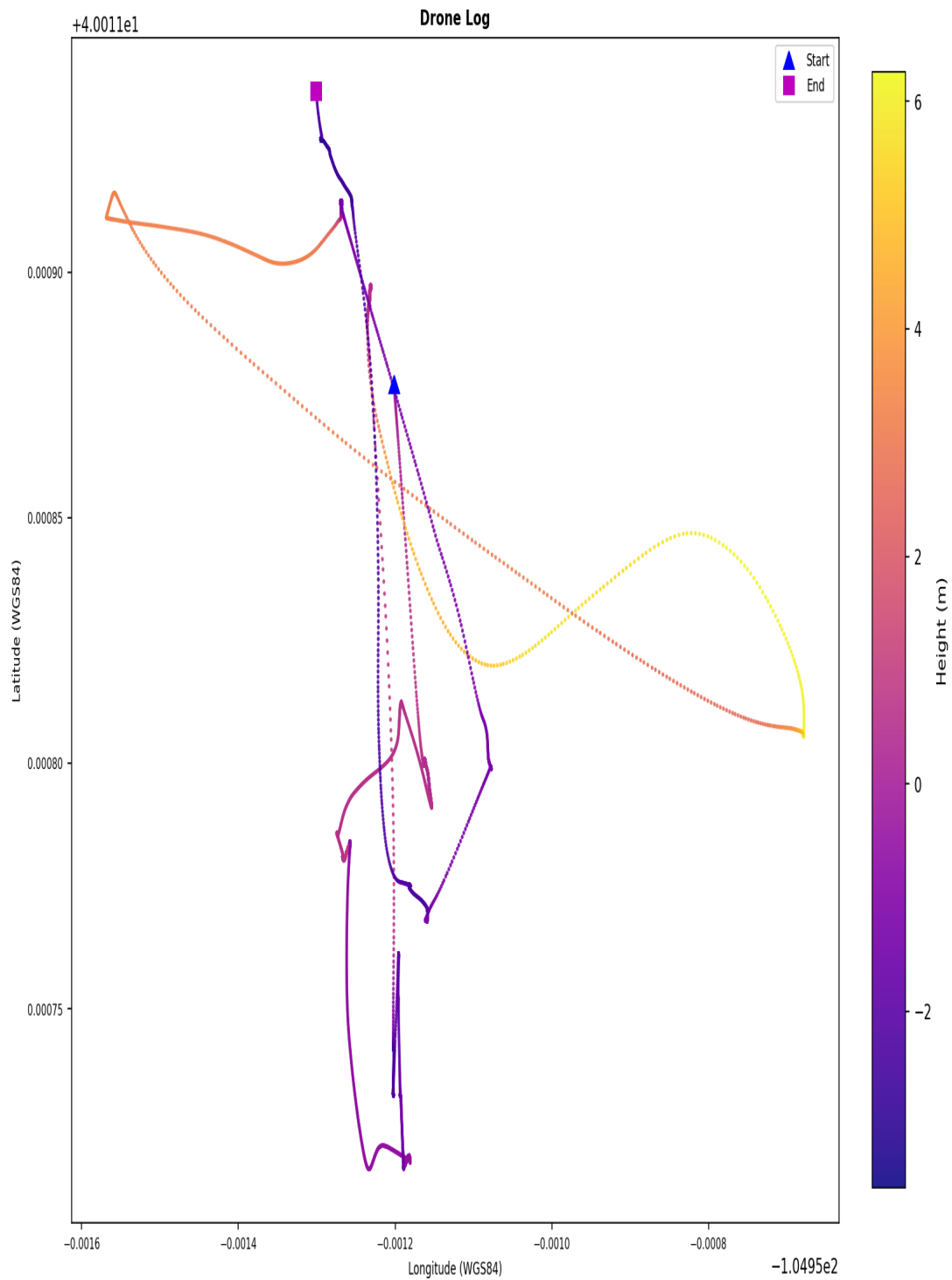
Drone Log Telemetry - flight_11



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_11



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY022.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY022.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 12

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: c62fb76dea07a70d9de9c19312224070d92dc27d5118d5d8aa5c0f5f602cee75
Drone Log Name: FLY023.csv
Start: 2018-01-30T22:17:05.072000+00:00
End: 2018-01-30T22:28:35.760000+00:00
Duration: 690.7 s

Flight Metrics

Peak Height: 34.4000692620227 m
Distance Travelled: 361.1119154406475 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

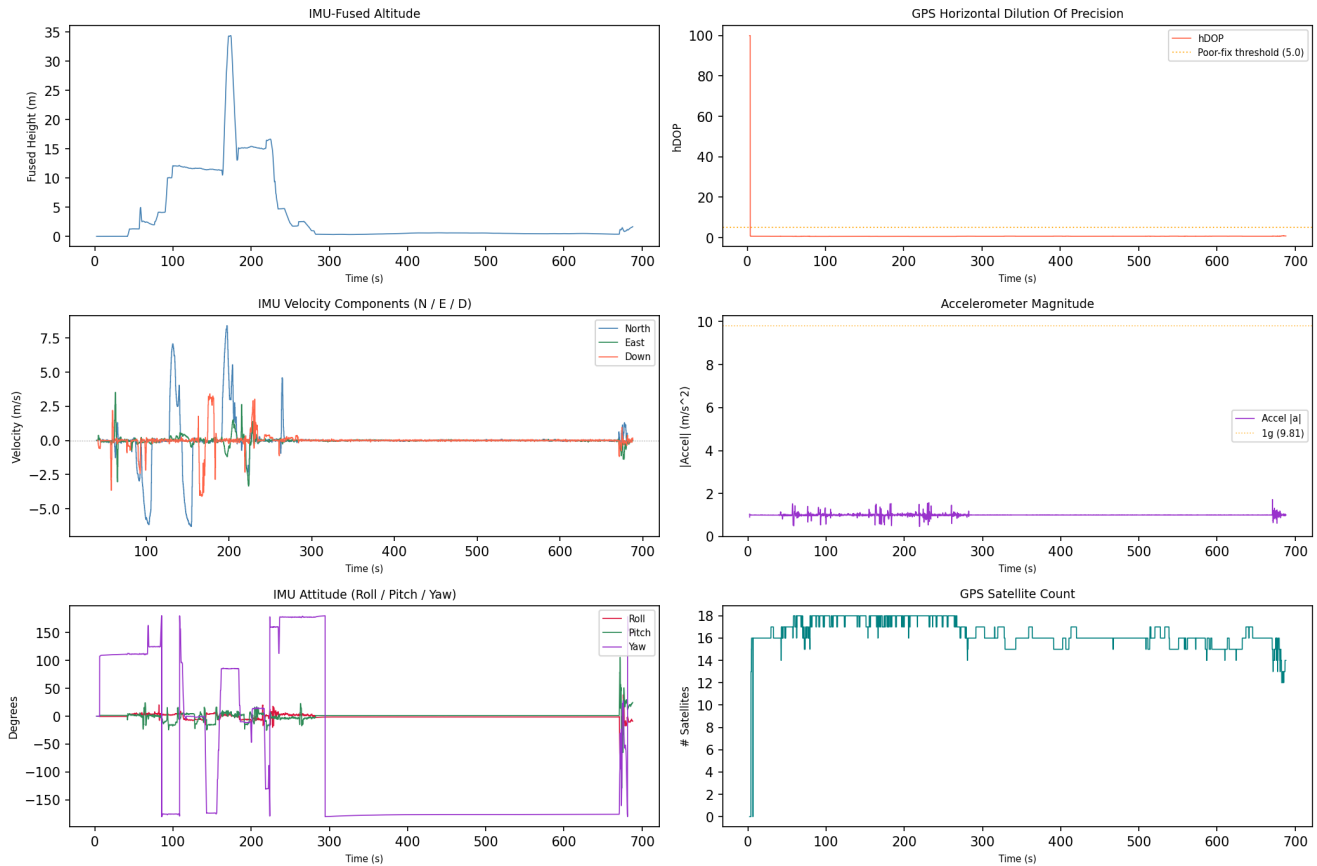
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. AutoLanding
7. FORCE_LANDING

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-01-30 22:17:11	drone_flight_storage:dl	Log started	MavicPro
2018-01-30 22:17:48	drone_flight_storage:dl	Motor turned on	on
2018-01-30 22:20:02	drone_flight_storage:dl	Reached peak height	34.4 m
2018-01-30 22:21:52	drone_flight_storage:dl	Motor turned off	off
2018-01-30 22:28:36	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_12



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

4. Flight Analysis - Flight 13

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: af03392cd1fda25b5895ef1096066102601e4e3087473481b09c45a73f0c97f3
Drone Log Name: FLY024.csv
Start: 2018-02-23T18:07:59.351000+00:00
End: 2018-02-23T18:33:09.146000+00:00
Duration: 1509.8 s

Flight Metrics

Peak Height: 21.29777336906465 m
Distance Travelled: 217.0243445184562 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. FORCE_LANDING
7. AutoLanding
8. GoHome
9. AutoTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:21:54	drone_flight_storage:dl	Log started	MavicPro
2018-02-23 18:22:39	drone_flight_storage:dl	Reached peak height	21.3 m
2018-02-23 18:24:32	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:27:34	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:28:43	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:28:47	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:28:58	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:29:01	drone_flight_storage:dl	Motor turned on	on

Timestamp (UTC)	Source	Event	Details
2018-02-23 18:29:50	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:30:04	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:30:31	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:30:43	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:31:23	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:31:24	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:31:36	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:31:42	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:33:02	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:33:09	drone_flight_storage:dl	Log ended	

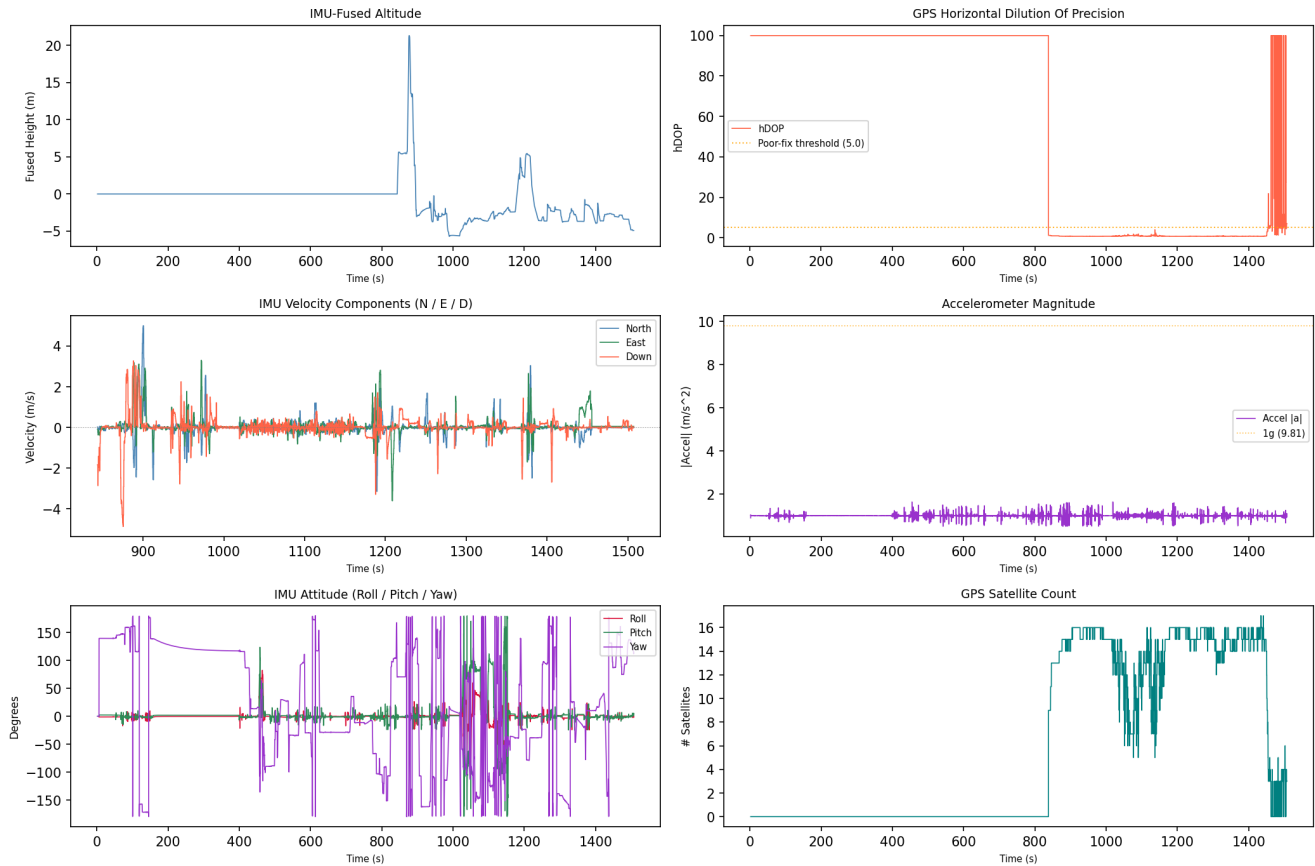
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	
controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cfcfb7b341bd62ed6b	
drone_sd:images	GPS within flight bounding box (lat 39.9612, lon 106.2165)
p5:96a25a259d992c3d34222683e21a8591c808ce0162178ffadd50edddcc03c9578	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896	
drone_sd:videos	GPS within flight bounding box (lat 39.9637, lon -106.2141)
p5:355b378ae4911bb1566e0254cf3c299c34c84c15b9f835fc4ed9a1d361b8a01c	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:0f0eb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24	

Drone Log - Sensor Telemetry

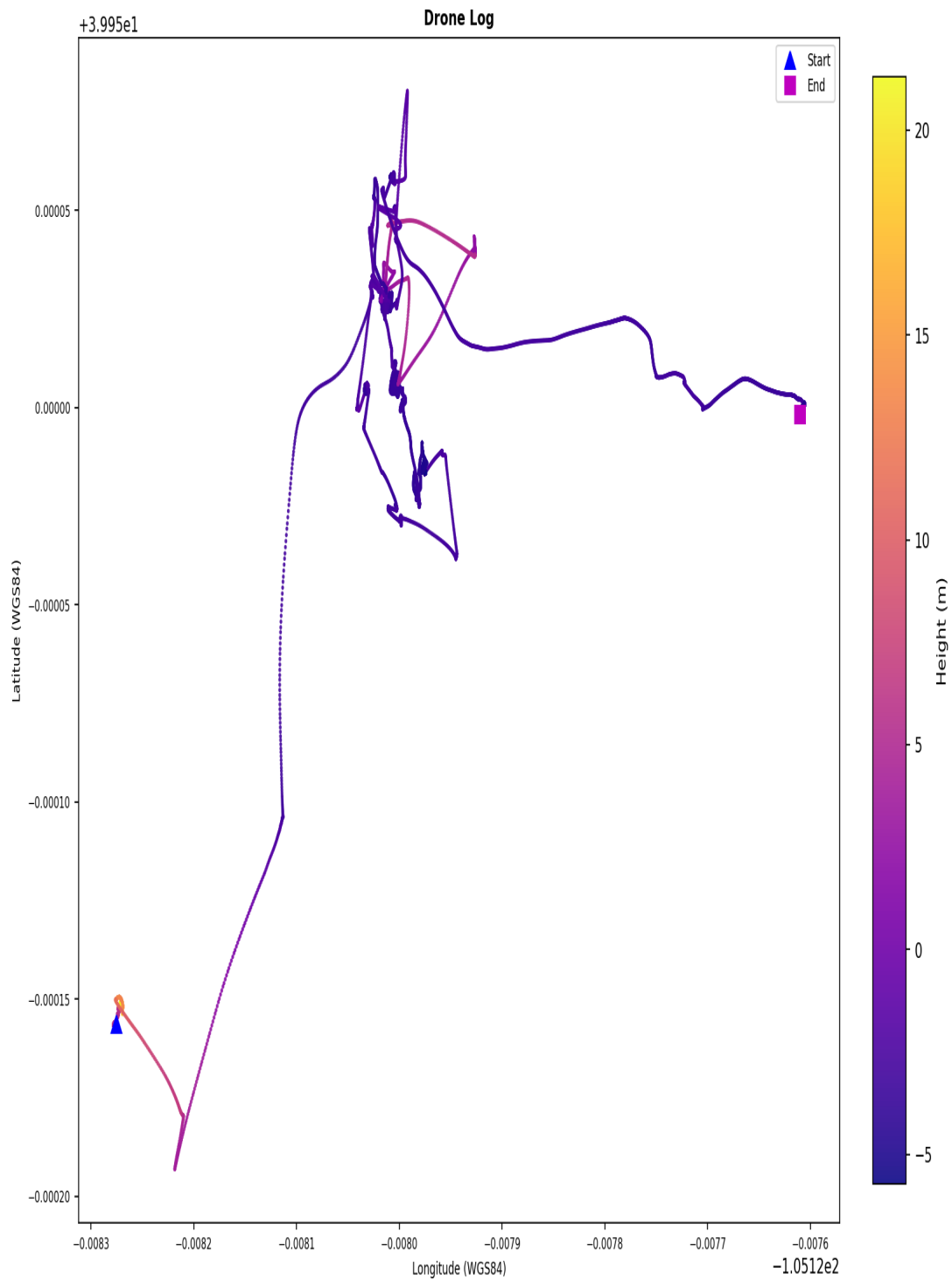
Drone Log Telemetry - flight_13



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hdOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hdOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hdOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_13



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY024.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY024.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 14

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 83f8743c528f9ed5b3a2cad50618fdb6be7fa9721ccee8e9353de4141b137873
Drone Log Name: FLY025.csv
Start: 2018-02-23T18:33:59.439000+00:00
End: 2018-02-23T18:41:27.083000+00:00
Duration: 447.6 s

Flight Metrics

Peak Height: 38.58680871214903 m
Distance Travelled: 510.03009223452574 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

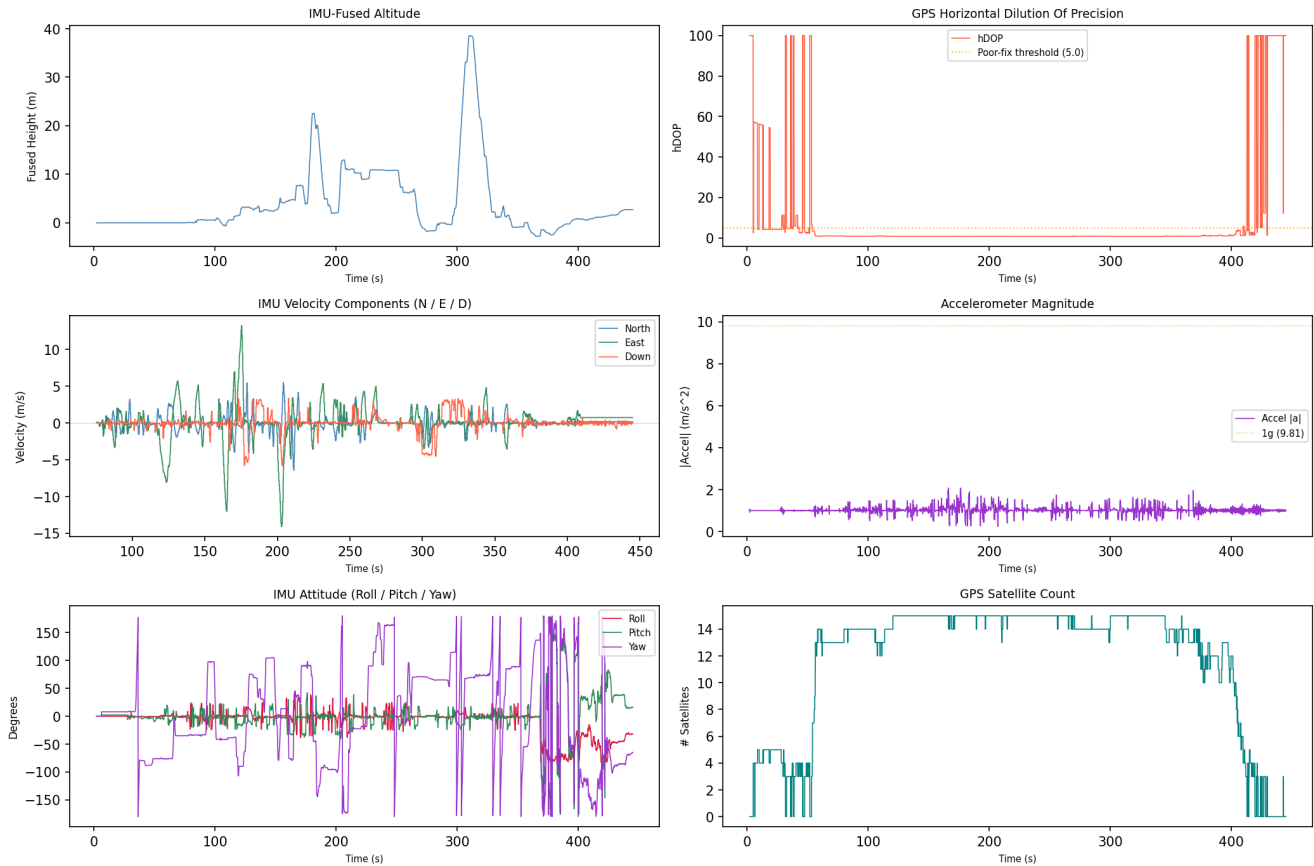
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. SPORT
7. FORCE_LANDING
8. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:34:32	drone_flight_storage:dl	Log started	MavicPro
2018-02-23 18:38:38	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:38:42	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:39:12	drone_flight_storage:dl	Reached peak height	38.6 m
2018-02-23 18:40:08	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:41:25	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

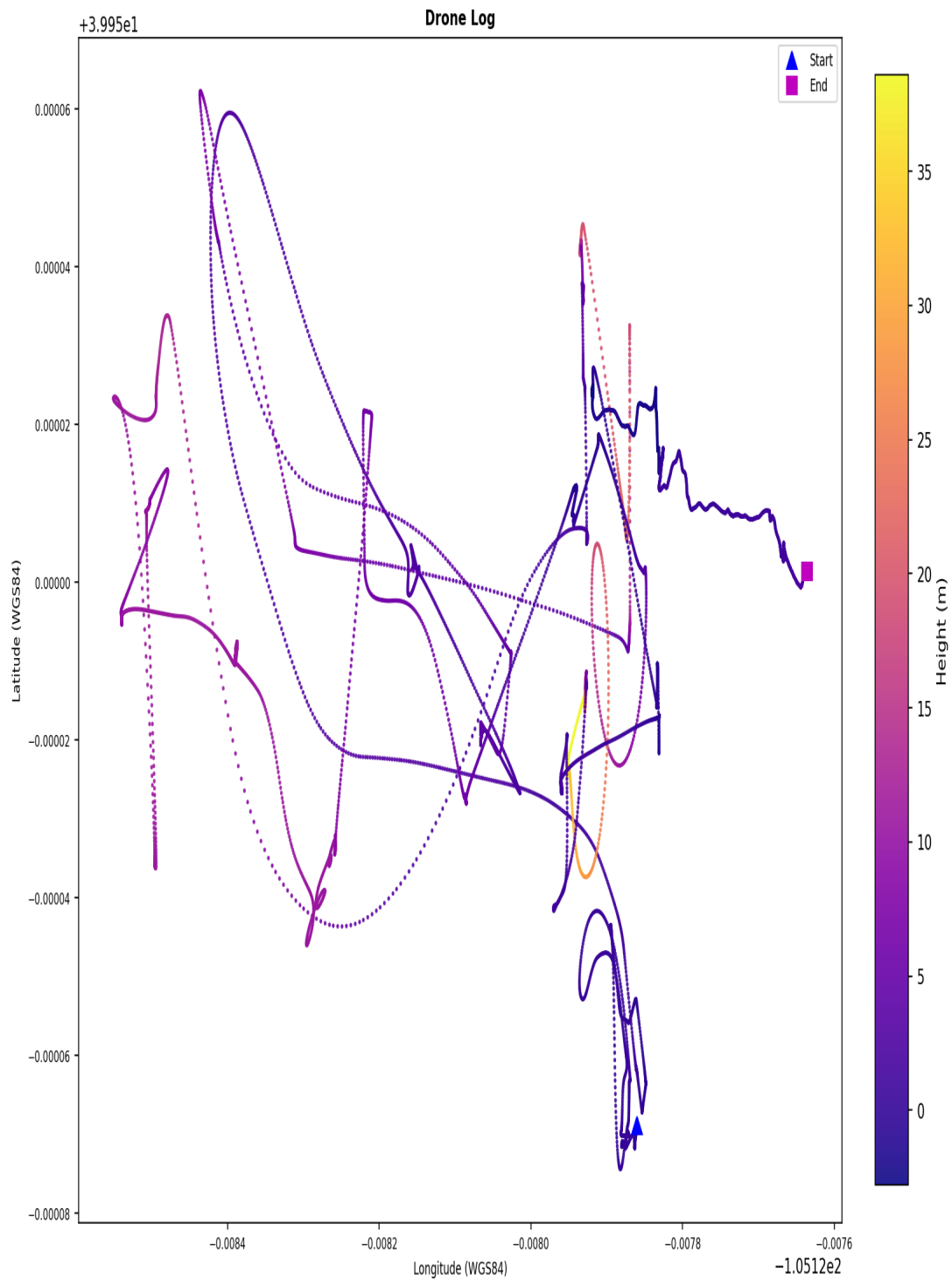
Drone Log Telemetry - flight_14



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_14



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY025.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY025.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 15

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 1815544cb41a4fc79e7a19f37debab195421e083f18dc322115310043f4970b7
Drone Log Name: FLY026.csv
Start: 2018-02-23T18:42:06.767000+00:00
End: 2018-02-23T18:55:17.987000+00:00
Duration: 791.2 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

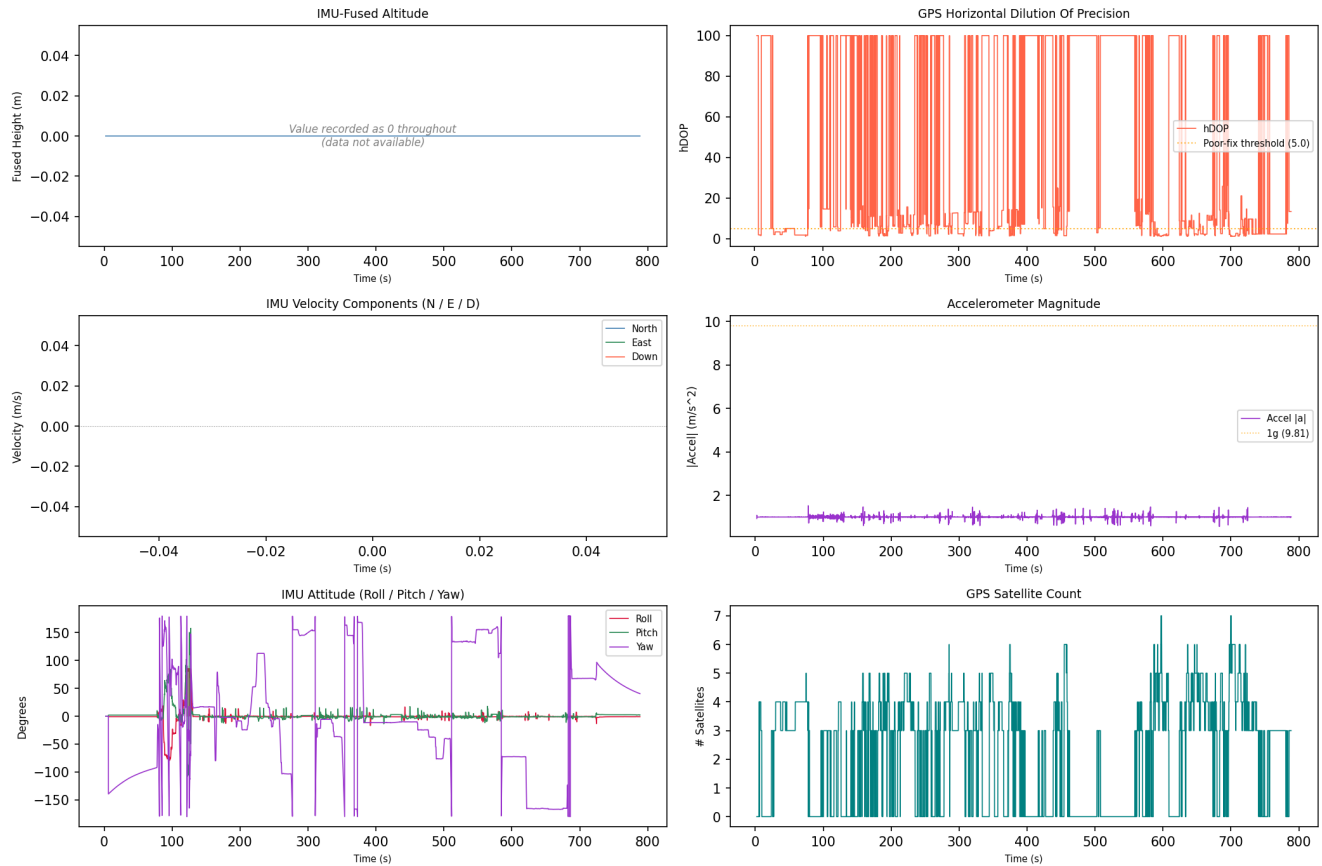
1. Manual
2. Atti
3. SPORT
4. FORCE_LANDING
5. GPS_Atti
6. ASST_TAKEOFF
7. AssitedTakeoff
8. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:43:03	drone_flight_storage:dl	Log started	MavicPro
2018-02-23 18:44:26	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:47:01	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:47:07	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:49:13	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:49:24	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:51:41	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:51:46	drone_flight_storage:dl	Motor turned on	on
2018-02-23 18:54:14	drone_flight_storage:dl	Motor turned off	off
2018-02-23 18:55:18	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_15



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 16

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 9b044a440ea23234bcdeba0035d371dbc285a5adc8ab80d252f93fcceec96911a
Drone Log Name: FLY027.csv
Start: 2018-02-25T22:30:25.583000+00:00
End: 2018-02-25T22:53:27.506000+00:00
Duration: 1381.9 s

Flight Metrics

Peak Height: 50.698751730424135 m
Distance Travelled: 2797.7317723025158 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

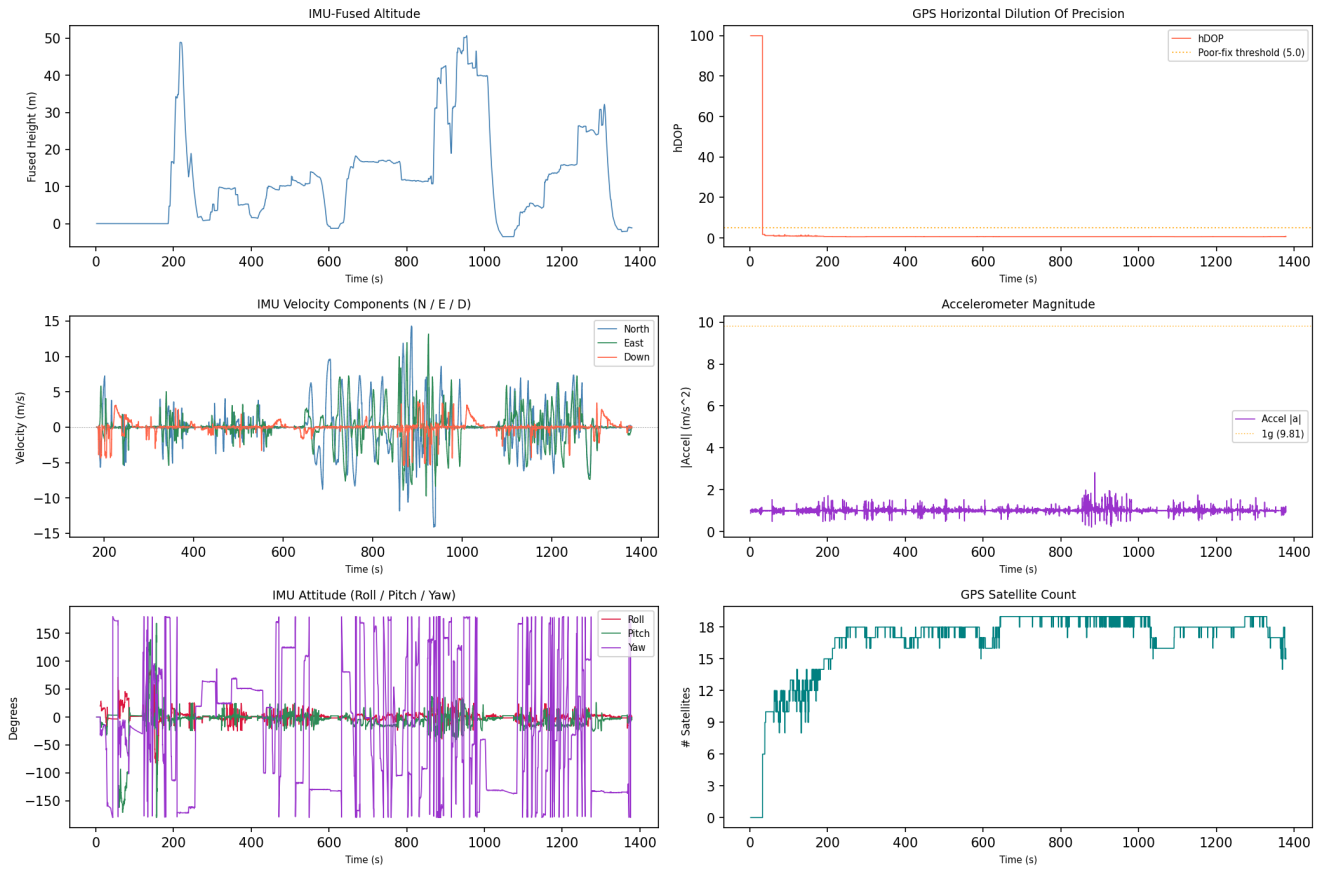
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. AutoLanding
7. FORCE_LANDING
8. SPORT
9. GoHome

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:31:01	drone_flight_storage:dl	Log started	MavicPro
2018-02-25 22:33:30	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:35:03	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:35:15	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:37:09	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:37:17	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:40:32	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:40:46	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:46:21	drone_flight_storage:dl	Reached peak height	50.7 m
2018-02-25 22:47:55	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:48:16	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:53:01	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:53:26	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

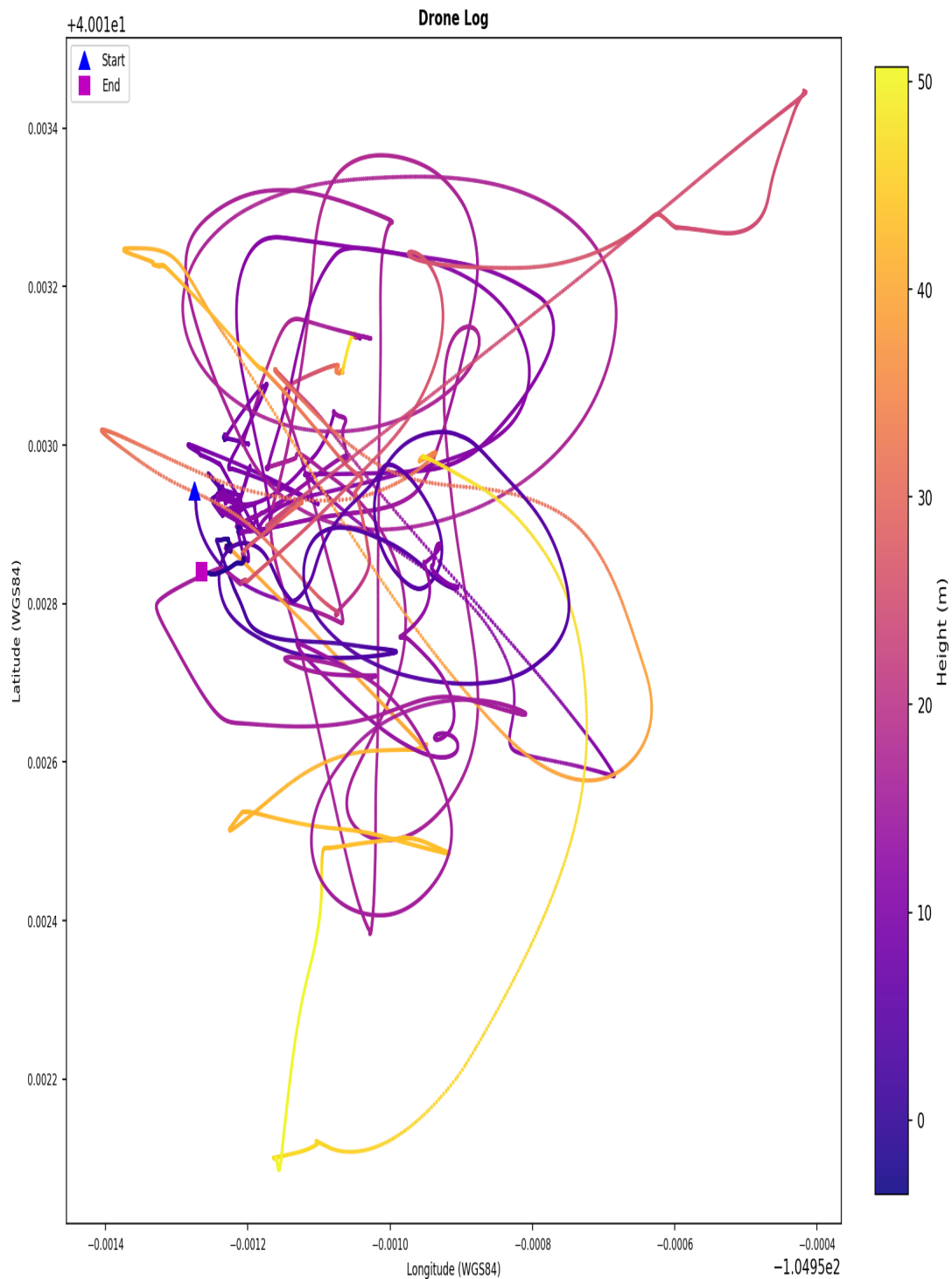
Drone Log Telemetry - flight_16



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_16



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY027.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY027.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 17

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: ba75cc66b5e06076204b37290f426bcff57d64874d8edb937d56311e4b8b9993
Drone Log Name: FLY028.csv
Start: 2018-02-25T22:53:35.712000+00:00
End: 2018-02-25T23:07:23.770000+00:00
Duration: 828.1 s

Flight Metrics

Peak Height: 113.00427983435897 m
Distance Travelled: 1009.4466269444176 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

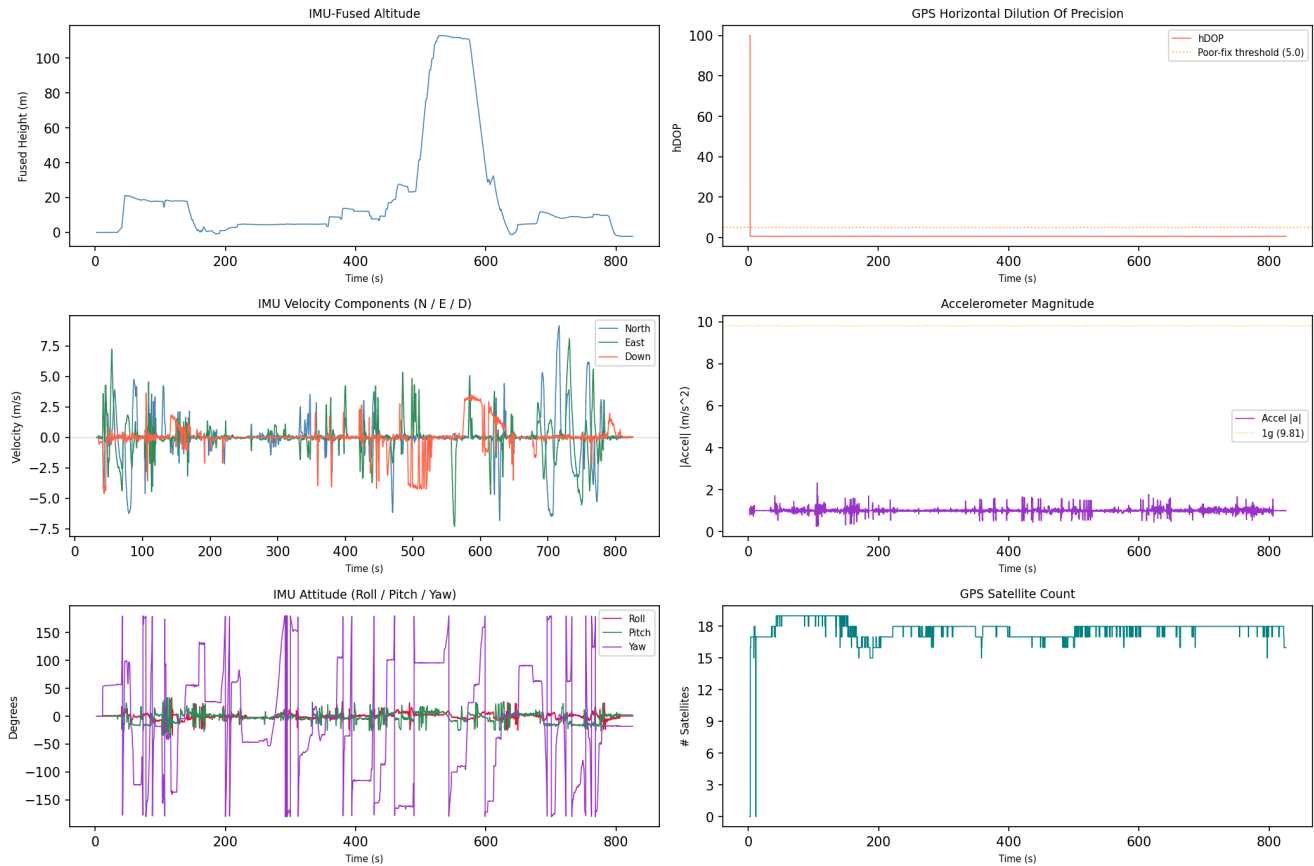
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. SPORT
7. GoHome
8. AutoLanding
9. FORCE_LANDING
10. Joystick
11. NaviSubMode_Tracking

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:53:41	drone_flight_storage:dl	Log started	MavicPro
2018-02-25 22:54:10	drone_flight_storage:dl	Motor turned on	on
2018-02-25 22:56:44	drone_flight_storage:dl	Motor turned off	off
2018-02-25 22:56:47	drone_flight_storage:dl	Motor turned on	on
2018-02-25 23:02:26	drone_flight_storage:dl	Reached peak height	113.0 m
2018-02-25 23:07:05	drone_flight_storage:dl	Motor turned off	off
2018-02-25 23:07:23	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

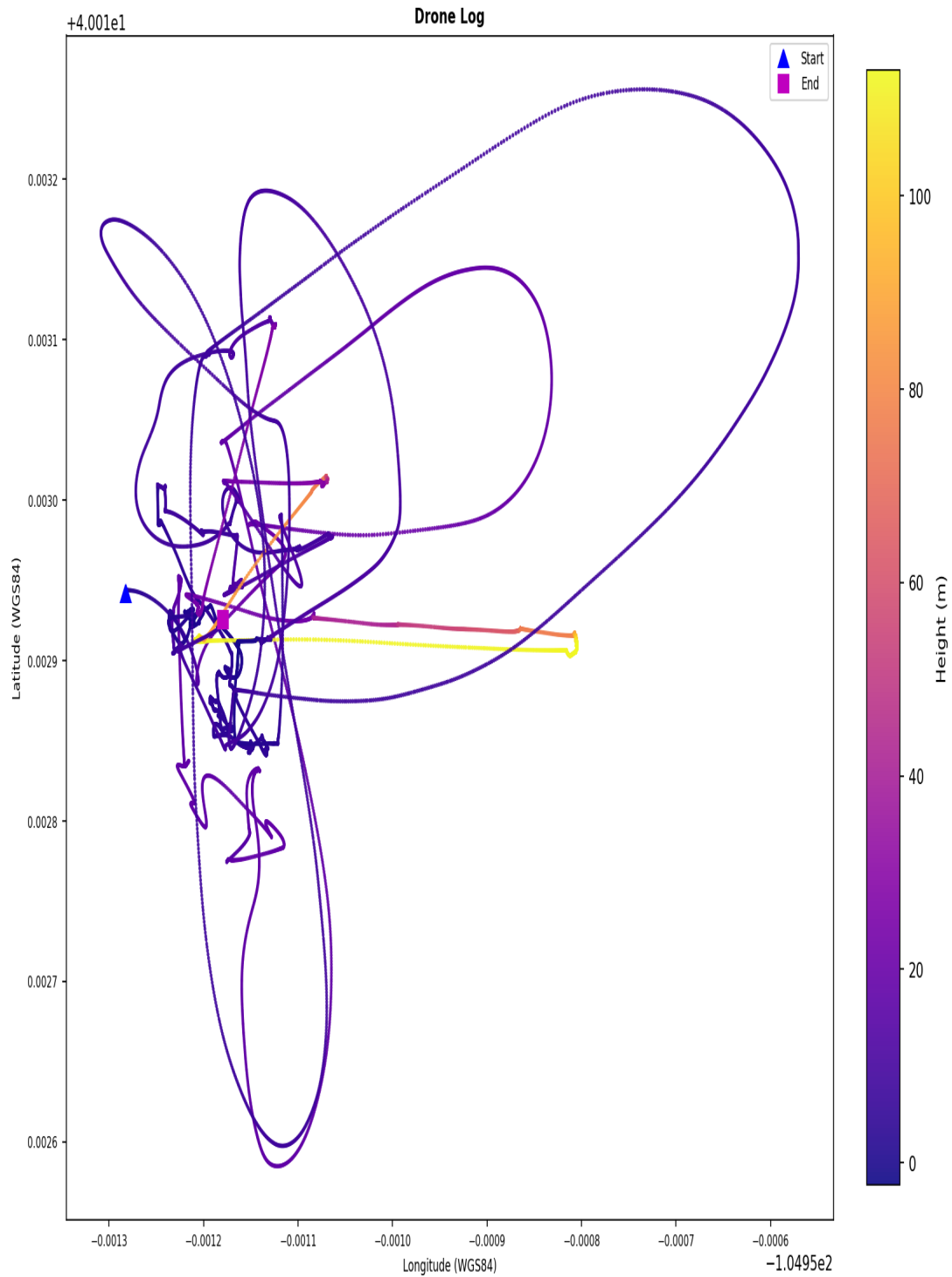
Drone Log Telemetry - flight_17



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_17



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY028.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY028.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 18

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 0e76364a79e2b78aef4e468d6e10b8875e74134715305ce1cbbf7ad169f07c38
Drone Log Name: FLY031.csv
Start: 2018-02-26T17:49:21.435000+00:00
End: 2018-02-26T18:12:19.233000+00:00
Duration: 1377.8 s

Flight Metrics

Peak Height: 50.27068717393614 m
Distance Travelled: 1149.3625435171891 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

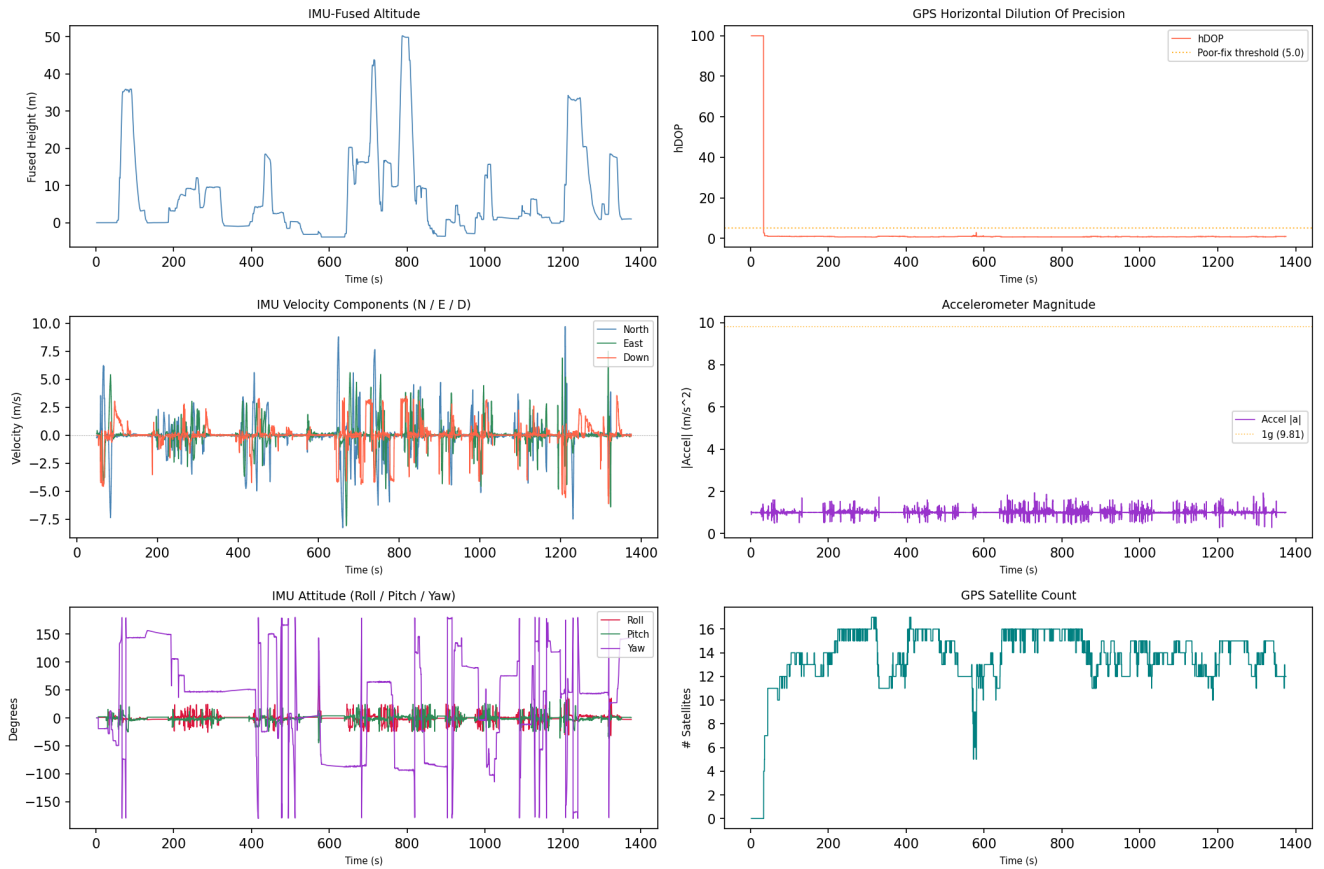
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. GoHome
7. AutoLanding
8. FORCE_LANDING
9. SPORT

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2018-02-26 17:49:58	drone_flight_storage:dl	Log started	MavicPro
2018-02-26 17:51:36	drone_flight_storage:dl	Motor turned off	off
2018-02-26 17:52:26	drone_flight_storage:dl	Motor turned on	on
2018-02-26 17:54:55	drone_flight_storage:dl	Motor turned off	off
2018-02-26 17:55:54	drone_flight_storage:dl	Motor turned on	on
2018-02-26 17:58:17	drone_flight_storage:dl	Motor turned off	off
2018-02-26 17:59:58	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:02:32	drone_flight_storage:dl	Reached peak height	50.3 m
2018-02-26 18:04:01	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:04:17	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:05:17	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:05:34	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:08:55	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:09:13	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:09:19	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:09:23	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:10:57	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:11:00	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:11:55	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:12:19	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

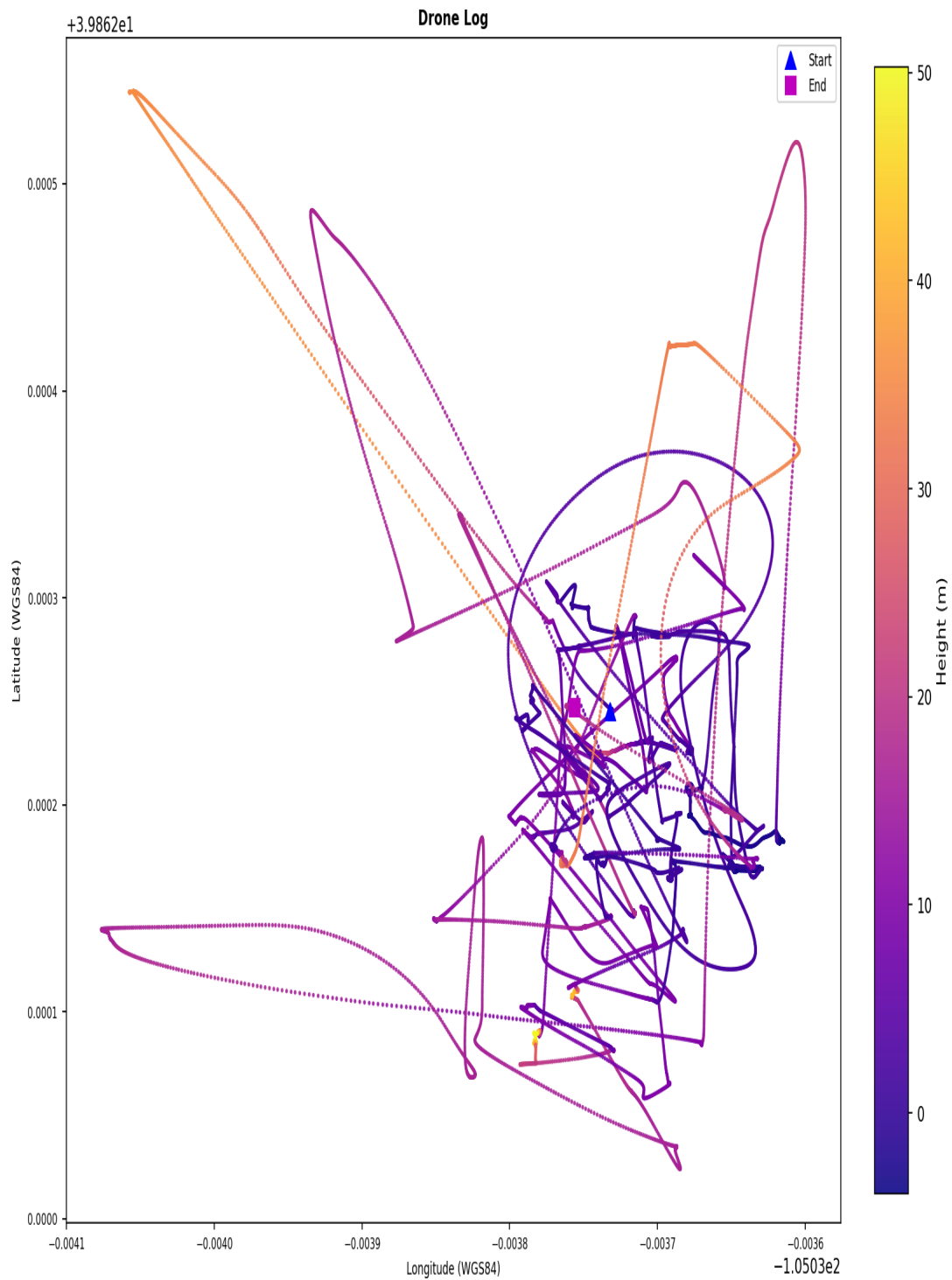
Drone Log Telemetry - flight_18



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_18



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY031.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY031.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 19

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 07e799551546a716f548867c33c06abc16c4c5f649629bb629cb854a401d2cee
Drone Log Name: FLY032.csv
Start: 2018-02-26T18:12:33.072000+00:00
End: 2018-02-26T18:16:13.272000+00:00
Duration: 220.2 s

Flight Metrics

Peak Height: 34.294127194318136 m
Distance Travelled: 295.21218627242365 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

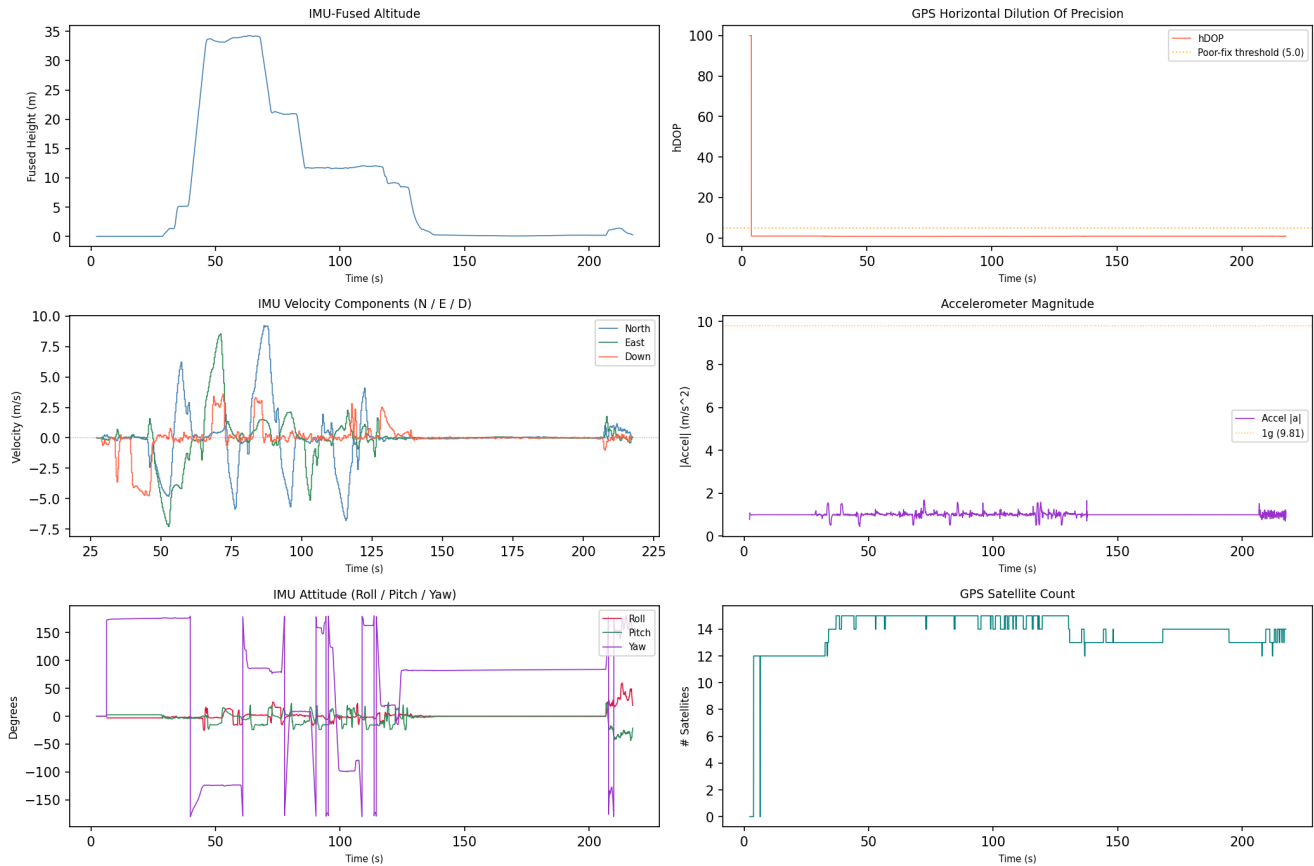
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. FORCE_LANDING
7. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:12:39	drone_flight_storage:dl	Log started	MavicPro
2018-02-26 18:13:02	drone_flight_storage:dl	Motor turned on	on
2018-02-26 18:13:39	drone_flight_storage:dl	Reached peak height	34.3 m
2018-02-26 18:14:54	drone_flight_storage:dl	Motor turned off	off
2018-02-26 18:16:13	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

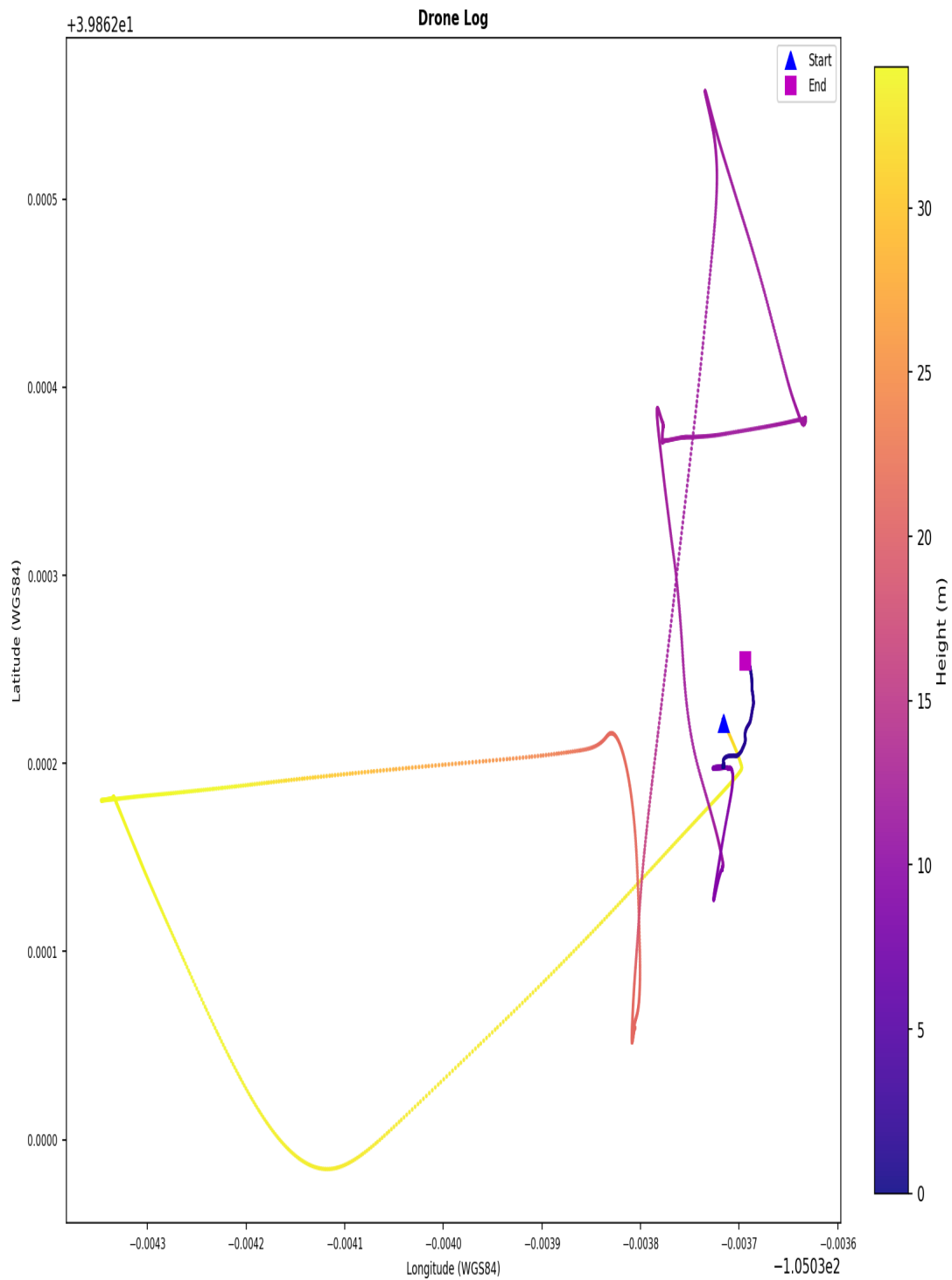
Drone Log Telemetry - flight_19



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_19



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY032.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY032.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 20

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 09e4913cc41b65b6115bc7a517cbaef567899a3ad8ce02b70140bdaa7b140a97
Drone Log Name: FLY034.csv
Start: 2018-02-28T03:15:36.238000+00:00
End: 2018-02-28T03:21:54.495000+00:00
Duration: 378.3 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

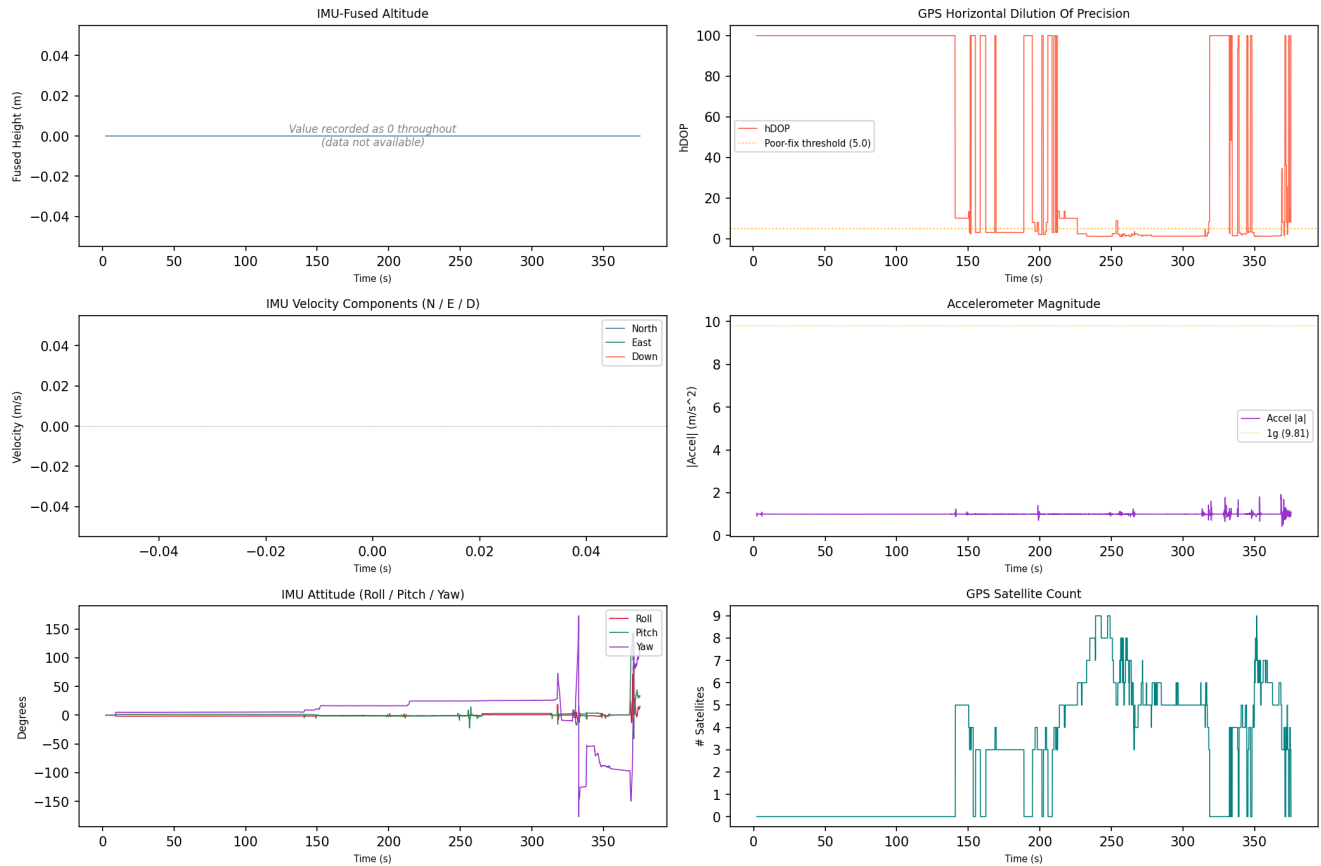
1. Manual
2. Atti
3. ASST_TAKEOFF
4. AssitedTakeoff
5. GPS_Atti
6. FORCE_LANDING
7. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2018-02-28 03:18:28	drone_flight_storage:dl	Log started	MavicPro
2018-02-28 03:20:05	drone_flight_storage:dl	Motor turned off	off
2018-02-28 03:20:40	drone_flight_storage:dl	Motor turned on	on
2018-02-28 03:20:59	drone_flight_storage:dl	Motor turned off	off
2018-02-28 03:21:22	drone_flight_storage:dl	Motor turned on	on
2018-02-28 03:21:33	drone_flight_storage:dl	Motor turned off	off
2018-02-28 03:21:54	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_20



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 21

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: f999e8dc71cc9cbe10e390e6dc5c3b56fd007c0260c08820135bf94c807b2b4
Drone Log Name: FLY035.csv
Start: 2018-03-08T20:58:36.158000+00:00
End: 2018-03-08T21:20:01.433000+00:00
Duration: 1285.3 s

Flight Metrics

Peak Height: 31.95914109695174 m
Distance Travelled: 722.0230950385726 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

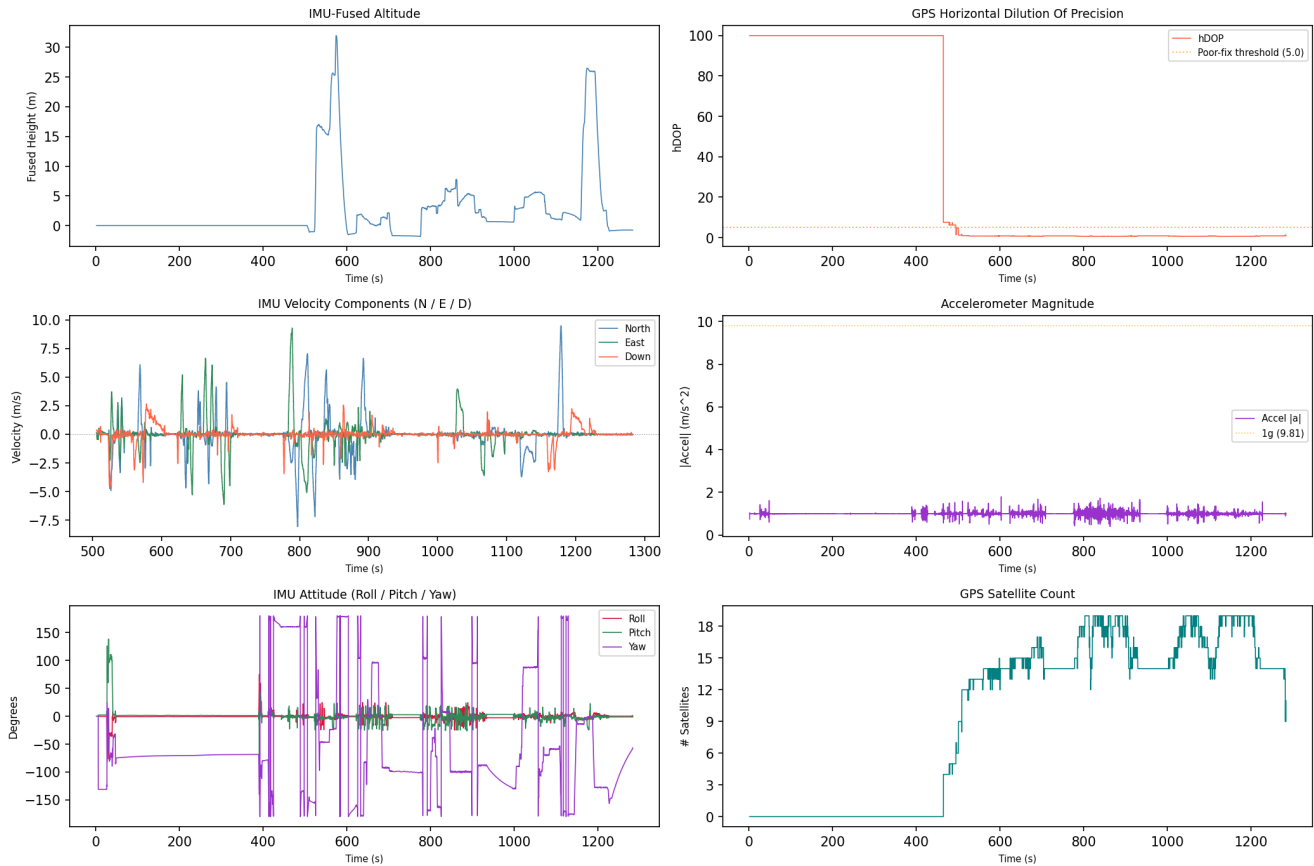
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. FORCE_LANDING
7. GoHome
8. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
-	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:06:45	drone_flight_storage:dl	Log started	MavicPro
2018-03-08 21:07:10	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:07:19	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:08:13	drone_flight_storage:dl	Reached peak height	32.0 m
2018-03-08 21:08:43	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:08:57	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:10:29	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:11:31	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:14:14	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:15:14	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:19:07	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:20:02	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

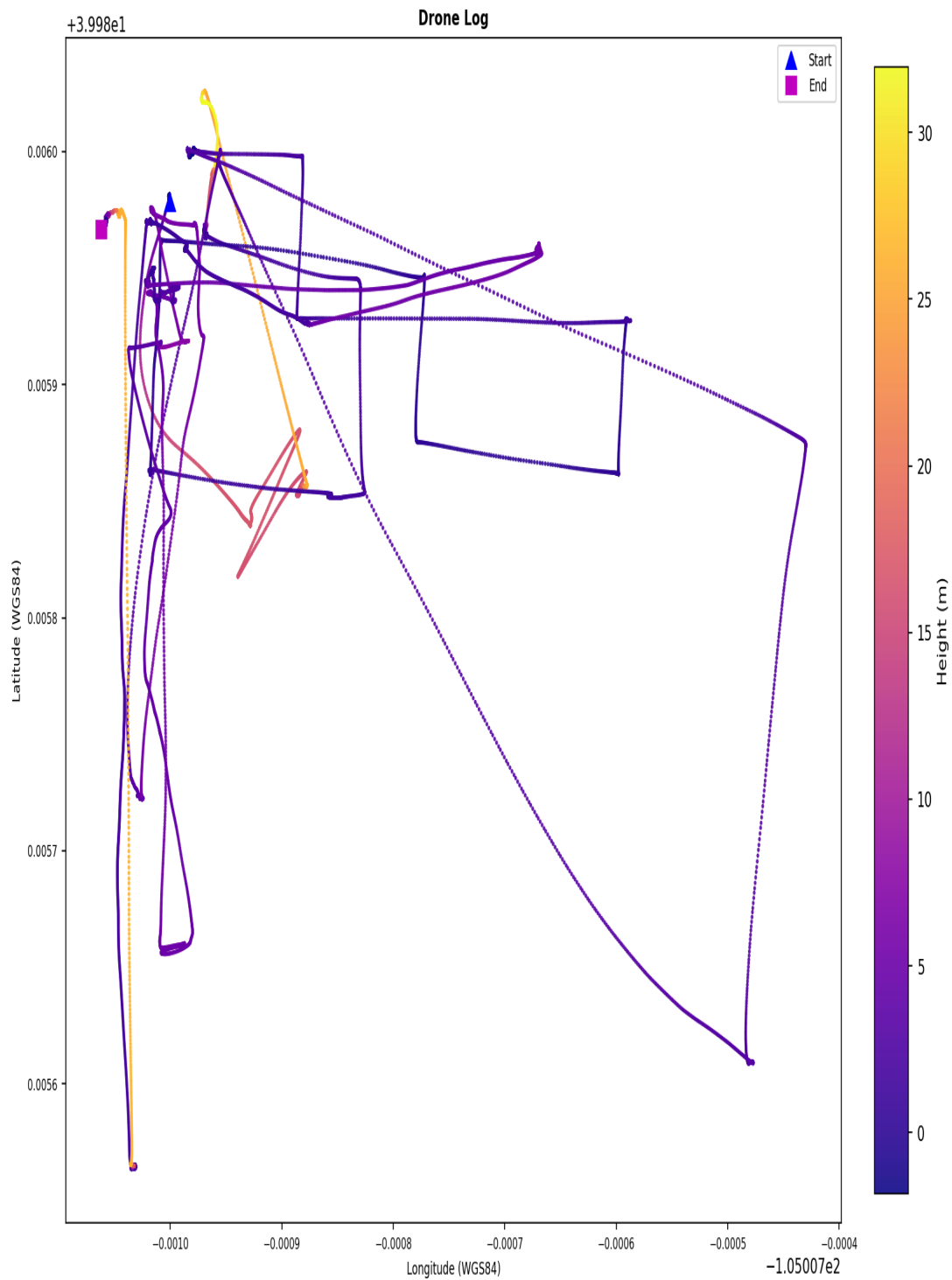
Drone Log Telemetry - flight_21



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_21



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY035.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY035.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 22

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: aed76edf128299f079bc5bfb3de6d445539fa71351668e7d5d27b04c452d581f
Drone Log Name: FLY036.csv
Start: 2018-03-08T21:20:10.810000+00:00
End: 2018-03-08T21:43:05.676000+00:00
Duration: 1374.9 s

Flight Metrics

Peak Height: 39.5921381701714 m
Distance Travelled: 3652.5404565626645 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

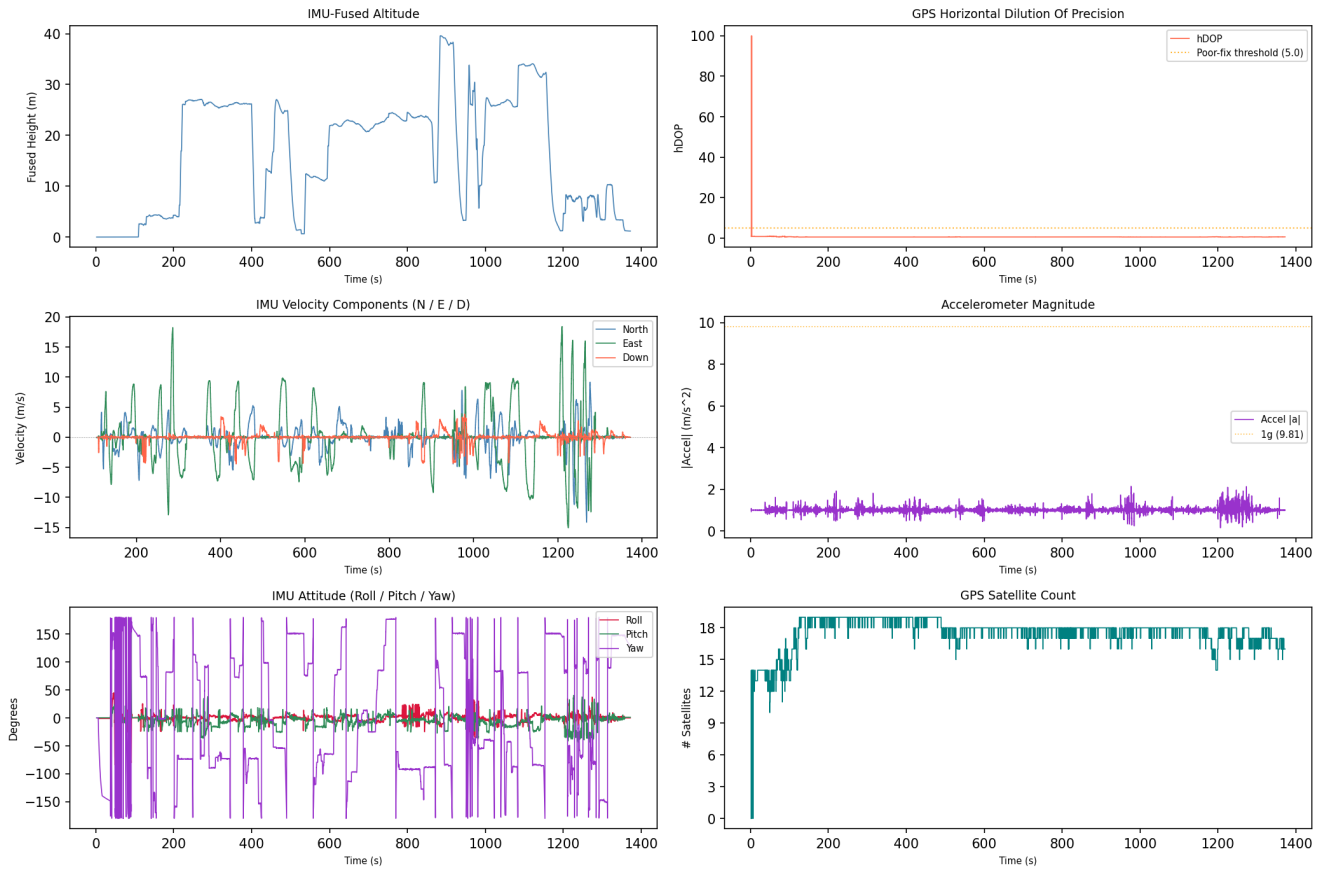
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. SPORT
7. GoHome
8. AutoLanding
9. FORCE_LANDING

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:20:16	drone_flight_storage:dl	Log started	MavicPro
2018-03-08 21:21:59	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:35:00	drone_flight_storage:dl	Reached peak height	39.6 m
2018-03-08 21:40:07	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:40:09	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:42:53	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:43:06	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

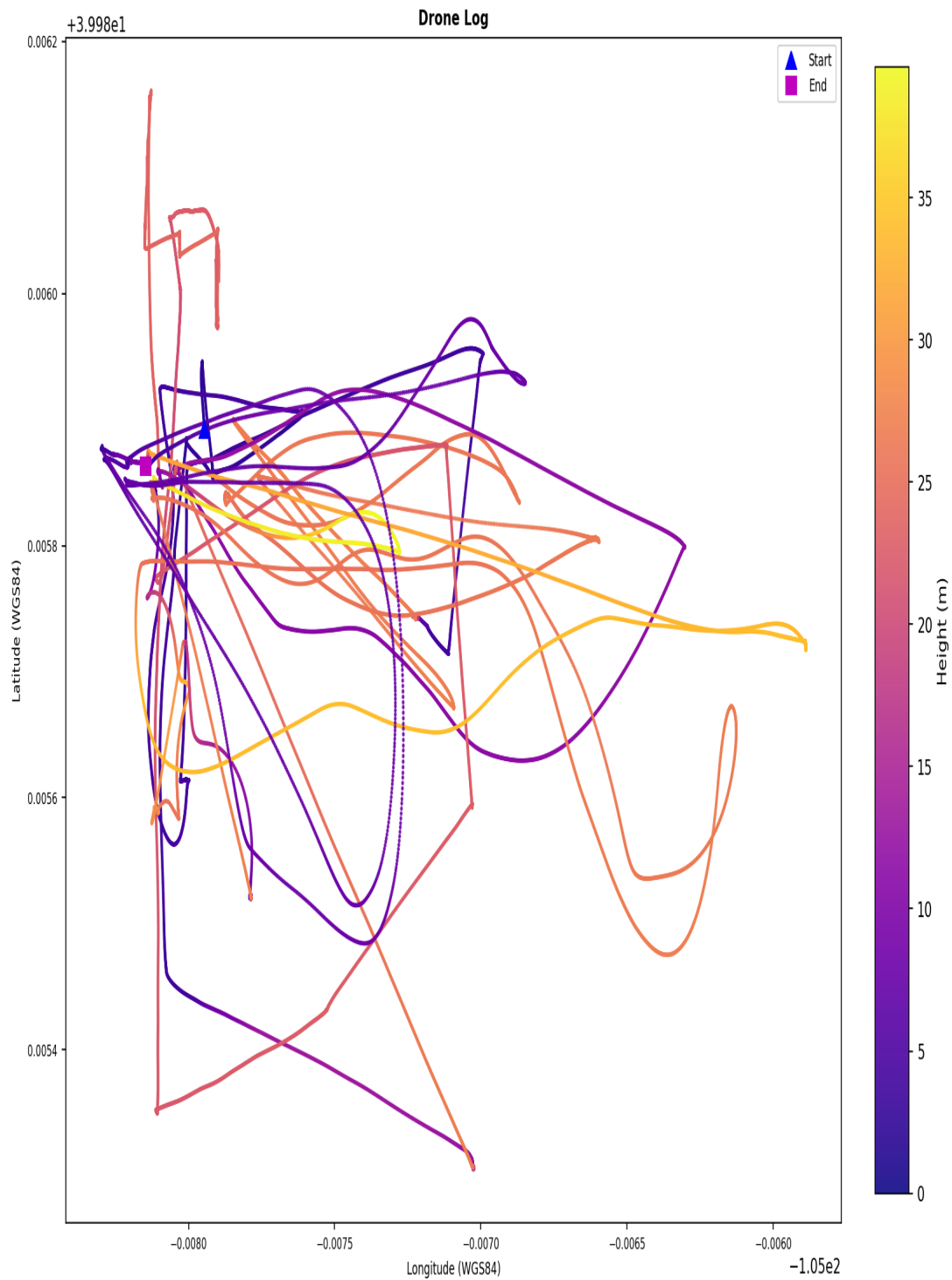
Drone Log Telemetry - flight_22



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_22



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY036.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 23

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: f3d07aaedf26400dfe0b55f4209999ca32bf3ef43955d446a0683747d0428367
Drone Log Name: FLY037.csv
Start: 2018-03-08T21:43:17.110000+00:00
End: 2018-03-08T22:04:26.112000+00:00
Duration: 1269.0 s

Flight Metrics

Peak Height: 114.520876463393 m
Distance Travelled: 4360.608606993383 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

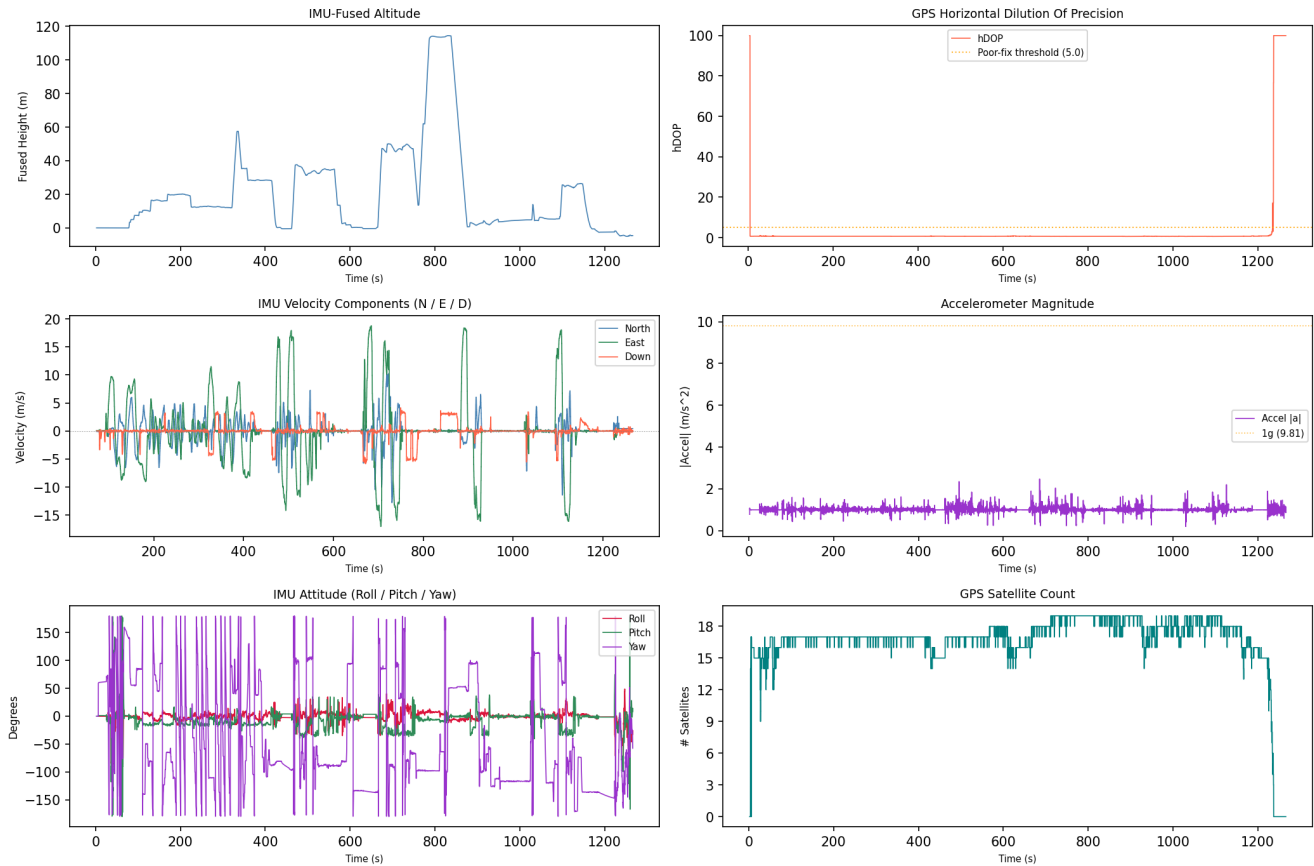
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. FORCE_LANDING
7. SPORT
8. AutoLanding
9. GoHome

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:43:23	drone_flight_storage:dl	Log started	MavicPro
2018-03-08 21:44:32	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:50:38	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:50:56	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:53:51	drone_flight_storage:dl	Motor turned off	off
2018-03-08 21:54:16	drone_flight_storage:dl	Motor turned on	on
2018-03-08 21:57:10	drone_flight_storage:dl	Reached peak height	114.5 m
2018-03-08 22:03:07	drone_flight_storage:dl	Motor turned off	off
2018-03-08 22:04:26	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

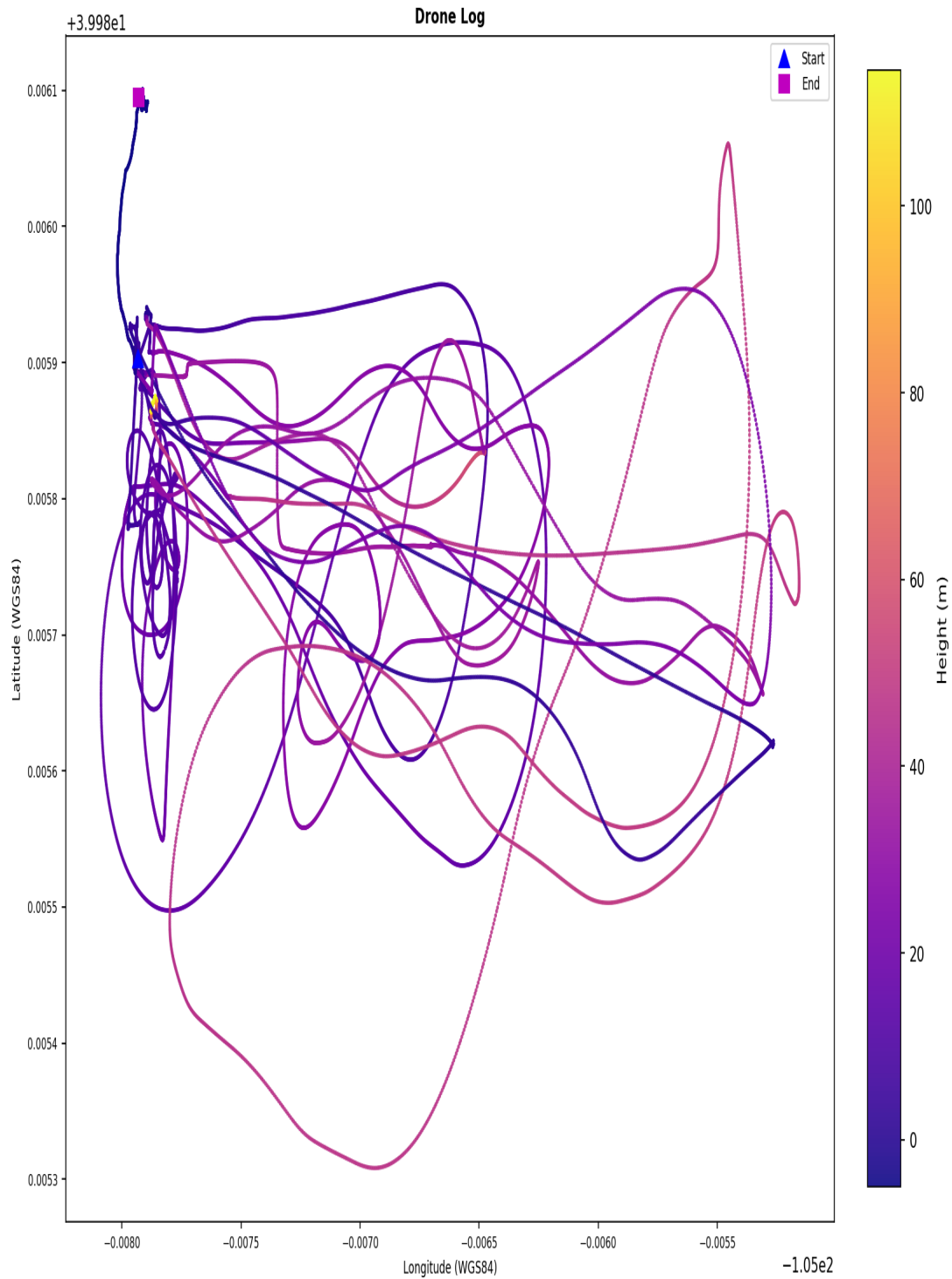
Drone Log Telemetry - flight_23



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_23



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY037.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 24

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 038ee1ef49b61b912bf876f6b85f61a956228bf0543ed094309634295b3fcb96
Drone Log Name: FLY038.csv
Start: 2018-03-08T23:43:56.555000+00:00
End: 2018-03-09T00:01:28.239000+00:00
Duration: 1051.7 s

Flight Metrics

Peak Height: 9.187671869442742 m
Distance Travelled: 556.0956060700404 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

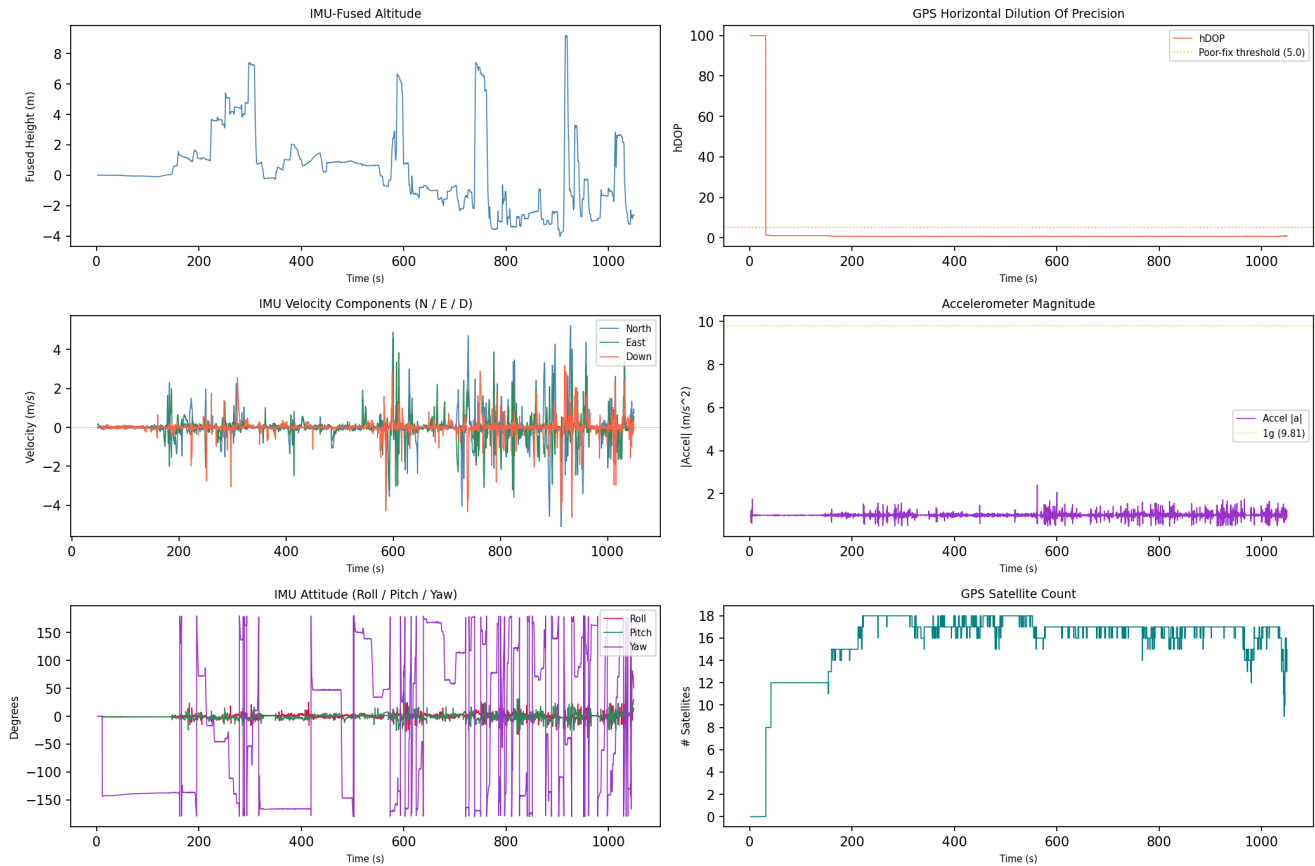
1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff
6. FORCE_LANDING
7. AutoLanding
8. SPORT

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:44:31	drone_flight_storage:dl	Log started	MavicPro
2018-03-08 23:44:46	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:44:55	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:46:19	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:49:26	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:49:40	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:53:19	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:53:26	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:55:48	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:55:52	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:59:04	drone_flight_storage:dl	Motor turned off	off
2018-03-08 23:59:07	drone_flight_storage:dl	Motor turned on	on
2018-03-08 23:59:14	drone_flight_storage:dl	Reached peak height	9.2 m
2018-03-09 00:00:05	drone_flight_storage:dl	Motor turned off	off
2018-03-09 00:00:19	drone_flight_storage:dl	Motor turned on	on
2018-03-09 00:01:17	drone_flight_storage:dl	Motor turned off	off
2018-03-09 00:01:27	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

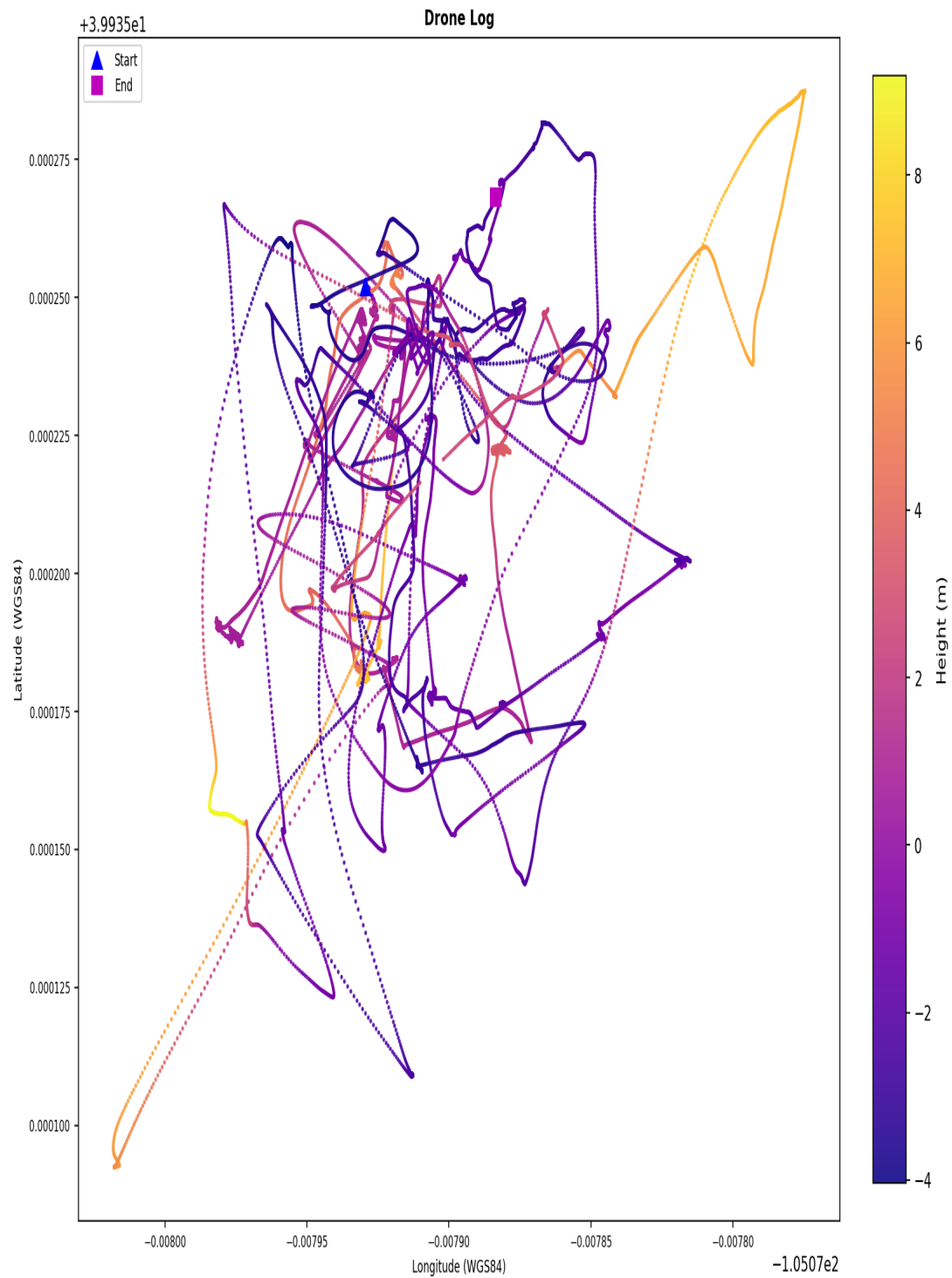
Drone Log Telemetry - flight_24



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_24



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY038.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY038.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 25

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 134436084d095412ac7d61f489b7ee2c15a88cd4b4d846785c0967e689395b6f
Drone Log Name: FLY039.csv
Start: 2018-03-09T00:02:01.087000+00:00
End: 2018-03-09T00:02:10.542000+00:00
Duration: 9.5 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

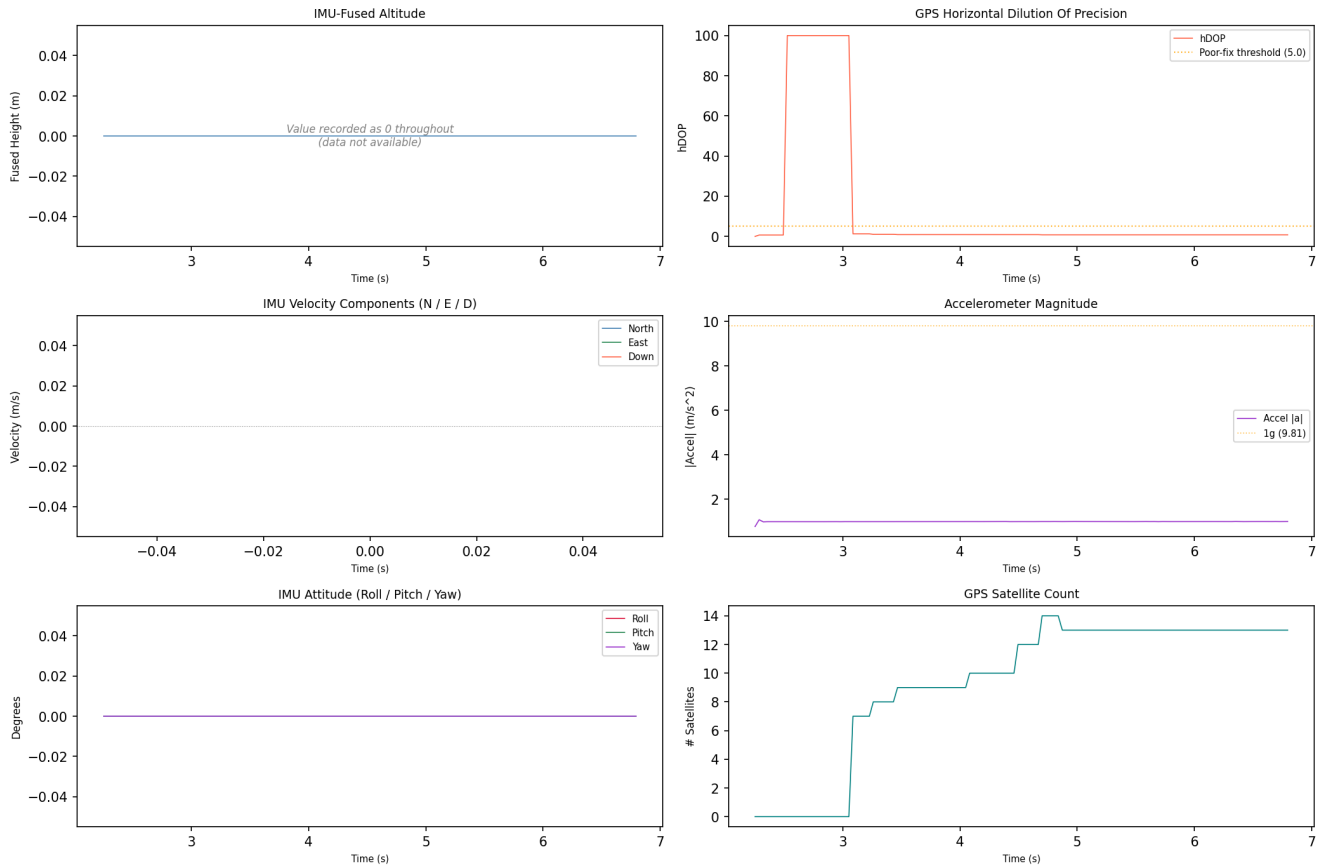
1. Manual
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
2018-03-09 00:02:06	drone_flight_storage:dl	Log started	MavicPro
2018-03-09 00:02:06	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-03-09 00:02:06	drone_flight_storage:dl	Motor turned off	off
2018-03-09 00:02:13	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_25



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 26

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	305.4 s
Median GPS Distance:	1.75 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	f36d1f9b2e295eccb028cfca3a19b6a8271452d872c73494631ce0dc7bed8845
Flight Record Name:	DJIFlightRecord_2018-06-19_[14-50-34].csv
Start:	2018-06-19T18:50:34.869000+00:00
End:	2018-06-19T18:55:40.243000+00:00
Duration:	305.4 s
Drone Log SHA-256:	9a760be61df4a375716f0485b44c3d4cb6a0fe21c6c6abc591f76768fdbfa434
Drone Log Name:	18-06-19-02-47-38_FLY057.csv
Start:	2018-06-19T18:46:00.048000+00:00
End:	2018-06-19T18:58:21.725000+00:00
Duration:	741.7 s

Flight Metrics

Peak Height:	32.196024004978064 m
Distance Travelled:	444.56 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	300.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. ASST_TAKEOFF
3. EngineStart
4. AutoTakeoff
5. AutoLanding
6. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_and:dl	Motor turned off	off
2018-06-19 18:49:18	ctrl_and:dl	Log started	MavicAir
2018-06-19 18:50:34	ctrl_and:dl	Motor turned on	on
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Log started	SN: 0K1DF313BD4PR1; App: 4.2.16; Mavic Air
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Motor turned on	on
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Record mode changed	No
2018-06-19 18:50:39	ctrl_and:fr (v4.2.16)	Record mode changed	Starting
2018-06-19 18:54:25	ctrl_and:dl	Reached peak height	32.2 m
2018-06-19 18:54:26	ctrl_and:fr (v4.2.16)	Reached peak height	28.5 m
2018-06-19 18:55:39	ctrl_and:dl	Motor turned off	off
2018-06-19 18:55:40	ctrl_and:fr (v4.2.16)	Log ended	Dist home: 0.7 m; Height: 0.0 m; Rec: 300.0 s
2018-06-19 18:55:40	ctrl_and:fr (v4.2.16)	Motor turned off	off
2018-06-19 18:58:23	ctrl_and:dl	Log ended	

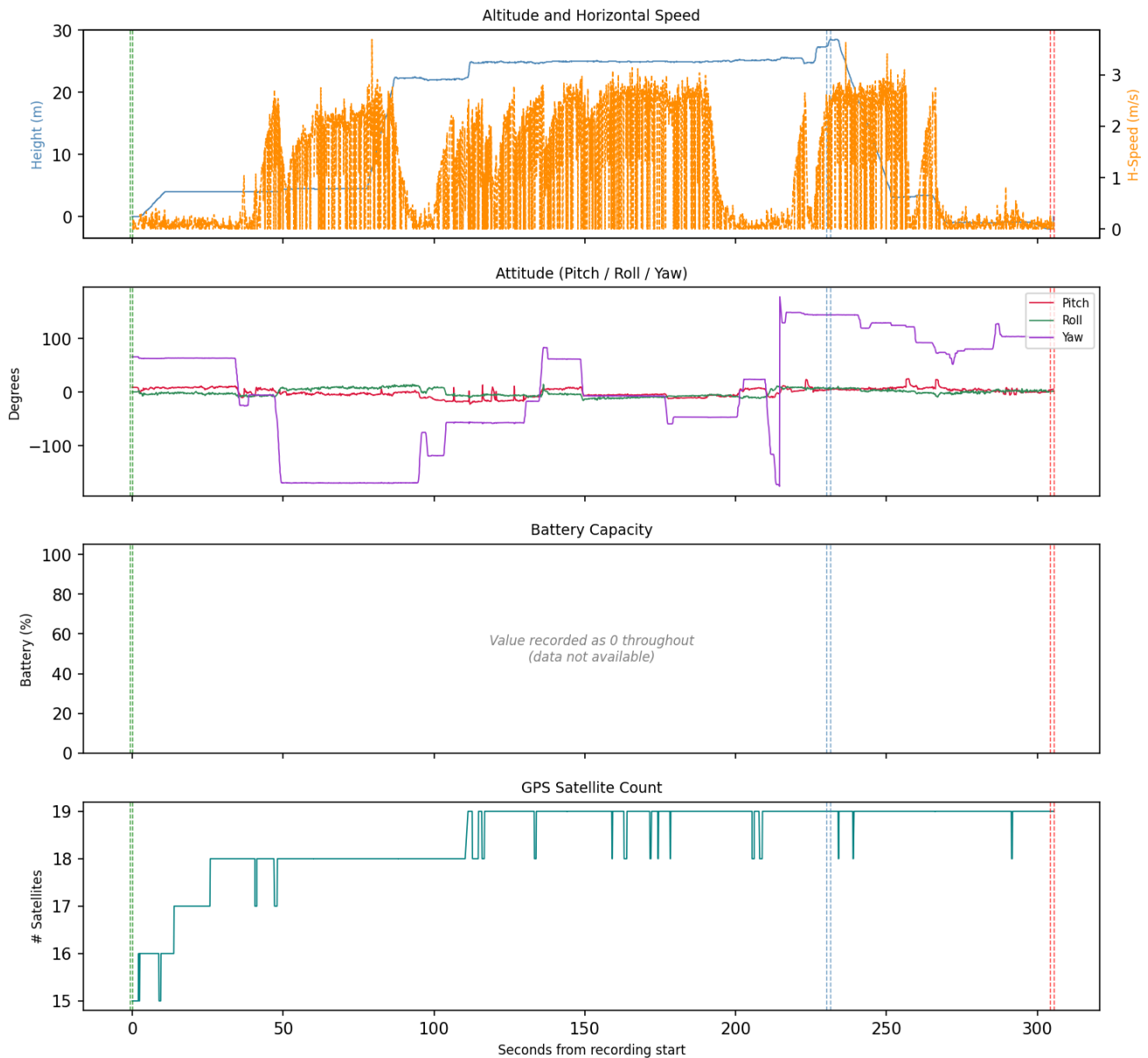
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	
controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cffcb7b341bd62ed6b	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896	
drone_sd:videos	GPS within flight bounding box (lat 39.9637, lon -106.2141)
p5:355b378ae4911bb1566e0254cf3c299c34c84c15b9f835fc4ed9a1d361b8a01c	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:0f0eb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24	

Flight Record - Sensor Telemetry

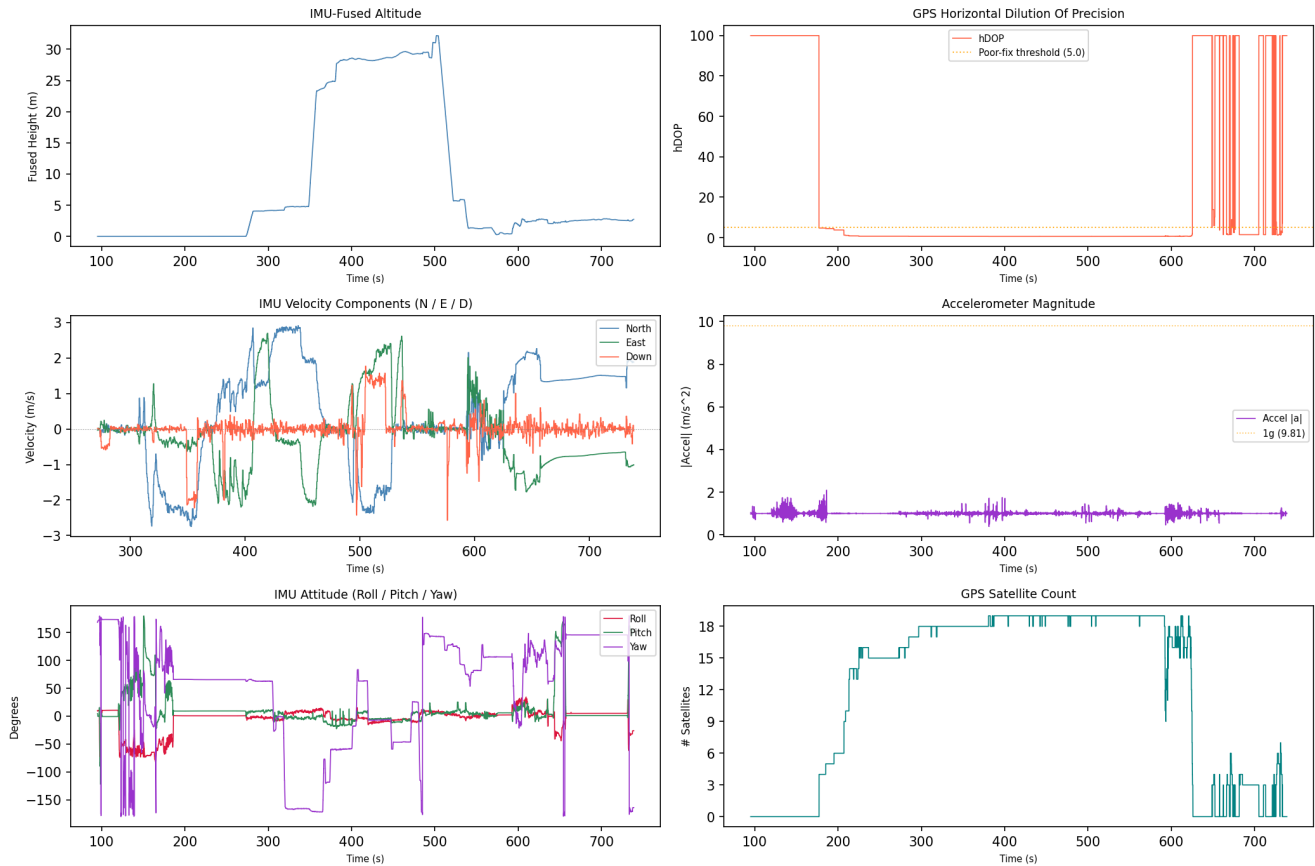
Flight Record Telemetry - flight_26



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

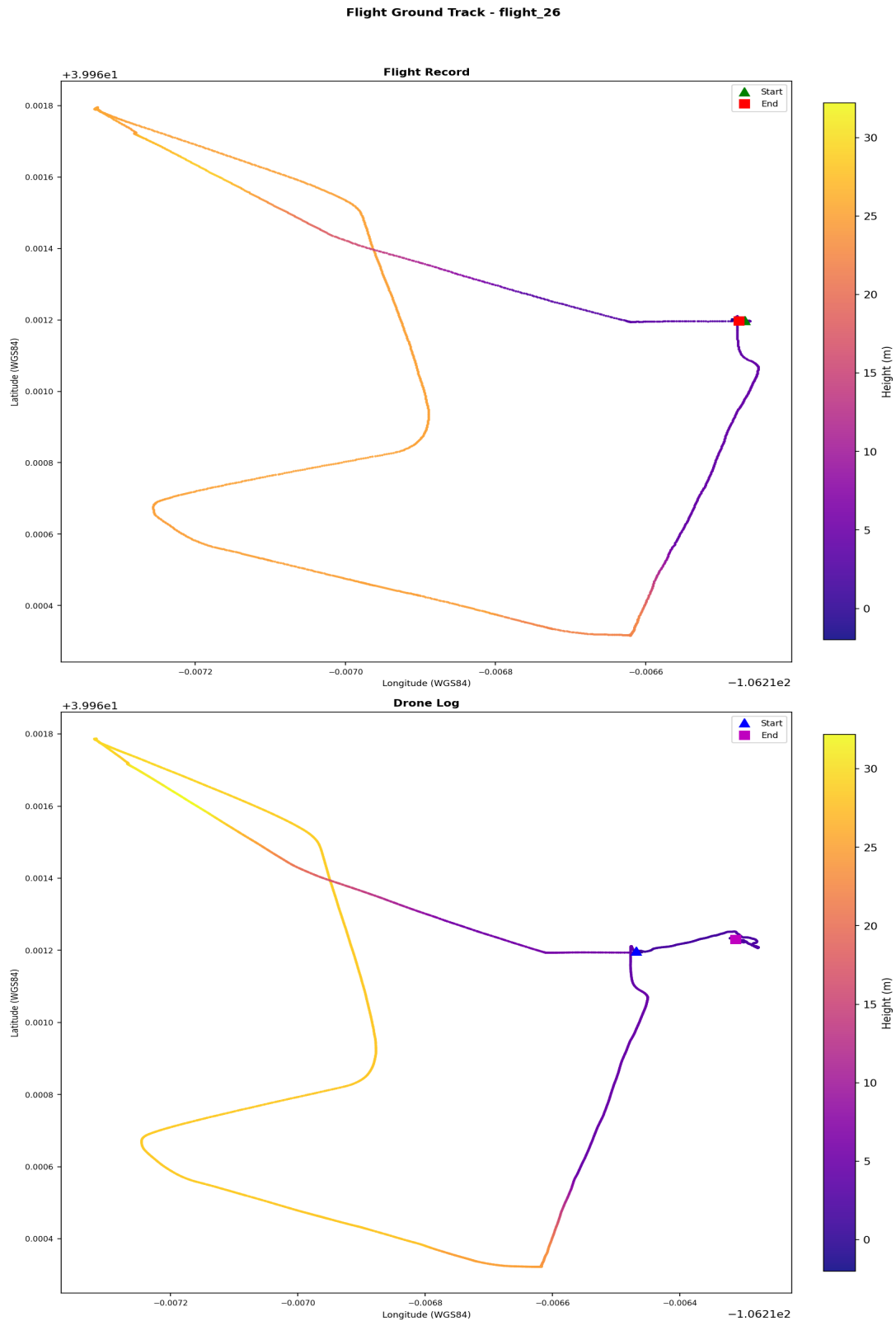
Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_26



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 18-06-19-02-47-38_FLY057.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (18-06-19-02-47-38_FLY057.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 27

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 223a0842cbf533238339fd2eec29bd01ec37e07942a7e967f465d3de0c25465a
Drone Log Name: 18-06-19-02-59-05_FLY058.csv
Start: 2018-06-19T18:58:38.544000+00:00
End: 2018-06-19T19:02:09.812000+00:00
Duration: 211.3 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. GPS_Atti
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_and:dl	Reached peak height	-0.0 m
-	ctrl_and:dl	Motor turned off	off
2018-06-19 19:00:23	ctrl_and:dl	Log started	MavicAir
2018-06-19 19:02:08	ctrl_and:dl	Log ended	

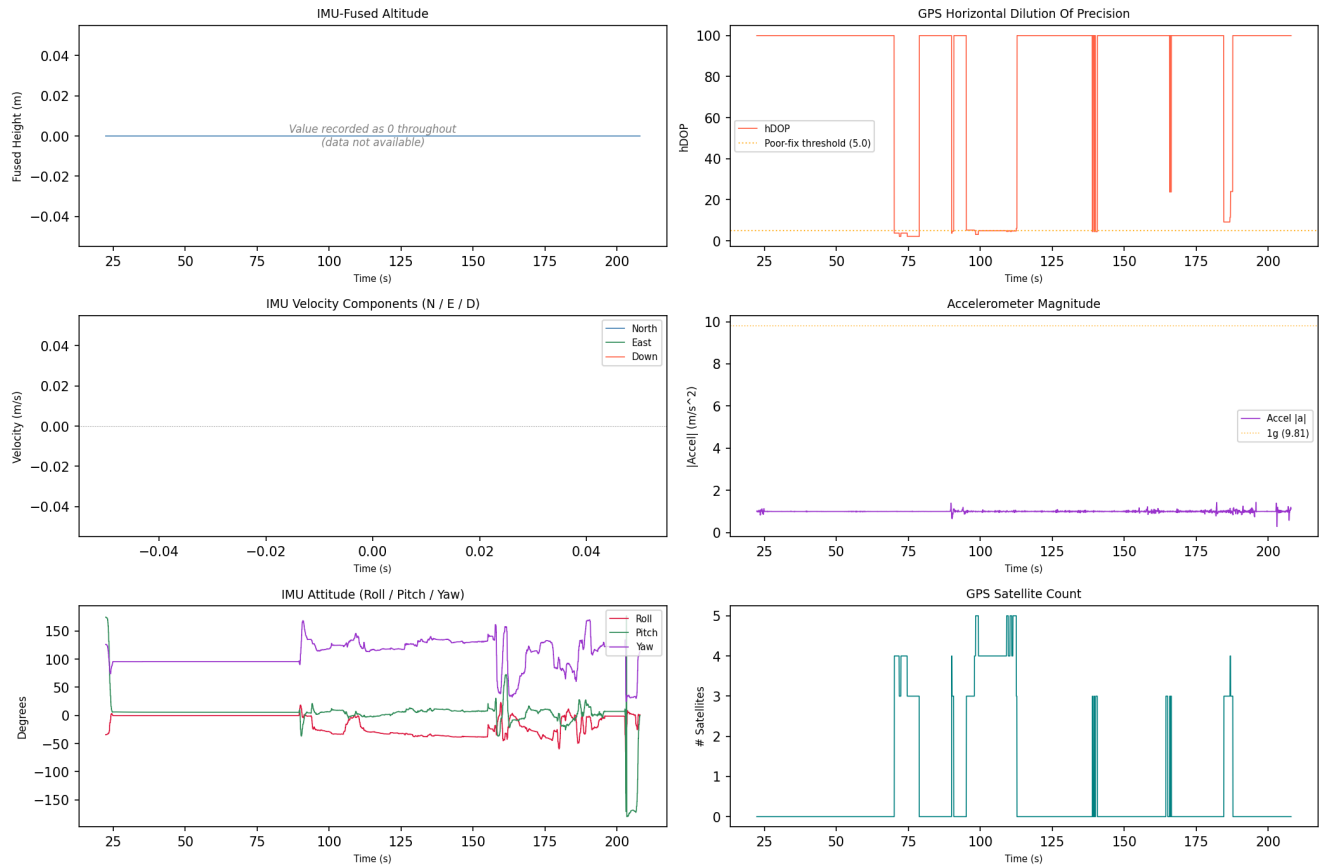
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	
controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cfcfb7b341bd62ed6b	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:0f0eb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_27



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 28

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: ba70b79a745566e8b5a37691b7c21341c8cdb4e9c45eb7852784d56acf13a155
Flight Record Name: DJIFlightRecord_2018-06-19_[18-48-38].csv
Start: 2018-06-19T22:48:38.356000+00:00
End: 2018-06-19T22:55:32.935000+00:00
Duration: 414.6 s

Flight Metrics

Peak Height: 22.0 m
Distance Travelled: 262.17 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 333.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-19 22:48:38	ctrl_and:fr (v4.2.16)	Log started	SN: 08RDE5F001007H; App: 4.2.16; Mavic Pro
2018-06-19 22:48:38	ctrl_and:fr (v4.2.16)	Motor turned on	on
2018-06-19 22:48:38	ctrl_and:fr (v4.2.16)	Record mode changed	No
2018-06-19 22:49:58	ctrl_and:fr (v4.2.16)	Record mode changed	Start
2018-06-19 22:49:59	ctrl_and:fr (v4.2.16)	Record mode changed	Starting
2018-06-19 22:51:00	ctrl_and:fr (v4.2.16)	Reached peak height	22.0 m
2018-06-19 22:55:32	ctrl_and:fr (v4.2.16)	Log ended	Dist home: 0.7 m; Height: 0.0 m; Rec: 333.0 s
2018-06-19 22:55:32	ctrl_and:fr (v4.2.16)	Motor turned off	off

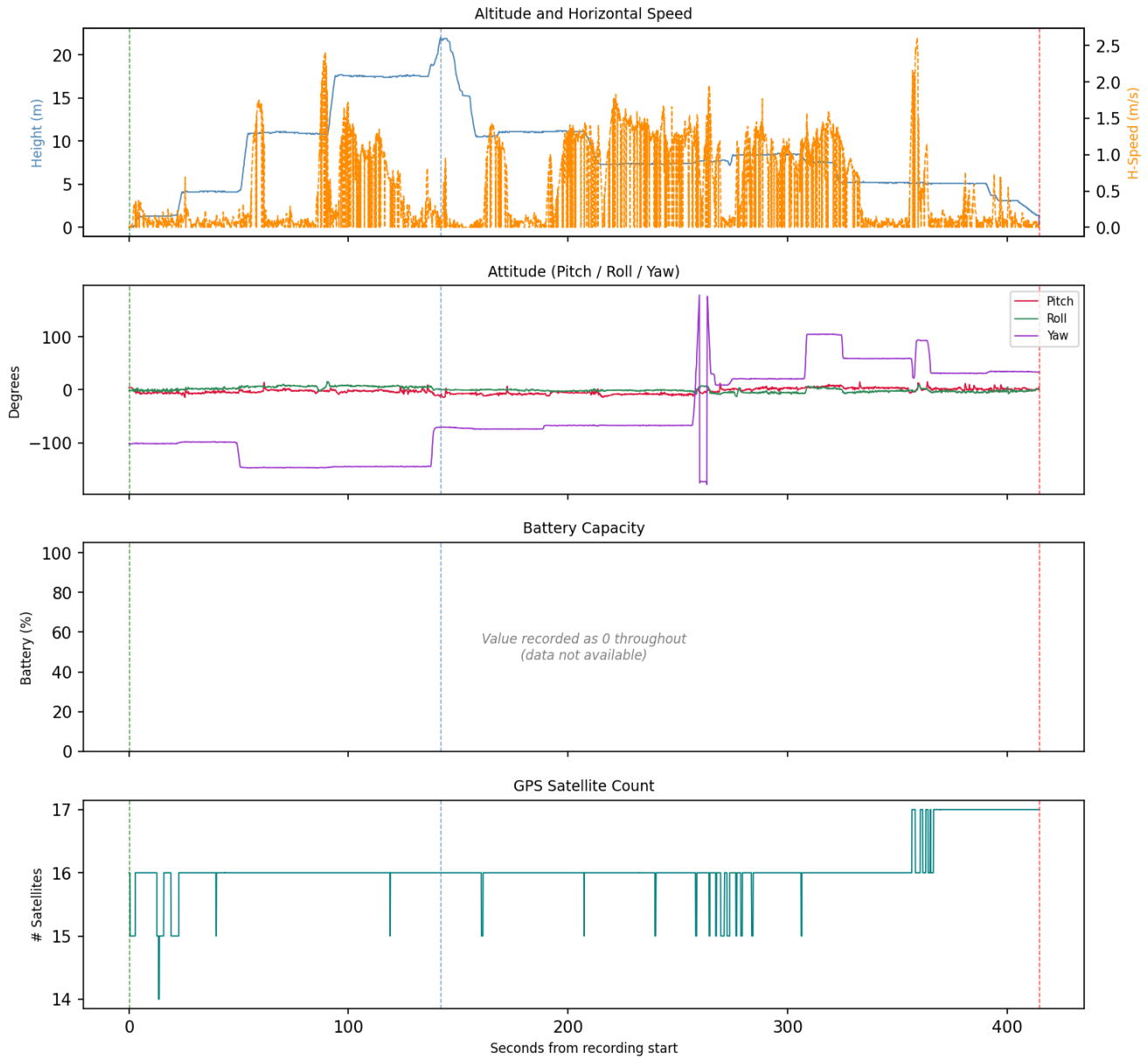
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	
controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cffcb7b341bd62ed6b	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2165)
p5:0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:0feb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24	

Flight Record - Sensor Telemetry

Flight Record Telemetry - flight_28



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	0	0	2	35	3	10	6
drone_sd	0	0	0	0	0	4	3
drone_flight_storage	0	0	62	0	0	0	0

Data Quality Notes per Category

Drone Logs (64 files):

- 63 file(s) with row anomalies
- 63 file(s) with empty columns
- 63 file(s) with constant-value columns

Flight Logs (35 files):

- 2 file(s) unreadable (binary)
- formats: 3 heading_log, 30 unknown

Flight Records (3 files):

- 2 file(s) with row anomalies
- 3 file(s) with empty columns
- 3 file(s) with constant-value columns

Images (14 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 14 file(s) missing (normalisable) EXIF timestamp
- 13 file(s) missing EXIF GPS coordinates

Videos (9 files):

- 6 file(s) contain(s) EXIF GPS coordinates
- 6 file(s) missing (normalisable) EXIF timestamp
- 3 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 63 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 63 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 63 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 2 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 3 heading_log, 30 unknown.
- Flight_records: 2 file(s) with row anomalies detected in telemetry data.
- Flight_records: 3 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 3 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone flight storage]: mmls: no partition table found in df020_internal.001; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
6. [p3 - controller android - account data]: no artefacts found
7. [p3 - controller android - databases]: no artefacts found
8. [p3 - controller ios]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage - drone logs]: DatCon reported an error processing: WPSettings.dat
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY007.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY008.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY011.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY012.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY013.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY015.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY016.csv
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY018.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY019.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY029.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY030.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY033.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY040.csv
26. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY041.csv
27. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY042.csv
28. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY043.csv
29. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY044.csv
30. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY045.csv
31. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY046.csv
32. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY047.csv
33. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY048.csv
34. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY049.csv
35. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY050.csv
36. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY051.csv
37. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY052.csv
38. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY053.csv
39. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY055.csv
40. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY056.csv
41. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY057.csv
42. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY058.csv
43. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY059.csv

44. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY060.csv
45. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY062.csv
46. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY063.csv
47. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY064.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Drone Logs

norm_18-06-19-02-47-38_FLY057.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 4 row(s) affected
- motor_airborne_off: 2614 row(s) affected
- gps_poor_fix: 3327 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, IMU_ATT(1):absoluteHeight:C (+1 more)
- contains_constant_value: 39 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:BatAuth:D, BatteryInfo:FullCap:D, BatteryInfo:LowVolThreshold:D (+34 more)
- gps_accuracy_poor: 2790 row(s) affected
- attitude_out_of_bounds: 513 row(s) affected

norm_18-06-19-02-59-05_FLY058.csv

- missing_gps: 4 row(s) affected
- gps_poor_fix: 3400 row(s) affected
- contains_no_value: 28 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+23 more)
- contains_constant_value: 90 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:comp_alti:D, AirSpeed:vel_level:D (+85 more)
- gps_accuracy_poor: 3276 row(s) affected
- attitude_out_of_bounds: 119 row(s) affected

norm_FLY004.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 651 row(s) affected
- gps_poor_fix: 651 row(s) affected
- contains_no_value: 19 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+14 more)
- contains_constant_value: 94 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+89 more)
- gps_accuracy_poor: 651 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY005.csv

- missing_gps: 8 row(s) affected
- altitude_negative: 116 row(s) affected
- motor_airborne_off: 1512 row(s) affected
- gps_poor_fix: 13 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 80 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+75 more)
- gps_accuracy_poor: 19 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY006.csv

- missing_gps: 31 row(s) affected
- altitude_negative: 3126 row(s) affected
- gps_poor_fix: 21 row(s) affected

- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 50 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+45 more)
- gps_accuracy_poor: 21 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY007.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 3925 row(s) affected
- gps_poor_fix: 7485 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+27 more)
- contains_constant_value: 93 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+88 more)
- gps_accuracy_poor: 7485 row(s) affected
- quaternion_invalid: 119 row(s) affected

norm_FLY008.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 8733 row(s) affected
- gps_poor_fix: 8733 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains_constant_value: 97 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+92 more)
- gps_accuracy_poor: 8733 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY009.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 2284 row(s) affected
- gps_poor_fix: 2284 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 103 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+98 more)
- gps_accuracy_poor: 2284 row(s) affected
- quaternion_invalid: 327 row(s) affected

norm_FLY010.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 574 row(s) affected
- gps_poor_fix: 574 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+21 more)
- contains_constant_value: 119 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+114 more)
- gps_accuracy_poor: 569 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY011.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 177 row(s) affected
- gps_poor_fix: 177 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 124 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+119 more)
- gps_accuracy_poor: 177 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY012.csv

- missing_gps: 12 row(s) affected

- zero_coordinate: 1327 row(s) affected
- gps_poor_fix: 1327 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 115 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+110 more)
- gps_accuracy_poor: 1327 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY013.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2097 row(s) affected
- gps_poor_fix: 2097 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 104 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+99 more)
- gps_accuracy_poor: 2097 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY014.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 411 row(s) affected
- gps_poor_fix: 411 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+22 more)
- contains_constant_value: 105 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+100 more)
- gps_accuracy_poor: 411 row(s) affected
- quaternion_invalid: 326 row(s) affected

norm_FLY015.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 35 row(s) affected
- gps_poor_fix: 35 row(s) affected
- contains_no_value: 39 column(s) - 4.3.0, ATTI_MINIO:pitch:C, ATTI_MINIO:roll:C, ATTI_MINIO:yaw:C, ConvertDatV3 (+34 more)
- contains_constant_value: 133 column(s) - ATTI_MINIO:s_agz0, ATTI_MINIO:s_cnt0, ATTI_MINIO:s_pgz0, ATTI_MINIO:s_qw0, ATTI_MINIO:s_qx0 (+128 more)
- gps_accuracy_poor: 35 row(s) affected
- quaternion_invalid: 40 row(s) affected

norm_FLY016.csv

- missing_gps: 7 row(s) affected
- zero_coordinate: 875 row(s) affected
- gps_poor_fix: 875 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 109 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+104 more)
- gps_accuracy_poor: 875 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY017.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 3220 row(s) affected
- gps_poor_fix: 13959 row(s) affected
- contains_no_value: 19 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+14 more)
- contains_constant_value: 102 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+97 more)
- gps_accuracy_poor: 15941 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY018.csv

- missing_gps: 11 row(s) affected

- zero_coordinate: 244 row(s) affected
- gps_poor_fix: 244 row(s) affected
- contains_no_value: 45 column(s) - 4.3.0, ATTI_MINI0:pitch:C, ATTI_MINI0:roll:C, ATTI_MINI0:yaw:C, ConvertDatV3 (+40 more)
- contains_constant_value: 140 column(s) - ATTI_MINI0:s_agz0, ATTI_MINI0:s_cnt0, ATTI_MINI0:s_pgz0, ATTI_MINI0:s_qw0, ATTI_MINI0:s_qx0 (+135 more)
- gps_accuracy_poor: 244 row(s) affected
- quaternion_invalid: 249 row(s) affected

norm_FLY019.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 8811 row(s) affected
- gps_poor_fix: 8811 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 71 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+66 more)
- gps_accuracy_poor: 8811 row(s) affected
- quaternion_invalid: 117 row(s) affected
- attitude_out_of_bounds: 5 row(s) affected

norm_FLY020.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 509 row(s) affected
- altitude_negative: 525 row(s) affected
- gps_poor_fix: 509 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 54 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+49 more)
- gps_accuracy_poor: 509 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY021.csv

- missing_gps: 8 row(s) affected
- altitude_negative: 6912 row(s) affected
- gps_poor_fix: 18 row(s) affected
- contains_no_value: 7 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, OA:frontDistance (+2 more)
- contains_constant_value: 49 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+44 more)
- gps_accuracy_poor: 18 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY022.csv

- missing_gps: 31 row(s) affected
- altitude_negative: 12113 row(s) affected
- gps_poor_fix: 21 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, OA:frontDistance (+1 more)
- contains_constant_value: 48 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+43 more)
- gps_accuracy_poor: 21 row(s) affected
- quaternion_invalid: 118 row(s) affected
- attitude_out_of_bounds: 30 row(s) affected

norm_FLY023.csv

- missing_gps: 33 row(s) affected
- motor_airborne_off: 381 row(s) affected
- gps_poor_fix: 19 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)

- contains_constant_value: 51 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+46 more)
- gps_accuracy_poor: 19 row(s) affected
- quaternion_invalid: 118 row(s) affected
- attitude_out_of_bounds: 37 row(s) affected

norm_FLY024.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 14159 row(s) affected
- altitude_negative: 16744 row(s) affected
- gps_poor_fix: 25801 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 44 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+39 more)
- gps_accuracy_poor: 25638 row(s) affected
- quaternion_invalid: 117 row(s) affected
- attitude_out_of_bounds: 1542 row(s) affected

norm_FLY025.csv

- missing_gps: 8 row(s) affected
- altitude_negative: 2075 row(s) affected
- motor_airborne_off: 939 row(s) affected
- gps_poor_fix: 1443 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 46 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+41 more)
- gps_accuracy_poor: 1454 row(s) affected
- quaternion_invalid: 120 row(s) affected
- attitude_out_of_bounds: 662 row(s) affected

norm_FLY026.csv

- missing_gps: 31 row(s) affected
- gps_poor_fix: 16521 row(s) affected
- contains_no_value: 17 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+12 more)
- contains_constant_value: 58 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+53 more)
- gps_accuracy_poor: 15818 row(s) affected
- quaternion_invalid: 120 row(s) affected
- attitude_out_of_bounds: 142 row(s) affected

norm_FLY027.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 886 row(s) affected
- altitude_negative: 3955 row(s) affected
- motor_airborne_off: 252 row(s) affected
- gps_poor_fix: 886 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 44 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+39 more)
- gps_accuracy_poor: 886 row(s) affected
- quaternion_invalid: 274 row(s) affected
- attitude_out_of_bounds: 1049 row(s) affected

norm_FLY028.csv

- missing_gps: 14 row(s) affected
- altitude_negative: 1364 row(s) affected
- gps_poor_fix: 17 row(s) affected

- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 44 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+39 more)
- gps_accuracy_poor: 17 row(s) affected
- quaternion_invalid: 276 row(s) affected

norm_FLY029.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 5686 row(s) affected
- gps_poor_fix: 5686 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 75 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+70 more)
- gps_accuracy_poor: 5686 row(s) affected
- quaternion_invalid: 130 row(s) affected

norm_FLY030.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 4065 row(s) affected
- gps_poor_fix: 6106 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 85 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+80 more)
- gps_accuracy_poor: 6106 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY031.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 372 row(s) affected
- altitude_negative: 9203 row(s) affected
- motor_airborne_off: 405 row(s) affected
- gps_poor_fix: 914 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 45 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+40 more)
- gps_accuracy_poor: 914 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY032.csv

- missing_gps: 31 row(s) affected
- motor_airborne_off: 183 row(s) affected
- gps_poor_fix: 20 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 50 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+45 more)
- gps_accuracy_poor: 20 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY033.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 6920 row(s) affected
- gps_poor_fix: 6920 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 69 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+64 more)
- gps_accuracy_poor: 6920 row(s) affected
- quaternion_invalid: 120 row(s) affected

norm_FLY034.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 1426 row(s) affected
- gps_poor_fix: 6408 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 61 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+56 more)
- gps_accuracy_poor: 6216 row(s) affected
- quaternion_invalid: 204 row(s) affected
- attitude_out_of_bounds: 59 row(s) affected

norm_FLY035.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 13201 row(s) affected
- altitude_negative: 5170 row(s) affected
- gps_poor_fix: 13615 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 47 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+42 more)
- gps_accuracy_poor: 14498 row(s) affected
- quaternion_invalid: 117 row(s) affected
- attitude_out_of_bounds: 281 row(s) affected

norm_FLY036.csv

- missing_gps: 8 row(s) affected
- motor_airborne_off: 422 row(s) affected
- gps_poor_fix: 17 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 41 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+36 more)
- gps_accuracy_poor: 17 row(s) affected
- quaternion_invalid: 117 row(s) affected
- attitude_out_of_bounds: 242 row(s) affected

norm_FLY037.csv

- missing_gps: 31 row(s) affected
- altitude_negative: 4542 row(s) affected
- gps_poor_fix: 874 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 40 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+35 more)
- gps_accuracy_poor: 910 row(s) affected
- quaternion_invalid: 118 row(s) affected
- attitude_out_of_bounds: 392 row(s) affected

norm_FLY038.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 857 row(s) affected
- altitude_negative: 12355 row(s) affected
- gps_poor_fix: 857 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 46 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+41 more)
- gps_accuracy_poor: 857 row(s) affected
- quaternion_invalid: 264 row(s) affected

- attitude_out_of_bounds: 22 row(s) affected

norm_FLY039.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 1 row(s) affected
- gps_poor_fix: 24 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ATTI_MINIO:pitch:C, ATTI_MINIO:roll:C, ATTI_MINIO:yaw:C, ConvertDatV3 (+27 more)
- contains_constant_value: 124 column(s) - ATTI_MINIO:s_agz0, ATTI_MINIO:s_cnt0, ATTI_MINIO:s_pgz0, ATTI_MINIO:s_qw0, ATTI_MINIO:s_qx0 (+119 more)
- gps_accuracy_poor: 16 row(s) affected
- quaternion_invalid: 133 row(s) affected

norm_FLY040.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 1331 row(s) affected
- gps_poor_fix: 1331 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 110 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+105 more)
- gps_accuracy_poor: 1331 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY041.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 6741 row(s) affected
- gps_poor_fix: 6741 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 108 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 6741 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY042.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2002 row(s) affected
- gps_poor_fix: 2002 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 109 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+104 more)
- gps_accuracy_poor: 2002 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY043.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 978 row(s) affected
- gps_poor_fix: 978 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 110 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+105 more)
- gps_accuracy_poor: 978 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY044.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 1012 row(s) affected
- gps_poor_fix: 1012 row(s) affected
- contains_no_value: 28 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+23 more)
- contains_constant_value: 106 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+101 more)
- gps_accuracy_poor: 1012 row(s) affected

- quaternion_invalid: 117 row(s) affected

norm_FLY045.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2734 row(s) affected
- gps_poor_fix: 2734 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 108 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 2734 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY046.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 3172 row(s) affected
- gps_poor_fix: 3172 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 105 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+100 more)
- gps_accuracy_poor: 3172 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY047.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 3672 row(s) affected
- gps_poor_fix: 3672 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 108 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 3672 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY048.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 11 row(s) affected
- zero_coordinate: 718 row(s) affected
- gps_poor_fix: 718 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 107 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+102 more)
- gps_accuracy_poor: 718 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY049.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2177 row(s) affected
- gps_poor_fix: 2177 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 109 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+104 more)
- gps_accuracy_poor: 2177 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY050.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 2391 row(s) affected
- gps_poor_fix: 2391 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 109 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+104 more)

- gps_accuracy_poor: 2391 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY051.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 898 row(s) affected
- gps_poor_fix: 898 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 116 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+111 more)
- gps_accuracy_poor: 898 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY052.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 40 row(s) affected
- gps_poor_fix: 40 row(s) affected
- contains_no_value: 39 column(s) - 4.3.0, ATTI_MINIO:pitch:C, ATTI_MINIO:roll:C, ATTI_MINIO:yaw:C, ConvertDatV3 (+34 more)
- contains_constant_value: 132 column(s) - ATTI_MINIO:s_agz0, ATTI_MINIO:s_cnt0, ATTI_MINIO:s_pgz0, ATTI_MINIO:s_qw0, ATTI_MINIO:s_qx0 (+127 more)
- gps_accuracy_poor: 40 row(s) affected
- quaternion_invalid: 45 row(s) affected

norm_FLY053.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 981 row(s) affected
- gps_poor_fix: 981 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 105 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+100 more)
- gps_accuracy_poor: 981 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY054.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 6915 row(s) affected
- altitude_negative: 397 row(s) affected
- gps_poor_fix: 6915 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 91 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:comp_alti:D, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D (+86 more)
- gps_accuracy_poor: 6910 row(s) affected
- quaternion_invalid: 3084 row(s) affected

norm_FLY055.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 3209 row(s) affected
- gps_poor_fix: 3209 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 84 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+79 more)
- gps_accuracy_poor: 3209 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY056.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 734 row(s) affected
- gps_poor_fix: 734 row(s) affected
- contains_no_value: 33 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+28 more)
- contains_constant_value: 110 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D,

AirSpeed:air_vby:D (+105 more)

- gps_accuracy_poor: 734 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY057.csv

- missing_gps: 6 row(s) affected
- zero_coordinate: 473 row(s) affected
- gps_poor_fix: 473 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 108 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 473 row(s) affected
- quaternion_invalid: 119 row(s) affected

norm_FLY058.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 221 row(s) affected
- gps_poor_fix: 221 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 115 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+110 more)
- gps_accuracy_poor: 221 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY059.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 428 row(s) affected
- gps_poor_fix: 428 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 118 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+113 more)
- gps_accuracy_poor: 428 row(s) affected
- quaternion_invalid: 118 row(s) affected

norm_FLY060.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 2144 row(s) affected
- gps_poor_fix: 2144 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 110 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+105 more)
- gps_accuracy_poor: 2144 row(s) affected
- quaternion_invalid: 119 row(s) affected

norm_FLY061.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 958 row(s) affected
- gps_poor_fix: 958 row(s) affected
- contains_no_value: 14 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):diffVelD:C, IMUCalcs(0):diffVelE:C, IMUCalcs(0):diffVelN:C (+9 more)
- contains_constant_value: 71 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+66 more)
- gps_accuracy_poor: 958 row(s) affected
- quaternion_invalid: 263 row(s) affected
- attitude_out_of_bounds: 72 row(s) affected

norm_FLY062.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 10956 row(s) affected
- gps_poor_fix: 10956 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)

- contains_constant_value: 86 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+81 more)
- gps_accuracy_poor: 10956 row(s) affected
- quaternion_invalid: 118 row(s) affected
- attitude_out_of_bounds: 94 row(s) affected

norm_FLY063.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 6922 row(s) affected
- gps_poor_fix: 8120 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+22 more)
- contains_constant_value: 106 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+101 more)
- gps_accuracy_poor: 8120 row(s) affected
- quaternion_invalid: 117 row(s) affected

norm_FLY064.csv

- missing_gps: 10 row(s) affected
- zero_coordinate: 22632 row(s) affected
- gps_poor_fix: 22632 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 103 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+98 more)
- gps_accuracy_poor: 22632 row(s) affected
- quaternion_invalid: 1 row(s) affected

Flight Logs**10-10-2017**

- format: heading_log | entries: 7

19-06-2018-08RDE5F001007H

- format: heading_log | entries: 3

19-06-2018-0K1DF313BD4PR1

- format: heading_log | entries: 7

log-2017-09-27.txt

- format: unknown | entries: 0

log-2017-09-27_1.txt

- format: unknown | entries: 0

log-2017-09-27_2.txt

- format: unknown | entries: 0

log-2017-09-27_3.txt

- format: unknown | entries: 0

log-2017-09-27_4.txt

- format: unknown | entries: 0

log-2017-09-27_5.txt

- format: unknown | entries: 0

log-2017-09-27_6.txt

- format: unknown | entries: 0

log-2017-09-27_7.txt

- format: unknown | entries: 0

log-2017-10-09.txt

- format: unknown | entries: 0

log-2017-10-09_1.txt

- format: unknown | entries: 0

log-2017-10-09_2.txt

- format: unknown | entries: 0

log-2017-10-09_3.txt

- format: unknown | entries: 0

log-2017-10-09_4.txt

- format: unknown | entries: 0

log-2017-10-09_5.txt

- format: unknown | entries: 0

log-2017-10-09_6.txt

- format: unknown | entries: 0

log-2017-10-09_7.txt

- format: unknown | entries: 0

log-2017-10-10.txt

- format: unknown | entries: 0

log-2017-10-10_2.txt

- format: unknown | entries: 0

log-2017-10-10_3.txt

- format: unknown | entries: 0

log-2017-10-10_4.txt

- format: unknown | entries: 0

log-2017-10-10_5.txt

- format: unknown | entries: 0

log-2017-10-10_6.txt

- format: unknown | entries: 0

log-2017-10-10_7.txt

- format: unknown | entries: 0

log-2017-10-10_8.txt

- format: unknown | entries: 0

log-2017-12-11.txt

- format: unknown | entries: 0

log-2017-12-11_1.txt

- format: unknown | entries: 0

log-2017-12-11_2.txt

- format: unknown | entries: 0

log-2017-12-12.txt

- format: unknown | entries: 0

log-2017-12-12_1.txt

- format: unknown | entries: 0

log-2017-12-12_2.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-10-10_[10-02-22].csv

- altitude_negative: 49 row(s) affected
- gps_poor_fix: 5 row(s) affected
- contains_no_value: 35 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh], CENTER_BATTERY.currentPV [V] (+30 more)
- contains_constant_value: 167 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+162 more)

norm_DJIFlightRecord_2018-06-19_[14-50-34].csv

- altitude_negative: 367 row(s) affected
- contains_no_value: 58 column(s) - APP_SER_WARN.warn, APP_TIP.tip, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh] (+53 more)
- contains_constant_value: 156 column(s) - APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState,

CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+151 more)

norm_DJIFlightRecord_2018-06-19_[18-48-38].csv

- contains_no_value: 55 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh], CENTER_BATTERY.currentPV [V] (+50 more)
- contains_constant_value: 161 column(s) - APP_TIP.tip, APP_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.encryptStatus (+156 more)

Images

screen_bf51868ae403be36_1529439024000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

screen_d1566e7e2a275f33_1529439020000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_33177fb665068406_1523611394000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_5be64acbef28816c_1523610990000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_6d9f39f0ca9decc3_1523610988000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_7f92b744bbc53b3a_1523610992000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_bf51868ae403be36_1529439024000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_d1566e7e2a275f33_1529439020000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_dd609e61934788b3_1523611494000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_ed8c1a297e1061b4_1524152656000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

2017_10_10_10_02_47.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2018_06_19_14_50_39.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2018_06_19_18_49_58.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0002.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Drone Serial Number inconsistency detected - conflicting values observed: 0670136230, 08RDE5F001007H, 0K1DF313BD4PR1. Manual verification required.
4. Drone Model inconsistency detected - conflicting values observed: Matrice 600 Pro, Mavic Air, Mavic Pro, MavicAir, MavicPro. Manual verification required.
5. Note: no controller device found - Device and account information.
6. DJI Application Version inconsistency detected - conflicting values observed: 3.1.11, 4.2.16. Manual verification required.
7. Note: no account e-mail found - Device and account information.
8. Note: no product type found - Device and account information.
9. Note: no firmware version found - Device and account information.
10. Note: no last country code found - Device and account information.
11. Note: no aircraft model code found - Device and account information.
12. Note: no controller serial number found - Device and account information.
13. Note: no last known latitude found - Device and account information.
14. Note: no last known longitude found - Device and account information.
15. Note: no account uid found - Device and account information.
16. Note: no account nickname found - Device and account information.
17. Note: no device uuid found - Device and account information.
18. 27 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_05, flight_06, flight_07, flight_08, flight_09, flight_10, flight_11, flight_12, flight_13, flight_14, flight_15, flight_16, flight_17, flight_18, flight_19, flight_20, flight_21, flight_22, flight_23, flight_24, flight_25, flight_27, flight_28). Independent corroboration is absent for these flight(s).

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

18-06-19-02-47-38_FLY057.DAT		Drone Logs	controller_android	extract_logical
SHA-256:	06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1			
SHA-1:	6f587093e526ec9f283190c1d33b04a0c206d470			

18-06-19-02-59-05_FLY058.DAT		Drone Logs	controller_android	extract_logical
SHA-256:	fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a			
SHA-1:	249331ad93c487a35571f531c4448907ce4e2ddd			

FLY004.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	58ed8baddcd6d687cffc8aef77d68f9db2521f1074f2c3caed8f8d48e0356b44			
SHA-1:	b8fb5a4f12ba0c6d0b8ea075c5b984c2fe3f5404			

FLY005.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	171b79ef5d7c469cdd35c9980fb1b49822e11f531023ac96d5f9523e1fbf240a			
SHA-1:	8b87cba821706b7115ce77d54694bfe09383195a			

FLY006.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	99db4012197bb754ba2999d22ea5fc44462a15761315a6a6d4ec16b128b1c1d8			
SHA-1:	165bef60517c70674a768453532d203b2bac2251			

FLY007.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e87a2bb1d25a9349645a591c5eb371e7a291053f544b6525bbadd0d0decf214a			
SHA-1:	faee17f7c52678200aa349c227dd571de07749c6			

FLY008.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	11b446da093910139fbf5746cca00c3e0848b43a117742878e010d1733e86c60			
SHA-1:	5f7261b1669f752c0847f385150a4bec59c2f4d3			

FLY009.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	bc40ad8a9f56e5b51d5af96f528162a5fb9f0e2ba4e37273f42ae0124f5aff00			
SHA-1:	054e08ffc29e45864fdc80964e37a0b46e021c4f			

FLY010.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	477e13d6d22da6a7f193e8a1a0d26d7d6970d99e942c188c1c48d5384bff5b2e			
SHA-1:	7f5b66b3ce93937ff9f7002dcd4ec7c92fb3b988			

FLY011.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c12bfa9f797e8b14b62dc4a8258f007a73f6bcce3bd3f55e6909deb0a25099d4			
SHA-1:	dcbd99c6e2061584baf25bbda76e4d54306b6851			

FLY012.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	40b091f4a73afe588e88d489f9e49404c617a8e13109905a5f9ddefae494defb			
SHA-1:	fc667c187e080ee1ef7f0b07c5d7ebf6ab4bcd5			

FLY013.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	0e3fa1b554ebe4f0392bac01cf486ceb420677d3a29052f5df2e90c9f967cbee			
SHA-1:	b886409c25d3edf7b56b20bb05642705dc6817d0			

FLY014.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	ae2eacc59fcad3ceb68c1cf9a50ef6f63cab2129e54b71b0043adaeb633e8444			
SHA-1:	4b5a4696a8b909122a0a3771f6b0a8529c990b82			

FLY015.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c665b4ad6b7e4a2ff7caf0ef0d6a333f94ea6ba3083fdaff93e298e61a425deb			
SHA-1:	1d42731ed0e0e8e23a754c88e0125056fa302848			

FLY016.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	d4fcb3c8efd8df0656ce4eea5512356e85b382bd00e06095f6278c630942c7bb			
SHA-1:	b1abec867ccfd5b5a22fffd2171df56144f5218e9			

FLY017.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	1ed328c1c0fa131070bc51c263219c0d810e2ddf976d5a1795354c68309a1bd7			
SHA-1:	980941a21da6255abde0b0df52a0839cb22be6ee			

FLY018.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	78058dc479a2e437ed23235f12bbadde6dbb8870f3a0abc64d0776ac48a8503c			
SHA-1:	e3d59d58c5cdeb9ef67dda6e0413faa6dc3b0d83			

FLY019.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	baadb87e4ff28c558967c0a25334a557e31a6a2d4e1e4919cddaed0b4b0f08b3			
SHA-1:	72f7eb608808afc486d2243182fdca8f2aace47d			

FLY020.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	a1476534bb0272153fb93fc870f9db80de52f17945a97b16103371632c48931c			
SHA-1:	d1eb8e838d5d893082bc3cc43752f27489066fa3			

FLY021.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	64e6e0b6bc27ff0dd27675114491b21a87f368c60426515931579ce87566d228			
SHA-1:	0017af5ca40a63fac24ccc7ad1f49939b622c6f			

FLY022.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e0b7c74464802fe363f29100443180c3995f31992af4e67196cc151b0e6dae4c			
SHA-1:	591b045ffdebdc6a4357e541042fb7aa8139409c			

FLY023.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	2f241fc6739862d269d07fcf563b82812f508e2f061fcb55fb02da4e9c2aa5b3			
SHA-1:	390c2d4c7cc6ff43ddb99239cd236da53c64c2ca			

FLY024.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	79184f68209fa10bfd31b3b58cbb09b15d536d092d4ae100e2e1e6f2567a2c15			
SHA-1:	8dcbe9180c542fa5096d808d251b9db5e1ca70eb			

FLY025.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	0b7a37036838f35eff28b507fac3d33e7475e1134594719368f7fe9823feb557			
SHA-1:	40ff5a56f87c1eee68ec7f2a7103fb0fa41b221c			

FLY026.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	1a0bc29d39be560e5c27ce7b1ad5786544e45ccf8ffef8810fc8edddaa6a04ed			
SHA-1:	dbe33971f996064b5cb23026bef6fb191575d579			

FLY027.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8ecdff0d45829c7ebdfc245fc0f28ab8dfd653b74a4dce70699799059f4c94ca			
SHA-1:	ab06c7e84595fda9f794d3baac70e1b57bed91c8			

FLY028.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e0d7364429b584e518b732d422b252e8e3082e69484f0d0bebc0daf9fd519dd			
SHA-1:	585a62a1474e2a43b9872e925d5b57cc7de3c02e			

FLY029.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	431a6722d58f63fe08d5af43d10652ee9bf9c8deafefa1f8618c5dd10a51268			
SHA-1:	3d72ef838cd90d0a2bd15f7328c89c022d77d230			

FLY030.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3ecf44b4290fc3be33f0077cdd07108e3c0eb138a4584cc214ff2bb514fef515			
SHA-1:	658fef652f3d38eb5b3d3e615722cc316b7771ad			

FLY031.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	15f94f68cc64b8ec695a47c195a82c3c12392346de66efb45217b3b1500d7106			
SHA-1:	36c95661c832dd41bdd3129974821ae90bcbf73f			

FLY032.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	5ff858c5d639582f680cd4797d16c3731330efe4a0efa8ed3b83ad7735d1f976			
SHA-1:	97650a0b1b9ac761f31464afae52eab63550dcea			

FLY033.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	dc0cbf35f0c25f1418db50d7f2fd04167436095121ee561f53b9886853a3c97d			
SHA-1:	b50c9eac935485d31e92cde6bdb3916612019faf			

FLY034.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	7e99045a65456b864d08d10c7cc007d4afc5525fb28bd95eb09944de7cbee0fd			
SHA-1:	b9dbee369b67c05718d8513aa74d28c524fe07c0			

FLY035.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	dce1f2fc0865178b0f31c37f1d290dc6b906520590dbda57584a9d6ffbb5c937			
SHA-1:	6cbc0592d121fb16d0376932654df8156b9c809e			

FLY036.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e2ba1311afff06737e0891d784d9c773835f52c790666fcd27b6307a77322de2			
SHA-1:	a68ef83d6742fe676c58cce57b72b1e69b9e36ca			

FLY037.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	b9759b4bc9755fb359419d4ffa857285bf37ee3c2aa3a3ed62e67bf8ed916c1d			
SHA-1:	a8f05b58b2a5470970c55cc73041c5d95e67e536			

FLY038.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	9065e6b3f84e0415c26ace13902d85f1b3446b0aef67d4431d4cf10a06c2d8d9			
SHA-1:	73b5233f3c015fd1c5a3af93d3920c0b053aae1a			

FLY039.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	6cf1153647818d0d36deac7e753aab2e4bbc209a3ae445f52c292b8452319068			
SHA-1:	b6c5ddbfc074dfb27fec80793df41c1e14d405b8			

FLY040.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	d49c618d838fb463e3c0b7680839214320a1f0b317d251b192caae5c3f8003c3			
SHA-1:	9101e7d70c8382402efbb539eab25dec4c462f7			

FLY041.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c0ca9b7c9916167dd78766a49e3b44846cc80a8ccd2c0ec32b2c9a1a311dc8cd			
SHA-1:	a2fc4a0fc0a13ab487c088b8b8899062a45c06bd			

FLY042.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e267a2ef07ded67689669bc0b10f262c2935478c8006eb3118725438c89be565			
SHA-1:	f8e09a7ece23d5bb0e8ea47d4d3b598d1ea05eb7			

FLY043.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	659fddaabae489b0b9e1a8f753cbbbef6b7503f159ad5c19e182cde811decae2			
SHA-1:	4edeb1317d4a68d3ed05b4aac256c2cc8cd41835			

FLY044.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	5135607952892cb3e79278799a42a1717612a86b2a91facd36d0064ecbc794b2			
SHA-1:	fbf65201eab257a1b0558a5253bed89a597fef2e			

FLY045.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c7f72acf014c1f62ac39df3cba2b4e6400355609c16453666d3565b770eb87d4			
SHA-1:	de91a2aec4c803d848e9dfa664fe97b0b219edbe			

FLY046.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	75ddc03d767e14f59cf830955b12cc8172cfba4632ee614f85276e945001dc98			
SHA-1:	60fbfd41444c054938994e8771107ce708f56099			

FLY047.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	bf0e35cd28673b6150baa7e1d12f1155f40b627a55f215b1b379a59b50142579			
SHA-1:	7ab4801d3d6d207271300db8ab1b5ca3b016f865			

FLY048.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	05347c1677905f87e8b7edb3975d98696717c83448e6003e7e6a0c97ba88ef21			
SHA-1:	82558578283e67294013f4f3ff8eeb76bd824fd2			

FLY049.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	eb9d529e861897950770a5a083130f3316b8cc665800968013c3fc697eff8e0f			
SHA-1:	7e2f6b7b80ee7966c7e2b9909a94a0d193b47245			

FLY050.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	2fe5d6e404d04c63fff70fac3f6de7557162303a988f021dbabb882467ec53ca9			
SHA-1:	c2e68512579c83f9e12a69e3e272afb554160172			

FLY051.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	6eb301232610bb23008c92bba93ee2f17f1da9b2d7e6b2fa073c0ce8bdb41044			
SHA-1:	352f2f94be712c4c619f743386ea25a2cd11b138			

FLY052.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	97c794a2e2f6a817f1547176dbf210eede97fc7ab8c89b76e5cb2489b84ddb44			
SHA-1:	8be8d3ffc83faa9e9512c3705dab8f1c5d0c687			

FLY053.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	0cd49fb88f83d9ec4764aad1171c39a995a11b4ac34d08e3c07f03c958b791ad			
SHA-1:	91a941e8df7f5f3674ab8eb97f320cf80941b328			

FLY054.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e9f3d7214dc01b959a9edd055fcda794b5136d34e97a4fc6a8196566ff1275f3			
SHA-1:	6baf64cd3e9557c4c130f78cdd27f6bd7dae3551			

FLY055.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	04bfac1d89630a1f6516e662534283f8c0fe4ddb85f8980a9cf522f66f955b7c			
SHA-1:	27d7acc6d6216c8f87004fed0946a33f7d0bb5ad			

FLY056.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3d257a76c05159a53b85b22c83abafc1269a55d083af4a2e4cec6bcdac616af0			
SHA-1:	7e372f99d9f50802b7b3c25c04fb517a7c633a57			

FLY057.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	24057640f6841e02154c5a65aa0d1ba25056ffcd4ee9c18e499b773e4f543434			
SHA-1:	3877e57a425d02ce19dd1f1c43852b0708a39204			

FLY058.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	4acf67971f8f57bd404450ab588e6196077e4b173d63a1b732f010c3ac73fc9b			
SHA-1:	a325ae13e63234d9663b248628877d22a6548925			

FLY059.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	9ac955b42ab0aa5c5fad12c016625efb5b01a2c51d20c886633fea57644ff7b8			
SHA-1:	b97cef9715fdb98f1ef49a4243c9dd2cb78ddaf1			

FLY060.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	602ff120a03b68b1dce095b59040ff29c66af2b43a68df1993e6fcf95c93b18f			
SHA-1:	210d0dc1240281e969d195af41b0e3b0fa3319b2			

FLY061.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	2eachb27e19d4b40672932f882cc8e050501ed25242b6ff7d6c6161475595f24b			
SHA-1:	4a7e88763eb3b7d8adb2123c8abc0b01f97cf5c1			

FLY062.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	162c978b5f5d833ac164178c8c75faa03ee4813734938a00bd7af75a51ee85f3			
SHA-1:	35b48a2f28eddb0b30dbf60ae2eda1fad18236d5			

FLY063.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	516e74c6b24853499b140df1d1fd32d78297bc00ac24912aa873471b2e62b3e6			
SHA-1:	9457ff3b0eccc4a4da840ebc1881625c60c97bef			

FLY064.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	85d7c0a7098d42e21b1e3ee4c807e2d0fa3cb5fe18b52b78f5249be8637d336e			
SHA-1:	50bf8e164144e715c170beaa6855bb57a5958642			

WPSettings.dat		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	e4687665625c0709d17645d2b53e2113f82608a549641962cb62f35b1a461bbf			
SHA-1:	4d48cf037778a7cf70c0a486096cbb0093c955f1			

10-10-2017		Flight Logs	controller_android	extract_logical
SHA-256:	48a35a1d275e0db3581766f72e8179401a4f74ea1bca7815942c659d86aea50a			
SHA-1:	a4d426fbf5f10d0b4375c5b4a082eb0819eaf4a0			

1529447164868		Flight Logs	controller_android	extract_logical
SHA-256:	9b78b9af59fd7ebb5b1c84e36f4d936774df9e8c2a348fe984eca6536a6e9850			
SHA-1:	c9d8d2954d50c98f0d093edbe1fc70e00a9c2203			

19-06-2018-08RDE5F001007H		Flight Logs	controller_android	extract_logical
SHA-256:	98afb1967cf1fb914334053461c16843d20c98c50f4494525a8e71a06785786			
SHA-1:	4c654b1253dd7d602f7db81cf9abc49932dda294			

19-06-2018-0K1DF313BD4PR1		Flight Logs	controller_android	extract_logical
SHA-256:	cb9671659ae2ff11b0fd3dd846e8702a81366479b04f3e2442300b2211bcff3e			
SHA-1:	3792e2ff77e3cecca7dd66823748139f459bd108			

log-2017-09-27.txt		Flight Logs	controller_android	extract_logical
SHA-256:	894f25c259709491d0a768c24bd0fc99ed9f4793d25d17c006024683817792bf			
SHA-1:	f15b75a35193a648e2a8c9a625aa8e45077f174e			

log-2017-09-27_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	9d2a843316232069c58ec9b794369f2d1e174ed5a9a681df9c3e92a63e21aaa9			
SHA-1:	7f860f235699bf30368d06735c86c4770163a883			

log-2017-09-27_2.txt		Flight Logs	controller_android	extract_logical
SHA-256:	2baea06c4aebba79a3d66fc9ecc1443785b551d9c7bf8ec01a764eb1cdeb501a			
SHA-1:	b5dc41909888ac0a72fbbb75389945e82d73d2b7			

log-2017-09-27_3.txt		Flight Logs	controller_android	extract_logical
SHA-256:	7a60de6d0104d8d1673a7e57f1e6d0f128785e032076426563ec469ebba23a39			
SHA-1:	382f017cb1d7ba79ef6457b288fb6ff5f9519652			

log-2017-09-27_4.txt		Flight Logs	controller_android	extract_logical
SHA-256:	d90e302e33743d784de0ef46a13a17beb71fd40b217e632195f1d1b19bfb870c			
SHA-1:	2774c8e36266f759189106d5f8ef05a9643022be			

log-2017-09-27_5.txt		Flight Logs	controller_android	extract_logical
SHA-256:	0038111eaa570a867f06b0d10f1fa2dff3c3ac967c5af10941aa0423526dd843			
SHA-1:	7488b69ac447860fc1d844d81fd4e42de4b84611			

log-2017-09-27_6.txt		Flight Logs	controller_android	extract_logical
SHA-256:	95fcab9567034a565b6bed21a8169e645846f97c8c033127d2a835733f9e78fc			
SHA-1:	10690bcf6ca5e428c11e1e10876cb70295a8b2aa			

log-2017-09-27_7.txt		Flight Logs	controller_android	extract_logical
SHA-256:	662ea13bf63cc9787d4d843305e204a3c6734350fcae3368f7edb1f17e17f585			
SHA-1:	202fb4e595461cdf82c1aa0da24abca3a45e4f3d			

log-2017-10-09.txt		Flight Logs	controller_android	extract_logical
SHA-256:	4961f9bd65d1da4df06918bb4aaa7933feb7c3c0189f03e7bbf477c9160c3750			
SHA-1:	2c678591d4c3d6f98c1e671dd0b95a9580151132			

log-2017-10-09_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	e36c5fa294f38c8e5ce12322e922365150ee5fbae22ac26db11781bdb3de94c8			
SHA-1:	755e6ee81cecb9cd2e51f01e32d3e3b3a73fa768			

log-2017-10-09_2.txt		Flight Logs	controller_android	extract_logical
SHA-256:	c8341a3abc580a827cffff3e9713e904b8d781a4a930007b67e61f5803b60e63e			
SHA-1:	b9c918529b0fcc1968a1edaa3f14ccfe855e671a			

log-2017-10-09_3.txt		Flight Logs	controller_android	extract_logical
SHA-256:	7724ba97cd4511eb2b0843aa8efdc5bd7192d2ff9c2fd3c5f3a4dbb3d81ada8f			
SHA-1:	e8745e71fe2a63c3f3817ffe18028dd672a0f0bf			

log-2017-10-09_4.txt		Flight Logs	controller_android	extract_logical
SHA-256:	bc0ee9e645bba2510f45462f13cdb2973e48c0abb880c79cfd53862229f7a633			
SHA-1:	1f83e5e1e0a0aba13153a410c74ab6ccad1dfc2f			

log-2017-10-09_5.txt		Flight Logs	controller_android	extract_logical
SHA-256:	d458d39b5a7a3c3cd2b3d277dd3bb1f6d628bf566bbe8daf26e98857485b4d71			
SHA-1:	210d44431515e4f8b6d3dd6648ab5591ca575a3f			

log-2017-10-09_6.txt		Flight Logs	controller_android	extract_logical
SHA-256:	2a24f2bd2096630c94171eaff4ee623d6dd060697250e81124b6bae7874fbd23			
SHA-1:	a398ba235179b487991c850d811463ce91a8569e			

log-2017-10-09_7.txt		Flight Logs	controller_android	extract_logical
SHA-256:	c578617197a4f0ff7f52391db1c29ac456712a3f5938f6315a8a4de174df353e			
SHA-1:	80e4508e292f6a494fb38d23182d7615b0b01eb0			

log-2017-10-10.txt		Flight Logs	controller_android	extract_logical
SHA-256:	87e7baf7d274ffff12e46e1de5db451f450cee31655b22f6c07c34591742774bc			
SHA-1:	082875e1cfd4f72f0a2b8bb40ca1c2accc868372			

log-2017-10-10_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	ae84503564a90d25c78355de393d7c7a7efa6d7731ddc9b7f41cabca39953187			
SHA-1:	3d50cf703e7e39252101d91a7c74cf587c853134			

log-2017-10-10_2.txt		Flight Logs	controller_android	extract_logical
SHA-256:	781f12c3c4e32cec55f04d45f6d1ea7cce95d23049f4ed66329cca98ea419270			
SHA-1:	a7f735c63f556d42ec358564daa0aaaf49c19df0			

log-2017-10-10_3.txt		Flight Logs	controller_android	extract_logical
SHA-256:	3a70004e5bccd9e5dd80f8d96d53dbab7bdfd31f3338f81964742584cb777fc56			
SHA-1:	ff114795961a33488379f9a7c4144503d899df19			

log-2017-10-10_4.txt		Flight Logs	controller_android	extract_logical
SHA-256:	f37f26fa51ad3027d858170b91eb4f5daf5dd824dcedb994e4f2d0b8dec8f474			
SHA-1:	8565ef7f3ced81d6102aa05193c15a900c0832f			

log-2017-10-10_5.txt		Flight Logs	controller_android	extract_logical
SHA-256:	c26f9650e93c2733f7316978f2cd3a370b2aedbc3db86bc4821fe6087cb487bc			
SHA-1:	d6dd690bc097b55ecad370a9f201a0cfd7e3d1dd			

log-2017-10-10_6.txt		Flight Logs	controller_android	extract_logical
SHA-256:	9e8e422fb7542b2d62fec5478852da23ecddf40dd08d177806da46cf3b619fac			
SHA-1:	4c68bf837ead4a5de9fd913b52cb3ad9b8559a2f			

log-2017-10-10_7.txt		Flight Logs	controller_android	extract_logical
SHA-256:	9876a533ddde0c5ad25489905bd3e0e35bce9a4d0f1f77aaebec2eaa9fb287ae			
SHA-1:	a44441dfa2f1e437087bc0d1fc6adf31d1ff89c7			

log-2017-10-10_8.txt		Flight Logs	controller_android	extract_logical
SHA-256:	4453dd1014b67ec7f35298f67ec559d96fb29e40aa8d5a16fa346469d011749a			
SHA-1:	9bdb40c1cf52b996ee1bdcf3230126cd3739f738			

log-2017-12-11.txt		Flight Logs	controller_android	extract_logical
SHA-256:	96f5a059fb10069371ed62066f00ca0661d1fc5a02c170a48a8cceb7d7447604			
SHA-1:	9a2d18560b053ddb0f03180a05f7715f0122483d1			

log-2017-12-11_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	8f97e22370768db84494a0db003fad1056d5a29e747f6c49dafa20ee6b594926			
SHA-1:	4ecab73d120e2fb79ec11c082a8ca3bd23dd8f0a			

log-2017-12-11_2.txt		Flight Logs	controller_android	extract_logical
SHA-256:	8bd579a33bd9246a9452fe243f851292d8b178607d625cd8946082d3aaf229a1			
SHA-1:	88b4283d8696d180cae205760f144935203b5cb9			

log-2017-12-12.txt		Flight Logs	controller_android	extract_logical
SHA-256:	bbe0eda5d1f58d893d5bdbb1355df2dcd5f0607e83c7a012e0b19ebe39999b7f			
SHA-1:	9e73a56dded6994d8bab87c6e0b36fb0c28fcc8d			

log-2017-12-12_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	ab4d29e99e718c141f9c9f85d211dc1827169f3fa6e65d749e4922c17c6be004			
SHA-1:	e7fdd9f6b2f537fb983df3316e23e321febd099d			

log-2017-12-12_2.txt		Flight Logs	controller_android	extract_logical
SHA-256:	87ed69dea6386ec3cc6b6cd7059bc7afe8444a6db49b91fab9fb23defba93c24			
SHA-1:	fc465cc7d2098e1532d58ac43d747785e2a2d424			

DJIFlightRecord_2017-10-10_[10-02-22].txt		Flight Records	controller_android	extract_logical
SHA-256:	f134bef34e6877ae5a030297f56ae6e12fe946b468c5143ef9976c08170295f8			
SHA-1:	53b8c7d8257402a8df7649c5549c70252b726138			

DJIFlightRecord_2018-06-19_[14-50-34].txt		Flight Records	controller_android	extract_logical
SHA-256:	0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a			
SHA-1:	66bc3fce709dff20df9fabcaa97a036d2c8f2c57			

DJIFlightRecord_2018-06-19_[18-48-38].txt		Flight Records	controller_android	extract_logical
SHA-256:	ab5bbc58a921d81c323a76c3a9d955fde58d503536facf43306e7e6bbb208c8d			
SHA-1:	b636a68a747801b43512f878469c0f8868418b9f			

screen_bf51868ae403be36_1529439024000.jp		Images	controller_android	extract_logical
SHA-256:	dcfa5ede56ef92757df00d80a915a118258d68bce1755828b85cd96c5be8bf0f			
SHA-1:	0fafc1f09c3820fabddf24a67f387b9d990774b2			

screen_d1566e7e2a275f33_1529439020000.jpg	Images	controller_android	extract_logical
SHA-256:	8e059c9f8536775e215e9d4fdd40f7943b12b6832de4bb96e9749cc3e0a9cf18		
SHA-1:	0593008249764bad01eb2096dd5fdb73ead6b487		

thumb_33177fb665068406_1523611394000.jpg	Images	controller_android	extract_logical
SHA-256:	e8d6ac3d4cda2bccff4dd3e64950e87fd68700cf80b37f2c45e9e0ac59a4d42a		
SHA-1:	55b6683fe7bf253b609ca32c4fcc3a33795ca21d		

thumb_5be64acbef28816c_1523610990000.jpg	Images	controller_android	extract_logical
SHA-256:	e6a060c7756c7e2fd99c850434b02c708a329ebf16a814e0fe0a77b15caaab24		
SHA-1:	ed1bfdcab3e02d87575cadd6c79fdd983021271		

thumb_6d9f39f0ca9decc3_1523610988000.jpg	Images	controller_android	extract_logical
SHA-256:	cbe209aa64bbb44340e4aed5d2e276fd1df75805f956307b24a83e79b1d6f90		
SHA-1:	5b90ca57716f15045a6a65c7215ca77036ff6af2		

thumb_7f92b744bbc53b3a_1523610992000.jpg	Images	controller_android	extract_logical
SHA-256:	32ff09733d14f4ecdd59171811fd50111cc00752ebdb528f04b75b08cf220874		
SHA-1:	874dff84aaa18baf06b4070276135346c328e178		

thumb_bf51868ae403be36_1529439024000.jpg	Images	controller_android	extract_logical
SHA-256:	4f5530dba1b433c1b0d223d2c7c954760a9428aab12427bb157787ea8adeb263		
SHA-1:	c4e2fb21853489b584209e98060f362ed807a4a6		

thumb_d1566e7e2a275f33_1529439020000.jpg	Images	controller_android	extract_logical
SHA-256:	e295a5518fc423a446ea5a9f2026e1b8cc944634e6ce3a1b78f41fe50fd5952f		
SHA-1:	381d1a6cc4335f148509621763576fcc5e08274e		

thumb_dd609e61934788b3_1523611494000.jpg	Images	controller_android	extract_logical
SHA-256:	8986d1b4a162ca4bbe0adbf12f01141b9d7673ba196436ee31e16fb6b540b2e2		
SHA-1:	3802402288db8466063ecaec81012031de544392		

thumb_ed8c1a297e1061b4_1524152656000.jpg	Images	controller_android	extract_logical
SHA-256:	924f798a8f815a2a8c3862bf40481f34194ce30fbd637c939e4aa66fcb108fd3		
SHA-1:	898544f8b343f3ad616898273468c7203082872e		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	7f699c964c4b84244597faec87b1d0eee052fac9e8d4f09a4af69ffa6b546111		
SHA-1:	f74fe88f25a19e43abf0aae394cdbeea06c89f58		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	42367baaac566d6f39211b1b298e884756ccc9156e246b49bc54f8e8e7ed2302		
SHA-1:	ef32dac387b442ab2e44b46de1d793fe48e48805		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	986f166d384eb384e7b6ca1e6b3d36caa32247667f8f33e30a6d965d8f3d272e		
SHA-1:	5c72706553e00450709e1bf12ee732646008cc9b		

DJI_0004.JPG	Images	drone_sd	extract_physical
SHA-256:	96a25a259d992c3d34222683e21a8591c808ce0162178ffadd50eddcc03c9578		
SHA-1:	6ee7575e65d9e29aee60fdb69a44d2653507957e		

2017_10_10_10_02_47.info		Videos	controller_android	extract_logical
SHA-256:	0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345			
SHA-1:	4ed6541a3da7544dded2162f25785a5a4b9a90d6			

2017_10_10_10_02_47.mp4		Videos	controller_android	extract_logical
SHA-256:	ce078b575399110fb022fac5c4802bf50dfe0f58e76976480c93aefee354e5dd			
SHA-1:	8be2153aba2e3419c8296774f6a64e61e52293c0			

2018_06_19_14_50_39.info		Videos	controller_android	extract_logical
SHA-256:	a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a			
SHA-1:	def346de1f2d68bb135f621930cfec9eca413581			

2018_06_19_14_50_39.mp4		Videos	controller_android	extract_logical
SHA-256:	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00			
SHA-1:	8afc6c44c37db8cb428875a43710b45f9549479e			

2018_06_19_18_49_58.info		Videos	controller_android	extract_logical
SHA-256:	a1124ad33ad282daf8c8eac97cf5d3b488835d8c31a010cffcb7b341bd62ed6b			
SHA-1:	35d033da145c19fd24d80a2ce449792226377fae			

2018_06_19_18_49_58.mp4		Videos	controller_android	extract_logical
SHA-256:	d64a60065fe82ea93cbbbe90de4ee962053c270673fc3c7cb0dafd23ece7870c			
SHA-1:	ef7174efc27be6c38a448018292541d5bf847356			

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	0cdb76c21c03e7dbb6c0807fc775a4305dbf83ca563bc949660eba20a44fe896			
SHA-1:	c2fabecaaa568f1b9a5ea557ae89516154748454			

DJI_0002.MOV		Videos	drone_sd	extract_physical
SHA-256:	355b378ae4911bb1566e0254cf3c299c34c84c15b9f835fc4ed9a1d361b8a01c			
SHA-1:	133647db986ab7697b1ee0cc14128cb5de8e47ec			

DJI_0003.MOV		Videos	drone_sd	extract_physical
SHA-256:	0f0eb0cad050d5d59d34d4c7858c28a245cd7a433e320da41cc8a279b38a1c24			
SHA-1:	d561d6fe9640620511497b22f1065153d289e480			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	2	4	2
Flight Logs	35	0	0
Flight Records	3	3	3
Images	10	10	0
Videos	6	6	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	4	4	0
Videos	3	3	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	62	122	61

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 14:27:27 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df020_external.001

2026-06-18 14:27:44 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df020_external.001

2026-06-18 14:27:44 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df020_internal.001

2026-06-18 14:27:45 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df020_internal.001

2026-06-18 14:29:00 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df020-2018-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-18 14:29:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 11269
out: DJI_0001.MOV

2026-06-18 14:29:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 11270
out: DJI_0002.MOV

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 11271
out: DJI_0003.MOV

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 11272
out: DJI_0004.JPG

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 12294
out: DJI_0001.THM

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 12295
out: DJI_0002.THM

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 12296
out: DJI_0003.THM

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 5
out: FLY044.DAT

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 6
out: FLY045.DAT

2026-06-18 14:29:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 8
out: FLY046.DAT

2026-06-18 14:29:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 9

out: FLY047.DAT

2026-06-18 14:29:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 10

out: FLY004.DAT

2026-06-18 14:29:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 11

out: FLY005.DAT

2026-06-18 14:29:21 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 12

out: FLY006.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 13

out: FLY007.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 14

out: FLY008.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 15

out: FLY009.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 16

out: FLY010.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 17

out: FLY011.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 18

out: FLY012.DAT

2026-06-18 14:29:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 19

out: FLY013.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 20

out: FLY014.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 21

out: FLY015.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 22

out: FLY016.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 23

out: FLY017.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 24

out: FLY018.DAT

2026-06-18 14:29:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 25

out: FLY019.DAT

2026-06-18 14:29:24 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 26

out: FLY020.DAT

2026-06-18 14:29:24 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 27

out: FLY021.DAT

2026-06-18 14:29:25 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 28

out: FLY022.DAT

2026-06-18 14:29:25 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 29

out: FLY023.DAT

2026-06-18 14:29:26 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 30

out: FLY024.DAT

2026-06-18 14:29:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 31

out: FLY025.DAT

2026-06-18 14:29:28 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 32

out: FLY026.DAT

2026-06-18 14:29:28 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 33

out: FLY027.DAT

2026-06-18 14:29:29 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 34

out: FLY028.DAT

2026-06-18 14:29:30 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 35

out: FLY029.DAT

2026-06-18 14:29:30 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 36

out: FLY030.DAT

2026-06-18 14:29:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 37

out: FLY031.DAT

2026-06-18 14:29:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 38

out: FLY032.DAT

2026-06-18 14:29:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 39

out: FLY033.DAT

2026-06-18 14:29:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 40

out: FLY034.DAT

2026-06-18 14:29:33 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 41

out: FLY035.DAT

2026-06-18 14:29:34 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 42

out: FLY036.DAT

2026-06-18 14:29:35 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 43

out: FLY037.DAT

2026-06-18 14:29:36 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 44

out: FLY038.DAT

2026-06-18 14:29:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 45

out: FLY039.DAT

2026-06-18 14:29:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 46

out: FLY040.DAT

2026-06-18 14:29:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 47

out: FLY041.DAT

2026-06-18 14:29:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 48

out: FLY042.DAT

2026-06-18 14:29:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 49

out: FLY043.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 50

out: FLY048.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 51

out: FLY049.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 52

out: FLY050.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 53

out: FLY051.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 54

out: FLY052.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 55

out: FLY053.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 56

out: FLY054.DAT

2026-06-18 14:29:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 57

out: FLY055.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 58

out: FLY056.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 59

out: FLY057.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 60

out: FLY058.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 61

out: FLY059.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 62

out: FLY060.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 63

out: FLY061.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 64

out: FLY062.DAT

2026-06-18 14:29:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 65

out: FLY063.DAT

2026-06-18 14:29:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 96964615

out: WPSettings.dat

2026-06-18 14:29:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_internal.001 69

out: FLY064.DAT

2026-06-18 14:30:23 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 18-06-19-02-47-38_FLY057.DAT

out: 18-06-19-02-47-38_FLY057.csv, 18-06-19-02-47-38_FLY057.kml

2026-06-18 14:30:40 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 18-06-19-02-59-05_FLY058.DAT

out: 18-06-19-02-59-05_FLY058.csv, 18-06-19-02-59-05_FLY058.kml

2026-06-18 14:30:41 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-10-10_[10-02-22].txt DJIFlightRecord_2017-10-10_[10-02-22].csv

out: DJIFlightRecord_2017-10-10_[10-02-22].csv

2026-06-18 14:30:42 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19_[14-50-34].txt DJIFlightRecord_2018-06-19_[14-50-34].csv

out: DJIFlightRecord_2018-06-19_[14-50-34].csv

2026-06-18 14:30:43 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19_[18-48-38].txt DJIFlightRecord_2018-06-19_[18-48-38].csv

out: DJIFlightRecord_2018-06-19_[18-48-38].csv

2026-06-18 14:30:43 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n screen_bf51868ae403be36_1529439024000.jpg

out: screen_bf51868ae403be36_1529439024000_exif.json

2026-06-18 14:30:43 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n screen_d1566e7e2a275f33_1529439020000.jpg

out: screen_d1566e7e2a275f33_1529439020000_exif.json

2026-06-18 14:30:44 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_33177fb665068406_1523611394000.jpg

out: thumb_33177fb665068406_1523611394000_exif.json

2026-06-18 14:30:44 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_5be64acbef28816c_1523610990000.jpg

out: thumb_5be64acbef28816c_1523610990000_exif.json

2026-06-18 14:30:44 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_6d9f39f0ca9decc3_1523610988000.jpg

out: thumb_6d9f39f0ca9decc3_1523610988000_exif.json

2026-06-18 14:30:45 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_7f92b744bbc53b3a_1523610992000.jpg

out: thumb_7f92b744bbc53b3a_1523610992000_exif.json

2026-06-18 14:30:45 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_bf51868ae403be36_1529439024000.jpg

out: thumb_bf51868ae403be36_1529439024000_exif.json

2026-06-18 14:30:45 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_d1566e7e2a275f33_1529439020000.jpg

out: thumb_d1566e7e2a275f33_1529439020000_exif.json

2026-06-18 14:30:45 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_dd609e61934788b3_1523611494000.jpg

out: thumb_dd609e61934788b3_1523611494000_exif.json

2026-06-18 14:30:46 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_ed8c1a297e1061b4_1524152656000.jpg

out: thumb_ed8c1a297e1061b4_1524152656000_exif.json

2026-06-18 14:30:46 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_10_10_10_02_47.mp4

out: 2017_10_10_10_02_47_exif.json

2026-06-18 14:30:46 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_14_50_39.mp4

out: 2018_06_19_14_50_39_exif.json

2026-06-18 14:30:47 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_18_49_58.mp4

out: 2018_06_19_18_49_58_exif.json

2026-06-18 14:30:47 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-18 14:30:48 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM

out: DJI_0002_exif.json

2026-06-18 14:30:48 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-18 14:30:48 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.JPG

out: DJI_0004_exif.json

2026-06-18 14:30:49 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV

out: DJI_0001_exif.json

2026-06-18 14:30:49 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MOV

out: DJI_0002_exif.json

2026-06-18 14:30:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV

out: DJI_0003_exif.json

2026-06-18 14:31:20 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-18 14:31:41 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-18 14:32:09 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-18 14:32:34 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-18 14:32:51 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-18 14:33:06 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-18 14:33:23 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

2026-06-18 14:33:37 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY011.DAT

out: FLY011.csv, FLY011.kml

2026-06-18 14:33:51 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY012.DAT

out: FLY012.csv, FLY012.kml

2026-06-18 14:34:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY013.DAT

out: FLY013.csv, FLY013.kml

2026-06-18 14:34:23 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY014.DAT

out: FLY014.csv, FLY014.kml

2026-06-18 14:34:39 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY015.DAT

out: FLY015.csv, FLY015.kml

2026-06-18 14:34:52 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY016.DAT

out: FLY016.csv, FLY016.kml

2026-06-18 14:35:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY017.DAT

out: FLY017.csv, FLY017.kml

2026-06-18 14:35:31 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY018.DAT

out: FLY018.csv, FLY018.kml

2026-06-18 14:35:49 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY019.DAT

out: FLY019.csv, FLY019.kml

2026-06-18 14:36:04 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY020.DAT

out: FLY020.csv, FLY020.kml

2026-06-18 14:36:18 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY021.DAT

out: FLY021.csv, FLY021.kml

2026-06-18 14:36:34 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY022.DAT

out: FLY022.csv, FLY022.kml

2026-06-18 14:36:49 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY023.DAT

out: FLY023.csv, FLY023.kml

2026-06-18 14:37:07 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY024.DAT

out: FLY024.csv, FLY024.kml

2026-06-18 14:37:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY025.DAT

out: FLY025.csv, FLY025.kml

2026-06-18 14:37:45 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY026.DAT

out: FLY026.csv, FLY026.kml

2026-06-18 14:38:00 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY027.DAT

out: FLY027.csv, FLY027.kml

2026-06-18 14:38:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY028.DAT

out: FLY028.csv, FLY028.kml

2026-06-18 14:38:38 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY029.DAT

out: FLY029.csv, FLY029.kml

2026-06-18 14:38:52 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY030.DAT

out: FLY030.csv, FLY030.kml

2026-06-18 14:39:10 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY031.DAT

out: FLY031.csv, FLY031.kml

2026-06-18 14:39:49 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY032.DAT

out: FLY032.csv, FLY032.kml

2026-06-18 14:40:11 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY033.DAT

out: FLY033.csv, FLY033.kml

2026-06-18 14:40:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY034.DAT

out: FLY034.csv, FLY034.kml

2026-06-18 14:40:47 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY035.DAT

out: FLY035.csv, FLY035.kml

2026-06-18 14:41:14 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY036.DAT

out: FLY036.csv, FLY036.kml

2026-06-18 14:41:35 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY037.DAT

out: FLY037.csv, FLY037.kml

2026-06-18 14:41:58 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY038.DAT

out: FLY038.csv, FLY038.kml

2026-06-18 14:42:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY039.DAT

out: FLY039.csv, FLY039.kml

2026-06-18 14:42:26 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY040.DAT

out: FLY040.csv, FLY040.kml

2026-06-18 14:42:38 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY041.DAT

out: FLY041.csv, FLY041.kml

2026-06-18 14:42:51 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY042.DAT

out: FLY042.csv, FLY042.kml

2026-06-18 14:43:03 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY043.DAT

out: FLY043.csv, FLY043.kml

2026-06-18 14:43:16 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY044.DAT

out: FLY044.csv, FLY044.kml

2026-06-18 14:43:31 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY045.DAT

out: FLY045.csv, FLY045.kml

2026-06-18 14:43:45 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY046.DAT

out: FLY046.csv, FLY046.kml

2026-06-18 14:43:58 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY047.DAT

out: FLY047.csv, FLY047.kml

2026-06-18 14:44:10 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY048.DAT

out: FLY048.csv, FLY048.kml

2026-06-18 14:44:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY049.DAT

out: FLY049.csv, FLY049.kml

2026-06-18 14:44:37 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY050.DAT

out: FLY050.csv, FLY050.kml

2026-06-18 14:44:53 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY051.DAT

out: FLY051.csv, FLY051.kml

2026-06-18 14:45:05 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY052.DAT

out: FLY052.csv, FLY052.kml

2026-06-18 14:45:17 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY053.DAT

out: FLY053.csv, FLY053.kml

2026-06-18 14:45:31 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY054.DAT

out: FLY054.csv, FLY054.kml

2026-06-18 14:45:45 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY055.DAT

out: FLY055.csv, FLY055.kml

2026-06-18 14:45:59 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY056.DAT

out: FLY056.csv, FLY056.kml

2026-06-18 14:46:11 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY057.DAT

out: FLY057.csv, FLY057.kml

2026-06-18 14:46:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY058.DAT

out: FLY058.csv, FLY058.kml

2026-06-18 14:46:40 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY059.DAT

out: FLY059.csv, FLY059.kml

2026-06-18 14:46:54 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY060.DAT

out: FLY060.csv, FLY060.kml

2026-06-18 14:47:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY061.DAT

out: FLY061.csv, FLY061.kml

2026-06-18 14:47:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY062.DAT

out: FLY062.csv, FLY062.kml

2026-06-18 14:47:42 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY063.DAT

out: FLY063.csv, FLY063.kml

2026-06-18 14:48:02 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY064.DAT

out: FLY064.csv, FLY064.kml

2026-06-18 14:48:07 datcon v4.3.0 rc=1

cmd: DatCon.4.3.0.exe WPSettings.dat

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.